From Ordinary Language to Computational Neuroscience

A Mathematical Foundation for Electrophysiological Analysis

A modular mathematical companion to neuronal

CON^{2}EPHYS Research Manuscript

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Independent exposition inspired by Halmos and Mendelson

Preface

This book begins with statements in ordinary English, not with calculus, matrix notation, or a list of methods. Its reader need not know what a set, function, derivative, matrix, probability, or oscillation is. We will name each mathematical object before asking the reader to manipulate it, and we will distinguish statements about abstract models from statements about nervous systems. The purpose is not to make mathematics superficial: it is to make the deepest methods accessible by building their prerequisites instead of assuming them.

We follow two complementary pedagogical inspirations. Halmos develops the set-theoretic language of mathematics through informal but precise exposition; Mendelson connects logic, sets, and successive number systems to the foundations of analysis. Neither book is copied here. Their themes guide an original scientific text culminating in signal processing, neuronal point processes, statistical inference, and population dynamics.

Every chapter includes formal mathematics, an electrophysiological interpretation, limitations, and exercises. A reader should practice proving elementary results as well as simulating advanced algorithms. Source-data quality, estimand definitions, and null hypotheses receive the same attention as derivations. The earlier edition presented a provisional `matlab/path` mapping. The repository is now verified; for actual MATLAB filenames and method status, consult the companion theorem crosswalk and `matlab/METHODS.tsv`.

Contents

Preface i

How to read this book xiii

I The language of mathematics 1

1 From ordinary English to mathematical statements 2
1.1 Why begin with sentences? . . . 2
1.2 Statements, questions, definitions, and hypotheses . . . 2
1.3 Translating ordinary English . . . 3
1.4 A claim is only as good as its scope . . . 3
1.5 Vagueness, ambiguity, and operational definitions . . . 3
1.6 The logical skeleton of a scientific sentence . . . 3
1.7 Words that hide quantifiers . . . 4
1.8 Model statements versus world statements . . . 4

2 Logic: the grammar of proof 6
2.1 And, or, and not . . . 6
2.2 Converse, inverse, and contrapositive . . . 6
2.3 Quantifiers and their order . . . 6
2.4 Necessary and sufficient conditions . . . 7
2.5 Formal proof laboratory: propositional and quantified logic . . . 7
2.6 Tautologies, contradictions, and valid arguments . . . 9
2.7 Useful equivalences . . . 9
2.8 Nested quantifiers and their negations . . . 10
2.9 Logic and uncertainty . . . 10

3 Definitions, proof, and mathematical maturity 11
3.1 The anatomy of an argument . . . 11
3.2 Direct proof and proof by contrapositive . . . 11
3.3 Contradiction, induction, and counterexamples . . . 11
3.4 Existence, uniqueness, and identifiability . . . 12
3.5 Writing a proof: from scratch work to final form . . . 12
3.6 Proof by cases and the pigeonhole principle . . . 12

3.7 Strong induction and well ordering . . . 13
3.8 Common errors in purported proofs . . . 13

4 Sets and the operations that create them 15
4.1 From a collection to a set . . . 15
4.2 Set operations from ordinary English . . . 15
4.3 Families, indexing, and power sets . . . 16
4.4 The significance for scientific data . . . 16
4.5 Algebra of sets . . . 16
4.6 Symmetric difference and indicator functions . . . 17
4.7 Partitions . . . 17
4.8 Russell's paradox in detail . . . 17

5 Ordered pairs, products, and relations 19
5.1 The order of information matters . . . 19
5.2 What is a relation? . . . 19
5.3 Composing and inverting relations . . . 19
5.4 Equivalence relations and partitions . . . 20
5.5 Closures of relations . . . 20
5.6 Orders . . . 21

6 Functions: inputs, outputs, and transformations 22
6.1 A function is a special relation . . . 22
6.2 Compositions and transformations . . . 22
6.3 Sequences are functions . . . 22
6.4 Functions of functions . . . 23
6.5 Images and preimages of sets . . . 23
6.6 Left and right inverses . . . 23
6.7 Composition, injectivity, and surjectivity . . . 24
6.8 Cardinality and countability . . . 24

7 Bridges from statements to relations and functions 26
Bridge 1: Set equality is equality of membership . . . 26
Bridge 2: Inclusion is a partial order . . . 26
7.1 Lean 4 verification: the first two bridges . . . 28
Bridge 3: Cartesian products define possible pairs . . . 29
7.2 Lean 4 verification: Cartesian products . . . 29
Bridge 4: Composition of relations is associative . . . 29
7.3 Lean 4 verification: relational composition . . . 30
Bridge 5: Equivalence classes form a partition . . . 30
7.4 Lean 4: equivalence classes form a partition . . . 30
Bridge 6: Images preserve unions; preimages preserve unions and intersections . . . 31
7.5 Lean 4: images and preimages . . . 31
Bridge 7: A two-sided inverse is equivalent to bijectivity . . . 32

7.6 Lean 4: two-sided inverses and bijections . . . 32
Bridge 8: Pairs of natural indices can be enumerated . . . 33
7.7 Lean 4: countability of natural-number pairs . . . 33

II The construction of number systems 34

8 Natural numbers, counting, and induction 35
8.1 Counting precedes measurement . . . 35
8.2 Successors and the Peano viewpoint . . . 35
8.3 Induction as a way to prove infinite families of claims . . . 35
8.4 Finite sets and combinatorics . . . 36
8.5 Proving the arithmetic laws from the definitions . . . 36
8.6 Recursion as a way to define sequences . . . 36
8.7 Divisibility, remainders, and modular indices . . . 37
8.8 Counting principles . . . 37

9 Integers: extending counting to differences 39
9.1 Why natural numbers are not enough . . . 39
9.2 Operations on classes . . . 39
9.3 Order and absolute value . . . 39
9.4 A neuroscience example . . . 40
9.5 Proving that the operations are well defined . . . 40
9.6 The ring axioms . . . 40
9.7 Divisibility and the greatest common divisor . . . 40

10 Rational numbers, ratios, and fields 42
10.1 Fractions as equivalence classes . . . 42
10.2 What does a ratio mean physically? . . . 42
10.3 Rationals are dense but incomplete . . . 42
10.4 Well-definedness and the field axioms . . . 43
10.5 Order and the Archimedean property . . . 43
10.6 Decimal expansions . . . 43
10.7 Weighted averages and ratio estimators . . . 43

11 The real number system, order, and completeness 45
11.1 The real line as an ordered field . . . 45
11.2 Bounds, supremum, and infimum . . . 45
11.3 Distances and the triangle inequality . . . 45
11.4 Interval notation and physical uncertainty . . . 46
11.5 Consequences of completeness . . . 46
11.6 Existence of square roots . . . 46
11.7 Nested intervals and uncountability . . . 47
11.8 Floating-point numbers are not real numbers . . . 47

12 Complex numbers: amplitude and phase **49**
12.1 Pairs with new arithmetic . . . 49
12.2 A plane with angles . . . 49
12.3 Why complex numbers are indispensable for signals . . . 49
12.4 The complex numbers form a field . . . 50
12.5 Euler's formula and de Moivre . . . 50
12.6 Roots of unity and the discrete Fourier basis . . . 50
12.7 Complex numbers as rotations and scalings . . . 51

13 Bridges between counting and number systems **52**
Bridge 9: Associativity of addition from recursion . . . 52
13.1 Lean 4: addition associativity by induction . . . 52
Bridge 10: Constructing integers from pairs of naturals . . . 53
13.2 Lean 4: integers as equivalence classes of pairs . . . 54
Bridge 11: Rational arithmetic is independent of representation . . . 54
Bridge 12: There is a rational between distinct reals . . . 55
Bridge 13: Archimedean property from completeness . . . 55
Bridge 14: A supremum can be approached from below . . . 55
Bridge 15: Floor partitions nonnegative time into half-open bins . . . 55
Bridge 16: Conjugation makes complex magnitude multiplicative . . . 56

III Calculus and analysis **57**

14 Sequences, limits, and continuity **58**
14.1 An infinite list is a function . . . 58
14.2 Cauchy sequences and completeness . . . 58
14.3 Functions of a real variable . . . 58
14.4 Why this matters for continuous-time recordings . . . 58
14.5 Algebra of limits . . . 59
14.6 Monotone sequences and the Bolzano–Weierstrass theorem . . . 59
14.7 Series . . . 60
14.8 Continuity on closed intervals . . . 60

15 Derivatives, integrals, and differential equations **62**
15.1 A derivative is a limit of ratios . . . 62
15.2 Integration as accumulation . . . 62
15.3 Differential equations: rules for evolving states . . . 62
15.4 A distinction between models and data . . . 63
15.5 Rules of differentiation . . . 63
15.6 The mean value theorem . . . 63
15.7 Taylor's theorem and the error of finite differences . . . 64
15.8 The exponential function . . . 64
15.9 Linear first-order equations in general . . . 64

16 Several variables, optimization, and geometry **66**
16.1 The gradient is a local direction of change . . . 66
16.2 Second derivatives and the Hessian . . . 66
16.3 Optimization as fitting a model . . . 66
16.4 The manifold idea starts locally . . . 67
16.5 Differentiability in several variables . . . 67
16.6 Convexity . . . 67
16.7 Constrained optimization and Lagrange multipliers . . . 68
16.8 Gradient descent and Newton's method . . . 68

17 Integration, function spaces, and transforms **69**
17.1 From vectors of numbers to functions . . . 69
17.2 Orthogonality and approximation . . . 69
17.3 Measure and probability: what does "almost everywhere" mean? . . . 69
17.4 Generalized signals and impulses . . . 70
17.5 The Riemann integral and its limits . . . 70
17.6 When can limits and integrals be exchanged? . . . 70
17.7 Inequalities in inner-product spaces . . . 70
17.8 Completeness of function spaces . . . 71
17.9 Convolution . . . 71

18 Bridges from number systems to calculus **73**
Bridge 17: A convergent sequence has only one limit . . . 73
Bridge 18: Limits respect addition and multiplication . . . 73
Bridge 19: Bounded monotone real sequences converge . . . 73
Bridge 20: Composition preserves continuity . . . 74
Bridge 21: A continuous curve cannot jump over an intermediate value . . . 74
Bridge 22: A continuous function has extrema on a compact interval . . . 74
Bridge 23: Sum and product differentiation rules . . . 75
Bridge 24: The chain rule follows from local linear approximation . . . 75
Bridge 25: Differentiability provides a first-order remainder . . . 75
Bridge 26: Integrating a rate recovers accumulated change . . . 75

IV Linear algebra and numerical models **77**

19 Vectors, matrices, and linear mappings **78**
19.1 Arrays are not merely spreadsheets . . . 78
19.2 Rank, kernels, and invertibility . . . 78
19.3 Inner products and projections . . . 78
19.4 Least squares as orthogonal projection . . . 79
19.5 Vector spaces, subspaces, and bases . . . 79
19.6 Linear maps and their matrices . . . 79
19.7 Row and column spaces; the four fundamental subspaces . . . 80

19.8 Norms and condition numbers . . . 80

20 Eigenvalues, singular values, and principal components 82
20.1 Eigenvectors as preserved directions . . . 82
20.2 Singular value decomposition . . . 82
20.3 Deriving PCA . . . 82
20.4 What PCA does not promise . . . 83
20.5 The spectral theorem . . . 83
20.6 The Rayleigh quotient and Courant–Fischer . . . 83
20.7 Proof of the SVD and its geometry . . . 84
20.8 Non-symmetric matrices and dynamics . . . 84

21 Tensors, reduced-rank models, and inverse problems 86
21.1 More than two indices . . . 86
21.2 Reduced-rank regression . . . 86
21.3 The forward model is not the inverse model . . . 86
21.4 Regularized reconstruction . . . 87
21.5 The Moore–Penrose pseudoinverse . . . 87
21.6 Ill-posedness and the Picard condition . . . 87
21.7 Tikhonov regularization through the SVD . . . 87
21.8 Tensor rank and uniqueness . . . 88

22 Bridges from vectors to neural population geometry 89
Bridge 27: Linear independence makes coordinate representations unique . . . 89
Bridge 28: Dimensions split into visible and invisible directions . . . 89
Bridge 29: Cauchy–Schwarz bounds the inner product . . . 89
Bridge 30: The triangle inequality follows from Cauchy–Schwarz . . . 90
Bridge 31: Orthogonal components add energy, not amplitude . . . 90
Bridge 32: Independent vectors can be orthogonalized . . . 90
Bridge 33: Projection solves constrained least squares . . . 90
Bridge 34: Covariance matrices are positive semidefinite . . . 91

V Probability, statistics, and time 92

23 Probability begins with sets of possible outcomes 93
23.1 Experiments and events . . . 93
23.2 Conditional probability . . . 93
23.3 Conditioning changes the question . . . 93
23.4 Consequences of the axioms . . . 94
23.5 Continuity of probability and sigma-algebras . . . 94
23.6 Independence of several events . . . 94
23.7 Conditional probability as a new probability measure . . . 95

24 Random variables, expectation, and statistical moments 96

24.1 A random variable is a function . . . 96
24.2 Means, variance, skewness, kurtosis . . . 96
24.3 Covariance and correlation . . . 96
24.4 The law of total variance . . . 97
24.5 Expectation as an integral and its linearity . . . 97
24.6 Fundamental inequalities . . . 97
24.7 Common distributions . . . 98
24.8 Independence of random variables and sums . . . 98
24.9 Moment generating functions . . . 98

25 Inference, likelihood, and stochastic processes 100
25.1 Data, models, estimators . . . 100
25.2 Time indexed random variables . . . 100
25.3 Conditional structure and hidden state . . . 100
25.4 Laws of large numbers and the central limit theorem . . . 101
25.5 Properties of estimators . . . 101
25.6 Maximum likelihood . . . 101
25.7 Stationary processes: examples . . . 102

26 Bridges from sets to random variables and spike counts 103
Bridge 35: Events form a sigma-algebra rather than an arbitrary list . . . 103
Bridge 36: Independence survives taking complements . . . 103
Bridge 37: Expectation is linear even for dependent variables . . . 103
Bridge 38: Indicator variables encode event counts . . . 104
Bridge 39: Conditional expectation obeys the tower property . . . 104
Bridge 40: Total variance separates within- and between-state effects . . . 104
Bridge 41: Covariance is bilinear under finite moments . . . 105
Bridge 42: Random Poisson intensity creates overdispersion . . . 105

VI Mathematics of electrophysiological signals 106

27 From a physical voltage to a sampled sequence 107
27.1 What electrodes actually measure . . . 107
27.2 Sampled time is a lattice . . . 107
27.3 The ideal sampling theorem and its assumptions . . . 107
27.4 Storage, quantization, and memory mapping . . . 107
27.5 Aliasing as an equivalence relation . . . 108
27.6 Quantization . . . 108
27.7 Resampling . . . 108
27.8 Timing and synchronization . . . 108
27.9 Sample-and-hold and the zero-order hold . . . 109

28 Fourier analysis, convolution, and digital filtering 110
28.1 The discrete Fourier transform . . . 110

28.2 Convolution and linear time-invariant systems . . . 110
28.3 FIR, IIR, and zero-phase filtering . . . 110
28.4 Cross-correlation and spectral duality . . . 111
28.5 Proof of DFT inversion and Parseval . . . 111
28.6 Symmetry and the meaning of negative frequencies . . . 111
28.7 Circular convolution . . . 111
28.8 Leakage and windows . . . 112
28.9 The uncertainty principle . . . 112
28.10 Filter design specifications . . . 112

29 Spectrograms, multitapers, and periodic-versus-aperiodic spectra 114
29.1 Why spectral estimates fluctuate . . . 114
29.2 Spectrograms and the time-frequency trade-off . . . 114
29.3 Multitaper estimation . . . 114
29.4 Spectral peaks over the background . . . 115
29.5 The periodogram is inconsistent . . . 115
29.6 Bias, variance, and degrees of freedom . . . 115
29.7 Slepian tapers . . . 116
29.8 The aperiodic component . . . 116

30 Analytic signals, circular statistics, and cross-frequency coupling 118
30.1 The analytic signal . . . 118
30.2 Filtering comes before interpretation . . . 118
30.3 Circular means and concentration . . . 118
30.4 Phase locking and phase-amplitude dependence . . . 119
30.5 The interpretation trap . . . 119
30.6 The Hilbert transform in the frequency domain . . . 119
30.7 When is amplitude-phase decomposition meaningful? . . . 119
30.8 Distributions on the circle . . . 120
30.9 Two PAC estimators and their assumptions . . . 120

31 Cross-spectra, coherence, and Granger prediction 122
31.1 From autocovariance to cross-covariance . . . 122
31.2 Common input and reference contamination . . . 122
31.3 Autoregressive models and direction of prediction . . . 122
31.4 Transfer entropy and conditional dependence . . . 123
31.5 The spectral density matrix . . . 123
31.6 Coherence under a common source . . . 123
31.7 Bias of estimated coherence . . . 124
31.8 The imaginary part of coherency . . . 124
31.9 Spectral Granger causality . . . 124

32 Extracellular potentials and source reconstruction 126
32.1 Potential is a field . . . 126

32.2 Current source density . . . 126
32.3 Assumptions should travel with the result . . . 126
32.4 From Maxwell's equations to the quasistatic approximation . . . 127
32.5 The point-source solution and superposition . . . 127
32.6 Uniqueness and nonuniqueness . . . 127
32.7 Inverse CSD and the role of boundaries . . . 128

33 Bridges from finite sequences to spectra 129
Bridge 43: Complex exponentials form an orthogonal finite family . . . 129
Bridge 44: The DFT admits an exact finite inverse . . . 129
Bridge 45: Finite Fourier energy has an exact accounting identity . . . 129
Bridge 46: Circular convolution becomes multiplication under the DFT . . . 130
Bridge 47: Time shifts rotate Fourier phase . . . 130
Bridge 48: Uniform samples cannot distinguish aliased sinusoids . . . 130

VII Point processes and neuronal populations 132

34 Spike trains as point processes 133
34.1 Events rather than amplitudes . . . 133
34.2 Poisson models and their scope . . . 133
34.3 Binning and the loss of exact timing . . . 133
34.4 Constructing the homogeneous Poisson process . . . 134
34.5 Inhomogeneous processes, thinning, and time rescaling . . . 134
34.6 Renewal processes and interval statistics . . . 134
34.7 Second-order statistics: autocorrelograms and the Bartlett spectrum . . . 135

35 Spike-train GLMs, resampling, and conditional hypotheses 136
35.1 A Poisson generalized linear model . . . 136
35.2 History filters and coupling . . . 136
35.3 Model assessment rather than fit alone . . . 136
35.4 Jitter and conditional resampling . . . 136
35.5 Concavity of the Poisson GLM log-likelihood . . . 137
35.6 Basis functions for filters . . . 137
35.7 Goodness of fit with time rescaling . . . 138
35.8 Validity of Monte Carlo tests . . . 138

36 Neuronal covariance and the geometry of population responses 139
36.1 One neuron versus a population . . . 139
36.2 Shared gain and overdispersion . . . 139
36.3 Latent factors and factor analysis . . . 139
36.4 Random-matrix caution . . . 140
36.5 Covariance from a latent factor model . . . 140
36.6 Noise correlations and coding . . . 140
36.7 Estimating correlation from spike counts . . . 140

36.8 Signal versus noise correlation from data . . . 141

37 State-space dynamics, neural manifolds, and rotations 142
37.1 A state is more than a snapshot . . . 142
37.2 Continuous dynamics and stability . . . 142
37.3 jPCA and rotational population trajectories . . . 142
37.4 Neural manifolds and coordinate nonuniqueness . . . 142
37.5 Nonlinear latent models . . . 143
37.6 Solving linear systems of differential equations . . . 143
37.7 Planar systems . . . 143
37.8 The Kalman filter . . . 144
37.9 Observability and identifiability . . . 144

VIII Scientific inference and reproducible methods 145

38 Hypothesis testing, resampling, and uncertainty 146
38.1 What is a hypothesis test? . . . 146
38.2 Type I error, power, and effect size . . . 146
38.3 Resampling methods . . . 146
38.4 Hierarchical data and pseudoreplication . . . 147
38.5 Bias, generalization, and validation . . . 147
38.6 The Neyman–Pearson lemma . . . 147
38.7 Confidence intervals and their duality with tests . . . 147
38.8 Multiple comparisons . . . 148
38.9 The bootstrap . . . 148

39 Numerical analysis and reproducible scientific computation 150
39.1 A formula is not yet an algorithm . . . 150
39.2 Conditioning and algorithmic stability . . . 150
39.3 Memory and complexity . . . 150
39.4 Testing the mathematical contract . . . 150
39.5 Scientific metadata as part of the result . . . 151
39.6 Floating-point error analysis . . . 151
39.7 Backward stability . . . 151
39.8 Complexity of core algorithms . . . 152
39.9 Randomness and reproducibility . . . 152

40 A complete path from first definitions to real datasets 153
40.1 Project I: two signals that look coupled but are not . . . 153
40.2 Project II: what a theta peak actually means . . . 153
40.3 Project III: a manufactured phase-amplitude relationship . . . 153
40.4 Project IV: spikes, jitter, and conditional inference . . . 154
40.5 Project V: latent neuronal dynamics . . . 154
40.6 A research conclusion requires four layers . . . 154

40.7 Project VI: a point-process model of a place cell . . . 154
40.8 Project VII: source mixing and laminar profiles . . . 154
40.9 Project VIII: population information and correlations . . . 155
40.10A checklist for each project . . . 155

A Notation, assumptions, and units **156**

B Provisional mathematical map to `matlab/path` **157**

C Selected answers, proof sketches, and further problems **158**
C.1 Selected answers . . . 158
C.2 Capstone proof and analysis problems . . . 158

How to read this book

The intended reader begins with ordinary English and curiosity about neural signals. Part I requires no mathematical notation beyond what it introduces. Parts II and III build number systems, limits, and finite-dimensional algebra before the book assumes calculus, probability, or complex phase. Later chapters use the same format: meaning in words, formal definition, calculation or derivation, a neuroscience connection, a failure mode, and exercises. A novice may work every exercise; an experienced reader can use the cross-reference path below to fill only missing prerequisites.

Suggested prerequisite chains

Hilbert and PAC
: English → functions → complex numbers → sequences/integrals → Fourier → analytic signal → circular inference.

PCA and manifolds
: Relations → functions → real numbers → vector spaces → covariance → SVD/PCA → dynamical systems.

Spike GLMs
: Sets → functions → counting → probability → conditional probability → Poisson processes → likelihood and GLM.

Granger analysis
: Covariance → stochastic processes → cross spectra → multivariate linear prediction → null models and validation.

Current source density
: Coordinates → derivatives → multivariable calculus → forward measurement physics → regularized inverse methods.

Technical vocabulary

An *estimand* is the quantity the analysis intends to estimate; an *estimator* is a data-dependent rule for estimating it. A *model* is a set of assumed mathematical relationships; a *measurement* is an observation through an instrument. *Association* is not automatically *causation*. *Prediction* need not identify an intervention effect. The word "significant" should be accompanied by the hypothesis and statistical criterion being used.

About the two foundational books

The structure draws inspiration from Halmos' conversational construction of sets and relations [22], and Mendelson's progression from logic and set theory to the number systems and real-number completeness [23]. The explanations, examples, and proofs here are written afresh for CON^{2}EPHYS and are not reproduced sections of either text.

Part I

The language of mathematics

Chapter 1

From ordinary English to mathematical statements

Reading goal 1.1. Read, question, and translate statements before manipulating symbols. No mathematical prerequisite is assumed.

1.1 Why begin with sentences?

A scientist looks at an experiment and makes claims: "this animal ran faster," "these spikes followed the stimulus," or "these electrodes recorded the same rhythm." At first, the claims are expressed in ordinary language. Mathematics begins when we decide exactly what would make a claim true, how it could be false, and what follows from it. The symbols are a shorthand for decisions already made in words. Learning a formula without learning its grammar is like memorizing a sentence in a language one cannot read.

Consider "every recorded neuron fired." Does this refer to every neuron in the brain, every electrode channel, or every neuron retained after spike sorting? The statement is incomplete until we specify its domain of discourse. Consider "some neurons were silent." The word *some* means *at least one*; it does not exclude the possibility that many neurons were silent. The word *or* in standard mathematics ordinarily permits both alternatives. When a manuscript says "the signal is oscillatory," we must also decide what counts as an oscillation. A label is not yet an operational test.

1.2 Statements, questions, definitions, and hypotheses

A *proposition* is a declarative statement to which a truth value can be assigned, given a specified interpretation. "Two is even" is a proposition. "Is two even?" is a question, while "Let n be the number of spikes" is an instruction introducing notation. "For every t in the recording interval, $x(t) > 0$" becomes a proposition once x and the recording interval are fixed. An *open sentence* such as "n is even" has a free variable and does not yet assert anything about a particular number.

A *definition* establishes how a term or symbol will be used; a *hypothesis* asserts a fact that may or may not hold; a *theorem* is a proposition established from previously stated assumptions by valid reasoning. In applied science we must further distinguish a mathematical theorem, which follows within a model, from an empirical assertion about the world, which requires observations.

1.3 Translating ordinary English

Let U be the set of units retained after quality control. Introduce the predicate $F(u)$ for "unit u fired in a designated 100-ms window." Then "every retained unit fired" becomes

$$\forall u \in U,\ F(u).$$

"At least one retained unit did not fire" becomes $\exists u \in U$ such that $\neg F(u)$. These are logical negations of one another. "No unit fired" instead means $\forall u \in U, \neg F(u)$. Notice the difference between a negative statement about *all* units and a statement that *not all* units fired.

In data analysis, "higher gamma amplitude when theta phase is near its peak" must be turned into an explicit comparison: define phase bins, summarize amplitudes, select a test statistic, and specify a null hypothesis. This transition from sentence to mathematical object is the beginning of sound methodology.

1.4 A claim is only as good as its scope

Suppose 40 of 50 detected units increase firing rate during a task. It is legitimate to say "80 percent of these detected units increased." It is not automatically legitimate to say "80 percent of neurons in this brain area increased." Electrode sampling, sorting errors, and selection rules change the population to which a claim refers. We will repeatedly separate *a set of observed objects*, *a mathematical model*, and *a physiological population*.

Worked example 1.2 (The ambiguity of "only")**.** Compare "only neurons in CA1 show a theta rhythm" with "neurons in CA1 show only a theta rhythm." The first restricts the location of units exhibiting theta; the second restricts the rhythms observed in CA1 units. Write down what observation would contradict each statement. The two claims can disagree even on the same recording.

1.5 Vagueness, ambiguity, and operational definitions

Ordinary language tolerates two different kinds of imprecision. A word is *ambiguous* when it has several distinct meanings ("rate" may mean spikes per second or spikes per trial). A word is *vague* when its meaning has no sharp boundary ("high" power, "many" spikes, "fast" ripples). Mathematics removes ambiguity by choosing one meaning, and removes vagueness by choosing a threshold or a comparison. Both choices are decisions, not discoveries.

An *operational definition* replaces a vague predicate by a rule that any analyst could apply to the same data and obtain the same verdict. For example, "the trial contains a ripple" may be replaced by: "the 150–250 Hz band-passed envelope exceeds the session median by five median absolute deviations for at least 15 consecutive milliseconds." This rule is reproducible, but it is not thereby *correct*: other reasonable thresholds yield other sets of ripples. A sound report states the rule and examines how conclusions change when the rule changes.

Remark 1.3. *A precise definition can be scientifically poor. Precision guarantees that two readers agree on what was computed; it does not guarantee that the computed object corresponds to the physiological phenomenon that motivated the word.*

1.6 The logical skeleton of a scientific sentence

Most claims in a methods or results section have a recurring skeleton:

1. a *domain*: the objects being talked about (units, trials, sessions, animals);
2. a *predicate* applied to those objects ("increased its rate", "was phase locked");
3. a *quantifier* stating how many objects satisfy the predicate (all, some, most, a fraction);
4. a *condition* restricting when the claim applies ("during running", "after stimulation");
5. a *standard of evidence* by which the predicate was judged (a test, a threshold, an estimate with uncertainty).

Making each component explicit is the first step toward mathematical form. Consider "place cells remap in the novel environment." Its domain is presumably units classified as place cells by some criterion; its predicate is a change of firing field; its quantifier is unstated (all? most? a significant fraction?); its condition is exposure to a novel arena; and its standard of evidence must specify how two firing maps are compared. A sentence missing any component is not false; it is *incomplete*.

1.7 Words that hide quantifiers

Several everyday words carry hidden logical structure:

- "*any*" usually means *every* in positive statements ("any unit can be recorded") but *at least one* in negative or interrogative ones ("no unit showed any change");
- "*unless*" means *if not*: "P unless Q" is $\neg Q \Rightarrow P$;
- "*only if*" marks a necessary condition: "P only if Q" is $P \Rightarrow Q$, not $Q \Rightarrow P$;
- "*the*" presupposes existence and uniqueness: "the peak frequency" assumes that exactly one peak exists;
- "*typically*" and "*tends to*" are statistical statements and require a distribution, not a quantifier.

The definite article deserves special attention. Writing "let f^* be the peak frequency of the spectrum" is meaningful only after establishing that the maximum exists and is attained at a single point. If two frequencies attain the same maximal power, the phrase "the peak" has no referent, and software will silently pick one (often the first index), which is an arbitrary convention rather than a property of the data.

Worked example 1.4 (Rewriting a results sentence)**.** Original: "Interneurons fire before pyramidal cells." A careful rewrite fixes each component. Let I be units classified as putative interneurons by waveform width below 0.4 ms; let P be the remaining units; for a pair $(i, p) \in I \times P$ let $\delta(i, p)$ be the lag of the peak of the cross-correlogram within ± 10 ms. The quantitative claim might be: "for 68% of pairs in $I \times P$, $\delta(i, p) < 0$, compared with 50% expected under a jitter null." The rewritten sentence is longer, but now a reader can check it, reproduce it, and disagree with it in a specific way.

1.8 Model statements versus world statements

A model statement has the form "*if* the data were generated by mechanism M with parameters θ, *then* quantity Q would have property R." It may be proved. A world statement has the form "quantity Q in this recording has property R." It is supported or contradicted by observations.

Much confusion in quantitative neuroscience comes from presenting a model statement in the grammar of a world statement. "The spectrum has a $1/f$ background" could mean that a fitted power law describes the estimate well over a range, or it could mean that the generative process has a specific scale-free structure. The first is descriptive; the second requires additional argument. Throughout this book we mark which kind of statement is being made.

Exercises

1. Translate "not every trial contains a ripple" into a quantified sentence and simplify its negation.
2. Explain why "all spikes occur in bursts" can be vacuously true for an empty spike train.
3. Identify the observational unit in each claim: neuron, trial, event, animal, and session.
4. Write one empirically falsifiable claim about coherence and one purely mathematical definition of coherence.
5. Rewrite "P unless Q" and "P only if Q" as implications, and give a neuroscience sentence of each form.
6. Give an operational definition of "burst" for a spike train, and name two alternative definitions that would classify some spikes differently.
7. Find a sentence in a paper you have read that uses "the" for an object whose uniqueness was not established. Rewrite it.
8. Decompose "most hippocampal neurons were modulated by speed" into domain, predicate, quantifier, condition, and standard of evidence.
9. Classify each of the following as a model statement or a world statement: "a Poisson process has Fano factor one"; "these units had Fano factors near one"; "cortical neurons fire like Poisson processes."

Chapter 2

Logic: the grammar of proof

Reading goal 2.1. Learn the connectives, truth conditions, quantifiers, and the difference between logical implication and scientific causation.

2.1 And, or, and not

Let P mean "the animal is running" and Q mean "theta power exceeds a specified threshold." The conjunction $P \wedge Q$ is true precisely when both claims are true. The disjunction $P \vee Q$ is false only when both are false; in mathematical usage it is *inclusive*. Negation $\neg P$ reverses the truth value of P. The entire truth table is small enough to check directly:

P	Q	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$
T	T	T	T	T
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

The conditional $P \Rightarrow Q$ is false only when P holds but Q does not. This is the *material implication*; it is not an assertion of a mechanism or a causal influence. Indeed, a false antecedent makes the material implication true without producing any biological event.

2.2 Converse, inverse, and contrapositive

From $P \Rightarrow Q$ one may infer the logically equivalent contrapositive $\neg Q \Rightarrow \neg P$. The converse $Q \Rightarrow P$ and inverse $\neg P \Rightarrow \neg Q$ are different claims. A low firing rate might be observed during inactivity, but inactivity need not be the only circumstance producing that rate. We may write $P \Leftrightarrow Q$ for a *biconditional*, meaning that both implications hold.

De Morgan's rules supply a useful bridge to data selection: $\neg(P \wedge Q) \Leftrightarrow (\neg P \vee \neg Q)$ and $\neg(P \vee Q) \Leftrightarrow (\neg P \wedge \neg Q)$. Thus the complement of trials satisfying both inclusion criteria is the collection failing at least one criterion.

2.3 Quantifiers and their order

The phrase "for every" is denoted $\forall$, and "there exists" by $\exists$. These operations are not interchangeable: $\forall i \, \exists t \, F(i,t)$ says each neuron fires at some time, possibly a different time for every neuron. The stronger statement $\exists t \, \forall i \, F(i,t)$ claims there is one time at which every neuron

fires. Neither form alone specifies simultaneity to a millisecond: that depends on how t and F are defined.

Negation propagates through quantifiers by the rules

$$\neg\forall x\, P(x) \;\Leftrightarrow\; \exists x\, \neg P(x), \qquad \neg\exists x\, P(x) \;\Leftrightarrow\; \forall x\, \neg P(x).$$

For a finite dataset, verifying a universal claim requires checking all included cases; refuting it requires only one counterexample.

2.4 Necessary and sufficient conditions

If $P \Rightarrow Q$, then P is *sufficient* for Q, while Q is *necessary* for P. An example: if a matrix A is invertible, its columns are linearly independent. In the square finite-dimensional case, the converse also holds; the statements are equivalent. Contrast this with "if the signal is a sinusoid, it has a spectral peak," whose converse may fail for many reasons. Whenever an algorithm claims to "detect oscillations," ask whether its acceptance rule is necessary, sufficient, both, or neither for the physiological interpretation.

2.5 Formal proof laboratory: propositional and quantified logic

The five mathematical statements below are shown alongside their Lean 4 implementations. A successful Lean build means the kernel accepts the propositions *as typed*; it does not establish the scientific interpretation of a null hypothesis.

English mathematics

LOGIC-001. De Morgan's conjunction law: for arbitrary propositions P, Q, $\neg(P \wedge Q) \iff \neg P \vee \neg Q$. If not both criteria hold, at least one fails. *Proof:* by excluded middle, either P is false, or P holds, in which case Q must be false. Conversely, either negated conjunct contradicts $P \wedge Q$.

LOGIC-002. De Morgan's disjunction law: $\neg(P \vee Q) \iff \neg P \wedge \neg Q$. To fail either-of-two inclusion rules is to fail both. *Proof:* $\neg(P \vee Q)$ separately contradicts each disjunct. Conversely, $\neg P \wedge \neg Q$ refutes every proposed witness for $P \vee Q$.

LOGIC-003. Contraposition: $(P \Rightarrow Q) \iff (\neg Q \Rightarrow \neg P)$. The backward direction is classical: assuming $\neg Q \Rightarrow \neg P$, if P held but Q did not, one would obtain $\neg P$, a contradiction; by excluded middle, Q must hold. The forward implication follows constructively: from $P \Rightarrow Q$ and $\neg Q$, assuming P would contradict $\neg Q$.

LOGIC-004. Quantifier negation: $\neg(\forall x\, R(x)) \iff \exists x\, \neg R(x)$. A counterexample disproves a universal claim. *Proof:* a failing witness contradicts the universal assertion. Conversely, if no witness failed, classical double-negation elimination would give $R(x)$ for every x, contradicting $\neg\forall x\, R(x)$.

LOGIC-005. Existential negation: $\neg(\exists x\, R(x)) \iff \forall x\, \neg R(x)$. *Proof:* if a witness existed, it would contradict the assumption that none exists. Conversely, if every candidate fails, an assumed witness immediately yields a contradiction.

Proof strategy. The statements about negating conjunction, contraposition and negating a universal quantifier use classical reasoning. The Lean proof invokes Mathlib tactics that elaborate these elementary steps; the file prints their axiom dependencies.

Lean 4 / Mathlib

```
import Mathlib

namespace Con2physFormal.Logic

-- LOGIC-001: the negation of a conjunction.
theorem deMorganAnd (P Q : Prop) :
    ¬ (P ∧ Q) ↔ ¬ P ∨ ¬ Q := by
  tauto

-- LOGIC-002: the negation of a disjunction.
theorem deMorganOr (P Q : Prop) :
    ¬ (P ∨ Q) ↔ ¬ P ∧ ¬ Q := by
  tauto

-- LOGIC-003: contraposition (classical converse).
theorem contrapositive (P Q : Prop) :
    (P → Q) ↔ (¬ Q → ¬ P) := by
  tauto

-- LOGIC-004: the negation of a universal quantifier.
theorem notForall {α : Type*} (P : α → Prop) :
    (¬ ∀ x, P x) ↔ ∃ x, ¬ P x := by
  push Not
  rfl

-- LOGIC-005: the negation of an existential quantifier.
theorem notExists {α : Type*} (P : α → Prop) :
    (¬ ∃ x, P x) ↔ ∀ x, ¬ P x := by
  push Not
  rfl

#print axioms deMorganAnd
#print axioms deMorganOr
#print axioms contrapositive
#print axioms notForall
#print axioms notExists

end Con2physFormal.Logic
```

Correspondence and provenance: each Lean declaration has the same quantification over propositions or predicates as the adjacent English statement. Mathlib provides the underlying logic and tactics. See the pinned dependency in `books/formal_foundations/lean/lakefile.lean`.

Worked example 2.2 (A null hypothesis in plain language)**.** Suppose H_0 states that the temporal positions of spikes inside specified jitter windows carry no additional information beyond their counts. Rejecting H_0 is evidence against this particular conditional model. It does *not* establish that a synapse connects the neurons, or that all alternative null models would also be rejected. Compare [1].

2.6 Tautologies, contradictions, and valid arguments

A compound formula is a *tautology* if it is true under every assignment of truth values to its atomic letters, and a *contradiction* if it is false under every assignment. The law of excluded middle $P \vee \neg P$ is a tautology; $P \wedge \neg P$ is a contradiction. Two formulas φ and ψ are *logically equivalent*, written $\varphi \equiv \psi$, when $\varphi \Leftrightarrow \psi$ is a tautology.

An *argument* consists of premises $\varphi_1, \dots, \varphi_k$ and a conclusion ψ. It is *valid* when every assignment making all premises true also makes ψ true; equivalently, when $(\varphi_1 \wedge \cdots \wedge \varphi_k) \Rightarrow \psi$ is a tautology. Validity concerns form, not content: a valid argument with a false premise can have a false conclusion, and an invalid argument can happen to have a true conclusion.

Proposition 2.3 (Four standard inference rules)**.** *The following argument forms are valid:*

1. modus ponens*: from P and $P \Rightarrow Q$, infer Q;*
2. modus tollens*: from $\neg Q$ and $P \Rightarrow Q$, infer $\neg P$;*
3. hypothetical syllogism*: from $P \Rightarrow Q$ and $Q \Rightarrow R$, infer $P \Rightarrow R$;*
4. disjunctive syllogism*: from $P \vee Q$ and $\neg P$, infer Q.*

Proof. We check (2); the others are similar. Suppose $\neg Q$ and $P \Rightarrow Q$ are both true. If P were true, then by the truth table of the conditional Q would be true, contradicting $\neg Q$. Hence P is false, i.e. $\neg P$ is true. For (3), suppose both premises are true and P is true; then Q is true by the first premise and R by the second, so $P \Rightarrow R$ cannot be falsified. For (4), if $P \vee Q$ is true and P is false, the only remaining way for the disjunction to hold is that Q is true. □

Two *invalid* forms are so common that they have names. *Affirming the consequent* infers P from Q and $P \Rightarrow Q$; *denying the antecedent* infers $\neg Q$ from $\neg P$ and $P \Rightarrow Q$. In both cases the assignment $P = F$, $Q = T$ makes the premises true and the conclusion false. A frequent scientific instance: "if the region generates gamma oscillations, the LFP shows a gamma peak; the LFP shows a gamma peak; therefore the region generates gamma." Volume conduction from another region, or a harmonic of a lower rhythm, also produces the peak.

2.7 Useful equivalences

The following equivalences can each be verified with a four- or eight-row truth table and will be used without comment later.

$$
\begin{aligned}
P \Rightarrow Q &\equiv \neg P \vee Q && \text{(material implication)}\\
P \Rightarrow Q &\equiv \neg Q \Rightarrow \neg P && \text{(contraposition)}\\
\neg(P \Rightarrow Q) &\equiv P \wedge \neg Q && \text{(negated conditional)}\\
P \wedge (Q \vee R) &\equiv (P \wedge Q) \vee (P \wedge R) && \text{(distributivity)}\\
(P \wedge Q) \Rightarrow R &\equiv P \Rightarrow (Q \Rightarrow R) && \text{(exportation)}\\
P \Leftrightarrow Q &\equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P) && \text{(biconditional)}
\end{aligned}
$$

The negated conditional is the most practically useful: to show that "P implies Q" fails, one must exhibit a case where P holds and Q does not. A case where P fails is irrelevant.

Proposition 2.4. *For any propositions, $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$.*

Proof. By material implication, $\neg(P \Rightarrow Q) \equiv \neg(\neg P \vee Q)$. By De Morgan's rule this is $\neg\neg P \wedge \neg Q$, and double negation gives $P \wedge \neg Q$. □

2.8 Nested quantifiers and their negations

Many definitions in analysis have three or more alternating quantifiers. The definition of a limit, met in Chapter 14, reads

$$\forall \varepsilon > 0 \; \exists N \; \forall n \geq N : \; |x_n - L| < \varepsilon.$$

Its negation is obtained mechanically by moving $\neg$ inward one quantifier at a time, swapping $\forall$ and $\exists$:

$$\exists \varepsilon > 0 \; \forall N \; \exists n \geq N : \; |x_n - L| \geq \varepsilon.$$

In words: there is a fixed tolerance that infinitely late terms keep violating. Note that the bounded quantifiers $\forall \varepsilon > 0$ and $\exists n \geq N$ keep their bounds under negation; only the quantifier type and the inner predicate change.

The *uniqueness quantifier* $\exists! x \, P(x)$ ("there is exactly one x with P") abbreviates

$$\exists x \, \big(P(x) \wedge \forall y \, (P(y) \Rightarrow y = x) \big).$$

Existence and uniqueness are proved separately: one exhibits an x, then shows that any two such objects coincide.

Worked example 2.5 (Negating a selectivity criterion)**.** Suppose a unit is called "direction selective" if there is a direction d^* such that for every other tested direction d, the mean rate at d^* exceeds the rate at d by at least 5 spikes/s. Symbolically, $\exists d^* \, \forall d \neq d^* : \; r(d^*) - r(d) \geq 5$. Its negation, "not direction selective", is $\forall d^* \, \exists d \neq d^* : \; r(d^*) - r(d) < 5$: every candidate preferred direction has at least one competitor within 5 spikes/s. This is a much weaker statement than "the unit responds equally to all directions," and the two must not be conflated in a results section.

2.9 Logic and uncertainty

Classical logic assigns each proposition exactly one of two truth values. Empirical claims usually come with uncertainty, and it is tempting to treat a p-value or posterior probability as a "degree of truth." This is a category mistake. A probability is assigned by a model to an event; the proposition "this event occurred" remains true or false. Probability theory (Part V) extends logic by attaching numbers to sets of outcomes, but every probabilistic statement is itself a classical proposition about those numbers, and proofs about probabilities use exactly the rules of this chapter.

Exercises

1. Construct a complete truth table for $(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow \neg P)$.
2. Negate "for every frequency there exists a channel with nonzero power."
3. Give a counterexample to the inference from $P \Rightarrow Q$ and Q to P.
4. Explain why statistical dependence is not equivalent to causal influence.
5. Use a truth table to show that $(P \Rightarrow Q) \wedge (Q \Rightarrow R) \Rightarrow (P \Rightarrow R)$ is a tautology.
6. Show that affirming the consequent is invalid by exhibiting a falsifying truth assignment.
7. Negate $\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \; (|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon)$ and state the result in words.
8. Write "exactly one channel exceeds the artifact threshold" using $\exists!$ and then without it.
9. Prove the exportation law $(P \wedge Q) \Rightarrow R \equiv P \Rightarrow (Q \Rightarrow R)$.

Chapter 3

Definitions, proof, and mathematical maturity

Reading goal 3.1. Learn to construct and audit arguments, identify hidden hypotheses, and write counterexamples.

3.1 The anatomy of an argument

A proof begins with an exact statement and a list of assumptions. It proceeds through steps justified by a definition, an axiom, or an earlier result. The conclusion is valid *conditional on those assumptions*. Mathematical evidence and empirical evidence answer different questions: a covariance matrix is positive semidefinite by construction, but a measured covariance need not describe only neuronal interactions.

3.2 Direct proof and proof by contrapositive

We say an integer n is even if $n = 2k$ for some integer k. To prove that the sum of two even integers is even, write $a = 2k$ and $b = 2m$. Then $a + b = 2(k + m)$; since $k + m$ is an integer, the sum is even. Notice the explicit use of the definition. An assertion such as "the result is obvious" hides exactly the step a beginner may need.

For the contrapositive method, consider: if n^2 is even, then n is even. Prove instead that odd $n = 2k + 1$ implies $n^2 = 4k(k + 1) + 1$, which is odd. This avoids assuming the conclusion.

3.3 Contradiction, induction, and counterexamples

Proof by contradiction begins with the negation of the proposed conclusion, then derives an impossibility. The classic irrationality argument for $\sqrt{2}$ assumes $\sqrt{2} = a/b$ with coprime integers a, b, derives that both must be even, and contradicts coprimality. We shall revisit this when constructing the real numbers.

A single valid counterexample refutes a universal proposition. To refute "every high-power spectral peak is neuronal," one can simulate a sinusoidal electrical artifact. To refute "correlation implies coupling," one can simulate two independently noisy signals influenced by a common source. These examples do not show that neuronal oscillations or coupling never occur; they refute the unrestricted inference.

3.4 Existence, uniqueness, and identifiability

Three questions recur throughout this book: Does the object exist? Is it uniquely determined? Can we infer it from the observations? The first two are mathematical; the third also depends on a measurement model. For example, different neuronal current distributions may produce the same measured extracellular potential, so a valid forward model does not guarantee a unique inverse solution.

Proposition 3.2. *If A and B are subsets of a fixed universe U, then $A = B$ if and only if both $A \subseteq B$ and $B \subseteq A$.*

Proof. If $A = B$, either inclusion follows immediately. Conversely, the first inclusion implies every element of A belongs to B, and the second implies every element of B belongs to A. The two sets therefore have exactly the same elements. □

3.5 Writing a proof: from scratch work to final form

Proofs are discovered backwards and written forwards. In scratch work one starts from the desired conclusion, unpacks its definitions, and asks what would suffice. The written proof reverses the direction: it begins with the hypotheses and proceeds by justified steps to the conclusion. A useful discipline is the *two-column check*: for every line of a draft proof, name the definition, axiom, earlier result, or logical rule that justifies it. A line without a justification is either a gap or an unstated assumption.

Good proofs also announce their structure. Phrases such as "we argue by contradiction", "we proceed by induction on n", or "it suffices to show" tell the reader which logical form is being used. The phrase "without loss of generality" (WLOG) is legitimate only when the omitted cases genuinely follow by symmetry; it should never hide a case that behaves differently.

3.6 Proof by cases and the pigeonhole principle

When a hypothesis is a disjunction, or when the domain splits naturally, one may prove the conclusion separately in each case.

Proposition 3.3. *For every integer n, the integer $n^2 + n$ is even.*

Proof. Either n is even or n is odd. If $n = 2k$, then $n^2 + n = 2k(2k + 1)$, which is even. If $n = 2k + 1$, then $n^2 + n = (2k + 1)(2k + 2) = 2(2k + 1)(k + 1)$, which is even. The two cases are exhaustive. □

Theorem 3.4 (Pigeonhole principle)**.** *If m objects are placed into k boxes and $m > k$, then some box contains at least two objects. More generally, some box contains at least $\lceil m/k \rceil$ objects.*

Proof. Suppose, for contradiction, that every box contains at most $\lceil m/k \rceil - 1$ objects. Since $\lceil m/k \rceil - 1 < m/k$, the total number of objects is at most $k(\lceil m/k \rceil - 1) < k \cdot m/k = m$, a contradiction. □

The principle has a direct data-analysis reading: if 1001 spikes are recorded in a 1-second window binned at 1 ms, at least one bin contains two spikes. A binary (0/1) spike representation at that resolution is then necessarily lossy.

3.7 Strong induction and well ordering

Ordinary induction proves $P(n+1)$ from $P(n)$. *Strong induction* allows the use of all previous cases: if $P(0)$ holds and $\big(P(0) \wedge \cdots \wedge P(n)\big) \Rightarrow P(n+1)$ for every n, then $P(n)$ holds for every $n \in \mathbb{N}_0$. It is equivalent to ordinary induction applied to the statement $Q(n) = P(0) \wedge \cdots \wedge P(n)$. A further equivalent form is the *well-ordering principle*: every nonempty subset of $\mathbb{N}_0$ has a least element.

Proposition 3.5. *Every integer $n \geq 2$ is a product of one or more primes.*

Proof. By strong induction. The integer 2 is prime. Let $n > 2$ and assume every integer m with $2 \leq m < n$ is a product of primes. If n is prime we are done. Otherwise $n = ab$ with $2 \leq a, b < n$; by hypothesis both a and b are products of primes, and so is their product. □

3.8 Common errors in purported proofs

Learning to *audit* arguments is as important as learning to write them. The following patterns recur.

1. *Proof by example.* Checking a universal claim for several values does not prove it. The polynomial $n^2 + n + 41$ is prime for $n = 0, 1, \ldots, 39$, but $40^2 + 40 + 41 = 41^2$.
2. *Circular reasoning.* A step that silently uses the conclusion. "The estimator is unbiased because its expected value equals the parameter" restates rather than proves.
3. *Division by zero.* Cancelling a factor that may vanish. From $ax = ay$ one may conclude $x = y$ only when $a \neq 0$.
4. *Wrong quantifier order.* Choosing an N that depends on n when the definition requires N to depend only on ε.
5. *Unjustified interchange of limits.* Exchanging a limit with a sum, integral, or derivative without a theorem that allows it. Chapter 17 gives counterexamples.

Worked example 3.6 (Auditing a methods argument)**.** Claim: "Because the shuffled-control coherence is lower than the observed coherence on every one of 20 sessions, coherence reflects genuine coupling." Audit: (i) the premise concerns 20 sessions, while "genuine coupling" is a claim about a mechanism; (ii) shuffling trials destroys trial-locked common input as well as coupling, so the control does not separate the two hypotheses; (iii) consistency across sessions shows reproducibility of the comparison, not uniqueness of its explanation. The argument is a valid rejection of the shuffled null, and an invalid inference to a specific mechanism.

Exercises

1. Prove that the intersection of two subsets is contained in each subset.
2. Give two distinct functions with the same value at zero.
3. Distinguish "the algorithm converges" from "the result estimates the intended biological quantity."
4. Write a three-line counterexample to "a zero Pearson correlation implies independence."
5. Prove by cases that $|ab| = |a||b|$ for real numbers a, b.

6. Use the pigeonhole principle to show that among any 13 people two were born in the same month.
7. Prove by strong induction that every amount of at least 12 cents can be paid with 4- and 5-cent coins.
8. Find the error: "Let $a = b$. Then $a^2 - b^2 = ab - b^2$, so $(a + b)(a - b) = b(a - b)$, hence $a + b = b$ and $2 = 1$."
9. Show that the well-ordering principle implies ordinary induction.

Chapter 4

Sets and the operations that create them

Reading goal 4.1. Use membership, inclusion, unions, intersections, set differences, power sets, and the idea of an admissible collection.

4.1 From a collection to a set

A set is a mathematical collection whose members are determined by a membership rule. If $U = \{a, b, c\}$, then $a \in U$ and $d \notin U$. In the ordinary language of mathematics, $a \in U$ means that *a belongs to U*; $A \subseteq U$ means every element of A belongs to U. These statements differ: $\{a\} \subseteq U$, but typically $\{a\} \notin U$. The empty set $\varnothing$ has no members; $\{\varnothing\}$ has one member, which happens to be the empty set.

Two sets are equal exactly when they have identical members (the extensional criterion). Mathematical set construction must take place within an explicit foundational framework; one cannot freely declare every grammatical property to specify a set of all possible objects. The unrestricted collection "all sets that are not members of themselves" generates Russell's paradox. Following the accessible axiomatic spirit of [22, 23], we allow subsets defined by conditions on an already established set and use conventional constructions for other sets.

4.2 Set operations from ordinary English

Let A be the trials in which the animal runs, and B those with a detected theta peak. Then $A \cap B$ consists of trials satisfying *both* criteria, while $A \cup B$ contains those satisfying *at least one*. Set difference $A \setminus B$ contains trials that are in A but not B. Relative complement A^c is meaningful only after choosing the universe U: here $A^c = U \setminus A$. We must never conflate the universe of trials in one session with all possible trials.

De Morgan's identities are

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

For $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 3, 5\}$, and $B = \{3, 4\}$, we have $A \cap B = \{3\}$ and $A \cup B = \{1, 3, 4, 5\}$. Work through the identities element by element before trusting the symbols.

4.3 Families, indexing, and power sets

A *family* $(A_i)_{i\in I}$ is an indexed collection of sets. The union $\bigcup_{i\in I} A_i$ collects elements present in at least one A_i; the intersection $\bigcap_{i\in I} A_i$ collects elements present in every A_i. The *power set* $\mathcal{P}(A)$ is the set of all subsets of A. For $A = \{u, v\}$ it is $\{\varnothing, \{u\}, \{v\}, \{u, v\}\}$. If A has n elements, its power set contains 2^n elements: each original element can independently be included or excluded.

4.4 The significance for scientific data

Many preprocessing operations change the set of data included in analysis. If U is the set of detected units, then U_{quality} might retain only units meeting refractory-violation and signal-to-noise thresholds. Estimating a correlation on U_{quality} is not the same question as estimating it on U. Record the inclusion criteria as mathematical predicates; this makes sampling assumptions auditable.

Worked example 4.2 (Counting selections)**.** Suppose four electrodes pass quality control. A subset represents the electrodes chosen for a multichannel analysis; there are $2^4 = 16$ possible subsets, including the empty selection. Requiring at least one electrode yields 15. Requiring exactly two yields $\binom{4}{2} = 6$. This distinction prepares us for combinatorics and multiple-comparison correction.

4.5 Algebra of sets

The operations of the previous sections obey identities that parallel the logical equivalences of Chapter 2. This parallel is not accidental: $x \in A \cap B$ means $x \in A \wedge x \in B$, $x \in A \cup B$ means $x \in A \vee x \in B$, and $x \in A^c$ means $\neg(x \in A)$. Every set identity can therefore be proved by translating to a logical equivalence about membership.

Proposition 4.3 (Laws of the algebra of sets)**.** *For subsets A, B, C of a universe U:*

1. *commutativity: $A \cup B = B \cup A$ and $A \cap B = B \cap A$;*
2. *associativity: $(A \cup B) \cup C = A \cup (B \cup C)$, and similarly for $\cap$;*
3. *distributivity: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;*
4. *identity and absorption: $A \cup \varnothing = A$, $A \cap U = A$, $A \cup (A \cap B) = A$;*
5. *complement: $A \cup A^c = U$, $A \cap A^c = \varnothing$, $(A^c)^c = A$.*

Proof. We prove the second distributive law; the others follow the same pattern. Let $x \in U$. Then $x \in A \cup (B \cap C)$ iff $x \in A \vee (x \in B \wedge x \in C)$. By the logical distributive law this is equivalent to $(x \in A \vee x \in B) \wedge (x \in A \vee x \in C)$, i.e. $x \in (A \cup B) \cap (A \cup C)$. Since the two sets have the same members, they are equal. □

Proposition 4.4 (De Morgan for families)**.** *For any indexed family $(A_i)_{i\in I}$ of subsets of U,*

$$\Big(\bigcup_{i\in I} A_i\Big)^c = \bigcap_{i\in I} A_i^c, \qquad \Big(\bigcap_{i\in I} A_i\Big)^c = \bigcup_{i\in I} A_i^c.$$

Proof. For the first identity, $x \notin \bigcup_i A_i$ iff $\neg\exists i\,(x \in A_i)$ iff $\forall i\,(x \notin A_i)$ iff $x \in \bigcap_i A_i^c$. The second follows by applying the first to the family (A_i^c) and taking complements. □

4.6 Symmetric difference and indicator functions

The *symmetric difference* $A \triangle B = (A \setminus B) \cup (B \setminus A)$ contains the elements in exactly one of the two sets. It measures disagreement between two classifications: if A and B are the spikes assigned to a unit by two sorting algorithms, $|A \triangle B|$ counts the spikes on which they disagree.

The *indicator function* of $A \subseteq U$ is $\mathbf{1}_A : U \to \{0, 1\}$, with $\mathbf{1}_A(x) = 1$ if $x \in A$ and 0 otherwise. Indicators turn set operations into arithmetic:

$$\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B, \qquad \mathbf{1}_{A^c} = 1 - \mathbf{1}_A, \qquad \mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B, \qquad \mathbf{1}_{A \triangle B} = |\mathbf{1}_A - \mathbf{1}_B|.$$

Summing the union identity over a finite universe gives the *inclusion–exclusion principle* $|A \cup B| = |A| + |B| - |A \cap B|$. Logical masks in numerical software are exactly indicator functions stored as arrays; an element-wise product of two masks is an intersection.

4.7 Partitions

A *partition* of a set U is a family of nonempty, pairwise disjoint subsets whose union is U. Trials partitioned by stimulus condition, spikes partitioned by unit identity, and time partitioned into bins are all partitions. If $\{B_1, \dots, B_k\}$ partitions a finite U, then for any $A \subseteq U$,

$$|A| = \sum_{j=1}^{k} |A \cap B_j|,$$

because the sets $A \cap B_j$ are disjoint and their union is A. This additivity, generalized from counting to probability and integration, is the root of the law of total probability (Chapter 23).

4.8 Russell's paradox in detail

Suppose there were a set R whose members are exactly the sets that are not members of themselves: $x \in R \Leftrightarrow x \notin x$. Taking $x = R$ yields $R \in R \Leftrightarrow R \notin R$, a contradiction. Hence no such set exists. The *axiom of separation* avoids the paradox by allowing $\{x \in A : P(x)\}$ only for an already given set A. Applied to any set A, the same argument shows that $\{x \in A : x \notin x\}$ is not a member of A; consequently, there is no set of all sets. In applied work this matters less as a danger than as a habit: every set in an analysis should be carved out of a specified universe.

Worked example 4.5 (Masks and exclusion rules)**.** A session has 400 trials. Let M be trials with movement artifacts (37 trials), L trials with lick contamination (52 trials), and suppose 15 trials have both. The excluded set is $M \cup L$, with $|M \cup L| = 37 + 52 - 15 = 74$ trials, leaving $|(M \cup L)^c| = 326$. Reporting "37 and 52 trials were excluded" would suggest 89 exclusions; inclusion–exclusion gives the correct count.

Exercises

1. Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ from element membership.
2. Determine whether $\varnothing \in \{\varnothing\}$ and whether $\varnothing \subseteq \{\varnothing\}$.
3. Describe the difference between a population of neurons and its sampled subset.
4. Given three data-exclusion rules, express the accepted trials with set operations.
5. Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
6. Show that $A \triangle B = \varnothing$ if and only if $A = B$, and that $\triangle$ is associative.

7. Derive inclusion–exclusion for three sets from the indicator identity.
8. Show that $\{x \in A : x \notin x\} \notin A$ for every set A.
9. Two sorters assign 1200 and 1100 spikes to a unit and agree on 1000. Compute $|A \triangle B|$ and a normalized disagreement $|A \triangle B| / |A \cup B|$.

Chapter 5

Ordered pairs, products, and relations

Reading goal 5.1. Understand how ordered observations lead to relations and why correlation, equality, and ordering are mathematically different relations.

5.1 The order of information matters

The collection $\{2, 7\}$ does not specify which element came first. An *ordered pair* $(2, 7)$ does. Formally, an ordered-pair construction must satisfy

$$(a, b) = (c, d) \quad \Longleftrightarrow \quad a = c \text{ and } b = d.$$

One set-theoretic realization is $(a, b) = \{\{a\}, \{a, b\}\}$, although the particular encoding is much less important than the stated property. A spike observation might be the pair (t, u) of timestamp and unit identifier; reversing it gives a different type of object.

The *Cartesian product* $A \times B$ contains all ordered pairs (a, b) with $a \in A$ and $b \in B$. If A has m elements and B has n, then $A \times B$ has mn elements. More generally, $A_1 \times \cdots \times A_k$ defines ordered tuples. A multichannel recording may be indexed by channel c, time sample n, and trial r: its index space is $C \times T \times R$.

5.2 What is a relation?

A *binary relation* from A to B is a subset $R \subseteq A \times B$. Writing aRb is shorthand for $(a, b) \in R$. One relation on the integers is $a \leq b$; another is "a divides b." In neuroscience, an empirically defined relation may connect two neurons if their estimated spike-count correlation is above a threshold. This makes a graph edge, not necessarily a synapse.

A relation $R \subseteq A \times A$ is *reflexive* if aRa for every a, *symmetric* if aRb implies bRa, and *transitive* if aRb and bRc imply aRc. A relation satisfying all three is an *equivalence relation*; it partitions A into disjoint equivalence classes. A *partial order* is reflexive, antisymmetric, and transitive. Equality is an equivalence relation; subset inclusion is a partial order; temporal precedence is an order relation that requires a convention for simultaneous events.

5.3 Composing and inverting relations

For $R \subseteq A \times B$ and $S \subseteq B \times C$, define

$$S \circ R = \{(a, c) : \exists b \in B,\ (a, b) \in R \ \wedge \ (b, c) \in S\}.$$

The inverse relation is $R^{-1} = \{(b, a) : (a, b) \in R\}$. Relation inversion is not yet the same as inversion of a function. For instance, a relation recording which neurons are active during which trials can be reversed to ask which trials contain activity from a neuron.

Worked example 5.2 (Why graph clustering needs care)**.** Let uRv mean "estimated Pearson correlation between u and v exceeds 0.5." The rule is symmetric whenever correlations are defined, and self-pairs have correlation one for nonconstant variables, but it need not be transitive: neuron 1 may correlate strongly with 2, 2 with 3, and 1 weakly with 3. Thus its connected components are not the same as equivalence classes generated by the direct threshold relation.

5.4 Equivalence relations and partitions

Equivalence relations and partitions are two descriptions of the same structure. Given an equivalence relation $\sim$ on A, the *equivalence class* of a is $[a] = \{x \in A : x \sim a\}$, and the *quotient set* is $A/\!\sim = \{[a] : a \in A\}$.

Theorem 5.3 (Fundamental theorem of equivalence relations)**.** *Let $\sim$ be an equivalence relation on A. Then*

1. *every $a \in A$ belongs to its own class $[a]$;*
2. *for $a, b \in A$, either $[a] = [b]$ or $[a] \cap [b] = \varnothing$;*
3. *consequently the classes form a partition of A.*

Conversely, every partition $\mathcal{P}$ of A defines an equivalence relation by $a \sim b$ iff a and b lie in the same block of $\mathcal{P}$, and its classes are exactly the blocks.

Proof. (1) follows from reflexivity. For (2), suppose $c \in [a] \cap [b]$, so $c \sim a$ and $c \sim b$. If $x \in [a]$ then $x \sim a$; by symmetry $a \sim c$; by transitivity $x \sim c$ and then $x \sim b$, so $x \in [b]$. Hence $[a] \subseteq [b]$, and the reverse inclusion follows by exchanging the roles of a and b. (3) The classes are nonempty by (1), cover A by (1), and are pairwise disjoint or equal by (2). For the converse, reflexivity and symmetry are immediate, and transitivity holds because the blocks are disjoint: if a, b share a block and b, c share a block, both blocks contain b and therefore coincide. $\square$

This theorem underlies all the number-system constructions of Part II: integers and rationals are equivalence classes of pairs. It also explains why "spike sorting" is mathematically a partition problem: an assignment of each spike to exactly one unit is the same as an equivalence relation "was fired by the same unit as."

5.5 Closures of relations

A relation that fails to be transitive can be enlarged to a transitive one. The *transitive closure* R^{+} of $R \subseteq A \times A$ is the smallest transitive relation containing R. Concretely, $aR^{+}b$ iff there is a finite chain $a = x_0 R x_1 R \cdots R x_k = b$ with $k \geq 1$. The *reflexive–symmetric–transitive closure* of R is the smallest equivalence relation containing R; its classes are the *connected components* of the undirected graph whose edges are the pairs in R.

Proposition 5.4. *The intersection of any nonempty family of transitive relations on A is transitive. Hence the transitive closure exists and equals the intersection of all transitive relations containing R.*

Proof. Let $(T_i)_{i\in I}$ be transitive and $T=\bigcap_i T_i$. If aTb and bTc, then for each i we have aT_ib and bT_ic, hence aT_ic. Thus aTc. The family of transitive relations containing R is nonempty, since $A\times A$ belongs to it, so its intersection is transitive, contains R, and is contained in every transitive relation containing R. □

The worked example on correlation thresholds above can now be stated precisely: clustering by connected components replaces the threshold relation by its equivalence closure, which may merge neurons that are not directly correlated at all. Long "chains" of moderately correlated units can join two unrelated groups.

5.6 Orders

A *partial order* $\preceq$ on A is reflexive, antisymmetric ($a\preceq b$ and $b\preceq a$ imply $a=b$), and transitive. It is a *total order* if, in addition, any two elements are comparable. The usual $\le$ on numbers is total; inclusion $\subseteq$ on the power set of a set with at least two elements is only partial, since $\{u\}$ and $\{v\}$ are incomparable. A *strict* order $<$ is irreflexive and transitive; from a partial order one obtains a strict one by $a<b \iff a\preceq b\wedge a\neq b$.

Orders appear in data whenever there is ranking: spike times are totally ordered, but the relation "neuron u tends to fire before neuron v" estimated from cross-correlograms need not even be transitive. Sequence-detection methods that assume a consistent total order of activation must test, not presuppose, that the estimated precedence relation has that structure.

Worked example 5.5 (Modular phase)**.** On the real line define $x\sim y$ iff $x-y=2\pi k$ for some integer k. This is an equivalence relation: $x-x=0$; if $x-y=2\pi k$ then $y-x=2\pi(-k)$; and $x-y=2\pi k$, $y-z=2\pi m$ give $x-z=2\pi(k+m)$. The quotient $\mathbb{R}/\!\sim$ is the circle of phases. Each class has exactly one representative in $(-\pi,\pi]$, which is the "wrapped" phase returned by `angle`. Arithmetic on representatives is well defined for addition (adding phases) but not for ordering: "phase α is less than phase β" depends on the representative chosen.

Exercises

1. List $A\times B$ for $A=\{a,b\}$, $B=\{0,1\}$, and explain why $A\times B\neq B\times A$ generally.
2. Construct a reflexive symmetric relation that is not transitive.
3. Prove the inverse of an equivalence relation is the same relation.
4. Interpret a directed graph edge as a relation; explain why its direction does not prove causal direction.
5. Show that "$a\equiv b \pmod 3$" is an equivalence relation on $\mathbb{Z}$ and list its classes.
6. Compute the transitive closure of $R=\{(1,2),(2,3),(3,4)\}$ on $\{1,2,3,4\}$.
7. Prove that the inverse of a partial order is a partial order.
8. Give a precedence relation among three neurons that is not transitive, and explain why no total order is consistent with it.
9. Let A be a finite set with a partition into k blocks. How many equivalence relations on A have exactly these classes?

Chapter 6

Functions: inputs, outputs, and transformations

Reading goal 6.1. Distinguish relations from functions; work with domain, codomain, image, inverse, and composition.

6.1 A function is a special relation

A function $f : A \to B$ assigns to each $a \in A$ exactly one value $f(a) \in B$. It may assign the same output to many inputs. The set A is its *domain*, B its *codomain*, and $f(A) = \{f(a) : a \in A\}$ its image or range. A relation that assigns two distinct outputs to the same input is not a function. These distinctions matter when saying "the neural signal determines the behavior": an observed relation between activity and behavior is not automatically a deterministic function.

A function is *injective* when different inputs have different outputs, *surjective* when every point in the codomain occurs as an output, and *bijective* when both conditions hold. Only a bijection $f : A \to B$ has a genuine inverse function $f^{-1} : B \to A$. Every relation has an inverse relation, but an arbitrary function does not have an inverse function.

6.2 Compositions and transformations

Given $f : A \to B$ and $g : B \to C$, their composition is $g \circ f : A \to C$, where $(g \circ f)(a) = g(f(a))$. Composition generally does not commute. In electrophysiology, one might take raw voltage x, apply a filter F, and then a nonlinear amplitude transform H: $H(F(x))$. Reversing the order may be meaningless or produce a different result.

6.3 Sequences are functions

A sequence $(x_n)_{n\in\mathbb{N}}$ is a function from an index set of natural numbers into a set of values. A finite sampled signal is a function $x : \{0, \ldots, N-1\} \to \mathbb{R}$. A continuously indexed signal is a function $x : I \to \mathbb{R}$ for some time interval I. A multichannel signal is $x : I \to \mathbb{R}^C$. The difference between continuous and sampled time is not a notational accident: sampling introduces an observation process and potentially destroys information.

6.4 Functions of functions

A *functional* maps functions to numbers. For example, energy over an interval can be defined as $E[x] = \int_I |x(t)|^2\,dt$. A *linear operator* maps functions to functions while respecting addition and scalar multiplication. Filtering, differentiation, and Fourier transformation are operators under suitable assumptions. These ideas will become precise after we construct number systems, vector spaces, and limits.

Worked example 6.2 (A map that loses information)**.** The square map $q : \mathbb{R} \to \mathbb{R}$, $q(x) = x^2$, is not injective: $q(-2) = q(2) = 4$. Its inverse relation associates 4 with both 2 and -2. Restricting the domain to $[0, \infty)$ and codomain to $[0, \infty)$ makes q bijective and its inverse is the nonnegative square root. Signal power similarly discards phase and sign information: it cannot reconstruct the original signal by itself.

6.5 Images and preimages of sets

For $f : A \to B$ and $S \subseteq A$, the *image* of S is $f(S) = \{f(a) : a \in S\}$; for $T \subseteq B$, the *preimage* is $f^{-1}(T) = \{a \in A : f(a) \in T\}$. The preimage is defined for every function, whether or not f is invertible; the notation $f^{-1}(T)$ does not presuppose an inverse function.

Proposition 6.3 (Preimages respect set operations)**.** *For $f : A \to B$ and subsets $T, T_i \subseteq B$:*

$$f^{-1}\Big(\bigcup_i T_i\Big) = \bigcup_i f^{-1}(T_i), \quad f^{-1}\Big(\bigcap_i T_i\Big) = \bigcap_i f^{-1}(T_i), \quad f^{-1}(B \setminus T) = A \setminus f^{-1}(T).$$

Images respect unions, $f(\bigcup_i S_i) = \bigcup_i f(S_i)$, but in general only $f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2)$.

Proof. $a \in f^{-1}(\bigcap_i T_i)$ iff $f(a) \in T_i$ for every i iff $a \in f^{-1}(T_i)$ for every i. The union and complement cases are identical in form. For images and intersections, if $b = f(a)$ with $a \in S_1 \cap S_2$ then $b \in f(S_1)$ and $b \in f(S_2)$. Equality can fail: with $f(x) = x^2$, $S_1 = \{-1\}$, $S_2 = \{1\}$, we have $f(S_1 \cap S_2) = \varnothing$ but $f(S_1) \cap f(S_2) = \{1\}$. □

This asymmetry explains why measurability (Part V) is defined through preimages: a function is measurable when preimages of measurable sets are measurable, precisely because preimages commute with all set operations.

6.6 Left and right inverses

A function $g : B \to A$ is a *left inverse* of $f : A \to B$ if $g \circ f = \mathrm{id}_A$, and a *right inverse* if $f \circ g = \mathrm{id}_B$.

Theorem 6.4. *Let A be nonempty and $f : A \to B$.*

1. *f has a left inverse if and only if f is injective.*
2. *f has a right inverse if and only if f is surjective (using the axiom of choice for infinite B).*
3. *If f has both a left inverse g and a right inverse h, then $g = h$, and this common function is the unique inverse f^{-1}.*

Proof. (1) If $g \circ f = \mathrm{id}$ and $f(a) = f(a')$, then $a = g(f(a)) = g(f(a')) = a'$. Conversely, if f is injective, fix $a_0 \in A$ and define $g(b)$ to be the unique a with $f(a) = b$ when $b \in f(A)$, and a_0 otherwise. (2) If $f \circ h = \mathrm{id}$, every $b = f(h(b))$ lies in the image. Conversely, if f is surjective, each preimage $f^{-1}(\{b\})$ is nonempty; choosing one element from each defines h. (3) $g = g \circ \mathrm{id}_B = g \circ (f \circ h) = (g \circ f) \circ h = \mathrm{id}_A \circ h = h$. □

A left inverse is a *decoder* that recovers the input whenever the output came from the encoder. A right inverse is a *generator* that produces, for each target output, some input achieving it. In neural decoding, a stimulus-to-response map that is not injective (two stimuli evoke the same response) has no perfect decoder; in neural control, a response space larger than the set of achievable responses has no right inverse.

6.7 Composition, injectivity, and surjectivity

Proposition 6.5. *Let $f : A \to B$ and $g : B \to C$. (i) If f and g are injective, so is $g \circ f$. (ii) If f and g are surjective, so is $g \circ f$. (iii) If $g \circ f$ is injective, then f is injective. (iv) If $g \circ f$ is surjective, then g is surjective.*

Proof. (i) $g(f(a)) = g(f(a'))$ gives $f(a) = f(a')$ and then $a = a'$. (ii) For $c \in C$ choose b with $g(b) = c$, then a with $f(a) = b$. (iii) If $f(a) = f(a')$ then $g(f(a)) = g(f(a'))$, so $a = a'$. (iv) Every $c = g(f(a))$ is a value of g. □

Statement (iii) has an important consequence for processing pipelines: if any early stage f discards information (is not injective), no later stage g can restore it. Rectification, thresholding, binning, and taking power are all non-injective stages.

6.8 Cardinality and countability

Sets A and B have the *same cardinality*, $|A| = |B|$, if there is a bijection between them. A set is *countable* if it is finite or has the cardinality of $\mathbb{N}_0$. The integers are countable via $0, 1, -1, 2, -2, \ldots$, and so are the rationals and every finite product of countable sets.

Theorem 6.6 (Cantor)**.** *For every set A there is no surjection $A \to \mathcal{P}(A)$. In particular $|A| < |\mathcal{P}(A)|$.*

Proof. Let $f : A \to \mathcal{P}(A)$ and set $D = \{a \in A : a \notin f(a)\}$. If $D = f(d)$ for some d, then $d \in D \iff d \notin f(d) = D$, a contradiction. Hence D is not in the image of f. □

Applied to $A = \mathbb{N}_0$, Cantor's theorem shows that the set of infinite binary sequences (equivalently, the set of all possible infinite spike rasters at fixed resolution) is uncountable. This is one reason why statements about "all possible spike trains" require measure theory rather than counting.

Exercises

1. Decide whether $f(x) = x^3$ from $\mathbb{R}$ to $\mathbb{R}$ is bijective.
2. State domain and codomain for a histogram of spike counts in 10-ms bins.
3. Find $f \circ g$ and $g \circ f$ for $f(x) = x + 1$ and $g(x) = x^2$.
4. Explain why a spectrogram generally does not contain sufficient information to uniquely recover a waveform without additional phase assumptions.
5. Prove that $f(f^{-1}(T)) \subseteq T$ with equality when f is surjective, and $S \subseteq f^{-1}(f(S))$ with equality when f is injective.
6. Give a function with a left inverse but no right inverse, and describe all its left inverses.
7. Show that $g \circ f$ injective does not imply g injective.

8. Construct an explicit bijection $\mathbb{N}_0 \to \mathbb{Z}$.
9. A spike detector maps a 30 kHz voltage trace to a list of threshold-crossing times. Identify the domain, codomain, and why the map is not injective.

Chapter 7

Bridges from statements to relations and functions

Reading goal 7.1. Turn statements in everyday language into small, verifiable arguments about sets, relations, and functions. You need only the preceding language chapters.

How to read a bridge. Each numbered bridge states exactly what is to be proved, spells out the argument, and connects it to a possible representation of recorded data. A small proof is not disposable: the same reasoning returns inside more advanced methods.

Bridge 1: Set equality is equality of membership

Proposition 7.2. *For sets A, B, we have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.*

Proof. If $A = B$, every element of either set belongs to the other. Conversely, $A \subseteq B$ says that $x \in A$ implies $x \in B$, and $B \subseteq A$ says the reverse. Thus for every object x, $x \in A$ exactly when $x \in B$. By extensionality the sets are equal. □

Example. A detector returning timestamps D and a manually verified set G agree exactly when no event belongs only to D and no event belongs only to G. Matching counts alone is insufficient.

Bridge 2: Inclusion is a partial order

Proposition 7.3. *The relation $\subseteq$ is reflexive, transitive and antisymmetric on the power set $\mathcal{P}(U)$.*

Proof. Every element of A belongs to A, so $A \subseteq A$. If $A \subseteq B$ and $B \subseteq C$, then any $x \in A$ belongs to B and then C, proving transitivity. If $A \subseteq B$ and $B \subseteq A$, Bridge 1 gives $A = B$. □

Example. Increasing a permissive spike detection threshold does not necessarily produce nested candidate sets unless the acceptance rule is itself monotone in that threshold; one should first prove monotonicity of the actual rule.

7.1 Lean 4 verification: the first two bridges

English: BRIDGE-001

For any sets A, B of elements of a common type, equality is equivalent to mutual inclusion:

$$A = B \iff (A \subseteq B \land B \subseteq A).$$

If $A = B$, both inclusions follow. Conversely, two-way membership gives equality by set extensionality.

English: BRIDGE-002

Set inclusion is reflexive ($A \subseteq A$), transitive ($A \subseteq B \subseteq C \Rightarrow A \subseteq C$), and antisymmetric ($A \subseteq B \land B \subseteq A \Rightarrow A = B$). For the final property, reuse BRIDGE-001.

Formal correspondence: `Set` α represents sets whose members belong to an arbitrary type α. The three order laws are stated explicitly in Lean. The code is checked separately by the pinned Lean toolchain.

Lean 4 / Mathlib

```
import Mathlib

namespace Con2physFormal.Bridges

universe u
variable {α : Type u}

-- BRIDGE-001: equality of sets is mutual inclusion.
theorem subset_antisymm_iff (A B : Set α) :
    A = B ↔ A ⊆ B ∧ B ⊆ A := by
  constructor
  · intro h
    constructor
    · exact h.le
    · exact h.ge
  · rintro ⟨hAB, hBA⟩
    exact Set.Subset.antisymm hAB hBA

-- BRIDGE-002: inclusion on the power set is a partial order.
theorem subset_reflexive (A : Set α) : A ⊆ A := by
  exact Set.Subset.refl A

theorem subset_transitive (A B C : Set α)
    (hAB : A ⊆ B) (hBC : B ⊆ C) : A ⊆ C := by
  exact hAB.trans hBC

theorem subset_antisymmetric (A B : Set α)
    (hAB : A ⊆ B) (hBA : B ⊆ A) : A = B := by
  exact (subset_antisymm_iff A B).mpr ⟨hAB, hBA⟩

-- BRIDGE-003: Cartesian product membership and empty factors.
universe v

theorem cartesian_membership {β : Type v} (A : Set α) (B : Set β)
    (a : α) (b : β) :
    (a, b) ∈ Set.prod A B ↔ a ∈ A ∧ b ∈ B := by
  rfl

theorem cartesian_empty_left {β : Type v} (B : Set β) :
    Set.prod (∅ : Set α) B = ∅ := by
  ext p
  change (p.1 ∈ (∅ : Set α) ∧ p.2 ∈ B) ↔ False
  simp

theorem cartesian_empty_right {β : Type v} (A : Set α) :
    Set.prod A (∅ : Set β) = ∅ := by
  ext p
  change (p.1 ∈ A ∧ p.2 ∈ (∅ : Set β)) ↔ False
  simp

-- BRIDGE-004: associativity of composition of typed relations.
-- relComp S R means R first, then S (as in function composition).
def relComp {X Y Z : Type*}
    (S : Y → Z → Prop) (R : X → Y → Prop) : X → Z → Prop :=
  fun x z => ∃ y, R x y ∧ S y z

theorem relation_composition_associative
    {X Y Z W : Type*}
    (R : X → Y → Prop) (S : Y → Z → Prop)
    (T : Z → W → Prop) :
    relComp (relComp T S) R = relComp T (relComp S R) := by
  funext x w
  apply propext
  change
    (∃ y, R x y ∧ (∃ z, S y z ∧ T z w)) ↔
    (∃ z, (∃ y, R x y ∧ S y z) ∧ T z w)
  constructor
  · rintro ⟨y, hxy, z, hyz, hzw⟩
    exact ⟨z, ⟨y, hxy, hyz⟩, hzw⟩
  · rintro ⟨z, ⟨y, hxy, hyz⟩, hzw⟩
    exact ⟨y, hxy, z, hyz, hzw⟩

-- BRIDGE-005: equivalence classes cover the carrier and overlap only when
      equal.
def equivalenceClass (r : α → α → Prop) (a : α) : Set α :=
  {b | r b a}

theorem equivalence_class_cover (r : α → α → Prop)
    (h : Equivalence r) (a : α) :
    a ∈ equivalenceClass r a := by
  exact h.refl a

theorem equivalence_classes_equal_of_overlap
    (r : α → α → Prop) (h : Equivalence r) (a b : α)
    (hoverlap : ∃ x, x ∈ equivalenceClass r a ∧
      x ∈ equivalenceClass r b) :
    equivalenceClass r a = equivalenceClass r b := by
  obtain ⟨x, hxa, hxb⟩ := hoverlap
  have hab : r a b := h.trans (h.symm hxa) hxb
  ext y
  constructor
  · intro hya
    exact h.trans hya hab
  · intro hyb
    exact h.trans hyb (h.symm hab)

-- BRIDGE-006: images preserve unions; preimages preserve intersections and
      unions.
theorem image_union_bridge {β : Type*} (f : α → β)
    (E F : Set α) :
    f '' (E ∪ F) = f '' E ∪ f '' F := by
  exact Set.image_union f E F
```

Bridge 3: Cartesian products define possible pairs

Proposition 7.4. *For sets A, B, the Cartesian product $A \times B$ consists exactly of the ordered pairs (a, b) with $a \in A$ and $b \in B$. If either factor is empty, the product is empty.*

Proof. By definition a pair is in $A \times B$ only when both coordinate conditions hold. When $A = \varnothing$, no first coordinate can satisfy the requirement, and similarly for $B = \varnothing$. □

Example. If U is a set of units and T a recording interval, $U \times T$ describes *possible* unit–time pairs; the observed spike relation is only a subset of this product.

7.2 Lean 4 verification: Cartesian products

For types α and β, and sets $A \subseteq \alpha$ and $B \subseteq \beta$, an ordered pair (a, b) belongs to $A \times B$ precisely when $a \in A$ and $b \in B$. If either set is empty, the product has no members. These are three distinct Lean theorems, representing the full statement of Bridge 3.

```
-- BRIDGE-003: Cartesian product membership and empty factors.
universe v

theorem cartesian_membership {β : Type v} (A : Set α) (B : Set β)
    (a : α) (b : β) :
    (a, b) ∈ Set.prod A B ↔ a ∈ A ∧ b ∈ B := by
  rfl

theorem cartesian_empty_left {β : Type v} (B : Set β) :
    Set.prod (∅ : Set α) B = ∅ := by
  ext p
  change (p.1 ∈ (∅ : Set α) ∧ p.2 ∈ B) ↔ False
  simp

theorem cartesian_empty_right {β : Type v} (A : Set α) :
    Set.prod A (∅ : Set β) = ∅ := by
  ext p
  change (p.1 ∈ A ∧ p.2 ∈ (∅ : Set β)) ↔ False
  simp

-- BRIDGE-004: associativity of composition of typed relations.
```

Bridge 4: Composition of relations is associative

Proposition 7.5. *If $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$, then $(T \circ S) \circ R = T \circ (S \circ R)$.*

Proof. A pair (a, d) belongs to the left side exactly when there are $b \in B$ and $c \in C$ such that aRb, bSc, and cTd. The same witnesses b, c characterize the right side. Both relations therefore contain exactly the same pairs. □

Example. A relation between units and channels followed by one between channels and brain regions can be composed. Composition alone does not make either assignment biologically correct.

7.3 Lean 4 verification: relational composition

For typed relations $R \subseteq X \times Y$, $S \subseteq Y \times Z$ and $T \subseteq Z \times W$, we define *composition* with the same order as function composition: $\mathrm{relComp}(S, R)$ relates x to z when some y satisfies xRy and ySz. The witnesses for either association are precisely the same y, z, so composition is associative.

```
-- BRIDGE-004: associativity of composition of typed relations.
-- relComp S R means R first, then S (as in function composition).
def relComp {X Y Z : Type*}
    (S : Y → Z → Prop) (R : X → Y → Prop) : X → Z → Prop :=
  fun x z => ∃ y, R x y ∧ S y z

theorem relation_composition_associative
    {X Y Z W : Type*}
    (R : X → Y → Prop) (S : Y → Z → Prop)
    (T : Z → W → Prop) :
    relComp (relComp T S) R = relComp T (relComp S R) := by
  funext x w
  apply propext
  change
    (∃ y, R x y ∧ (∃ z, S y z ∧ T z w)) ↔
    (∃ z, (∃ y, R x y ∧ S y z) ∧ T z w)
  constructor
  · rintro ⟨y, hxy, z, hyz, hzw⟩
    exact ⟨z, ⟨y, hxy, hyz⟩, hzw⟩
  · rintro ⟨z, ⟨y, hxy, hyz⟩, hzw⟩
    exact ⟨y, hxy, z, hyz, hzw⟩

-- BRIDGE-005: equivalence classes cover the carrier and overlap only when equal.
```

Bridge 5: Equivalence classes form a partition

Proposition 7.6. *If $\sim$ is reflexive, symmetric and transitive on A, the classes $[a] = \{b \in A : b \sim a\}$ cover A and any two classes are identical or disjoint.*

Proof. Reflexivity places each a in $[a]$, so the classes cover A. If $x \in [a] \cap [b]$, symmetry and transitivity give $a \sim b$. For any $y \sim a$, transitivity gives $y \sim b$, and conversely, hence $[a] = [b]$. □

Example. Grouping timestamps by identical half-open time-bin index defines equivalence classes. Grouping by "is within 2 ms" generally fails transitivity and is not an equivalence relation.

7.4 Lean 4: equivalence classes form a partition

Let r be a reflexive, symmetric, and transitive relation on a type α. We define the class of a by $[a] = \{b : r(b, a)\}$. Reflexivity proves that every a lies in its own class, so the classes cover the carrier.

If x lies in both $[a]$ and $[b]$, symmetry followed by transitivity gives $a\, r\, b$. Consequently every element of $[a]$ lies in $[b]$, and conversely. Thus intersecting classes are equal. This establishes the two characteristic partition properties without assuming that the relation is equality.

```
-- BRIDGE-005: equivalence classes cover the carrier and overlap only when equal.
def equivalenceClass (r : α → α → Prop) (a : α) : Set α :=
  {b | r b a}
```

```
theorem equivalence_class_cover (r : α → α → Prop)
    (h : Equivalence r) (a : α) :
    a ∈ equivalenceClass r a := by
  exact h.refl a

theorem equivalence_classes_equal_of_overlap
    (r : α → α → Prop) (h : Equivalence r) (a b : α)
    (hoverlap : ∃ x, x ∈ equivalenceClass r a ∧
      x ∈ equivalenceClass r b) :
    equivalenceClass r a = equivalenceClass r b := by
  obtain ⟨x, hxa, hxb⟩ := hoverlap
  have hab : r a b := h.trans (h.symm hxa) hxb
  ext y
  constructor
  · intro hya
    exact h.trans hya hab
  · intro hyb
    exact h.trans hyb (h.symm hab)

-- BRIDGE-006: images preserve unions; preimages preserve intersections and unions.
theorem image_union_bridge {β : Type*} (f : α → β)
```

Bridge 6: Images preserve unions; preimages preserve unions and intersections

Proposition 7.7. *For $f : A \to B$ and subsets $E, F \subseteq A$, $f(E \cup F) = f(E) \cup f(F)$. For $C, D \subseteq B$, $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$ and $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$.*

Proof. A value y belongs to $f(E \cup F)$ exactly when some input in E or F maps to y. This is the union of the images. An input x belongs to $f^{-1}(C \cap D)$ exactly when $f(x)$ belongs to both C and D. The union case follows by replacing "both" with "at least one". □

Example. A trial-to-condition mapping permits selecting all trials whose condition lies in a specified set via a preimage. Unlike an inverse function, the preimage is defined even for many-to-one mappings.

7.5 Lean 4: images and preimages

For any function $f : A \to B$, images preserve unions:

$$f(E \cup F) = f(E) \cup f(F).$$

Preimages preserve both intersections and unions:

$$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D), \qquad f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D).$$

The inverse-image notation does not assume that f has an inverse function; it describes the set of inputs meeting a membership criterion.

```
-- BRIDGE-006: images preserve unions; preimages preserve intersections and unions.
theorem image_union_bridge {β : Type*} (f : α → β)
    (E F : Set α) :
    f '' (E ∪ F) = f '' E ∪ f '' F := by
  exact Set.image_union f E F
```

```
theorem preimage_inter_bridge {β : Type*} (f : α → β)
    (C D : Set β) :
    f ⁻¹' (C ∩ D) = f ⁻¹' C ∩ f ⁻¹' D := by
  exact Set.preimage_inter

theorem preimage_union_bridge {β : Type*} (f : α → β)
    (C D : Set β) :
    f ⁻¹' (C ∪ D) = f ⁻¹' C ∪ f ⁻¹' D := by
  exact Set.preimage_union

-- BRIDGE-007: a two-sided inverse exists exactly for bijections.
theorem inverse_iff_bijective {β : Type*} (f : α → β) :
    (∃ g : β → α, Function.LeftInverse g f ∧
```

Bridge 7: A two-sided inverse is equivalent to bijectivity

Proposition 7.8. *A function $f : A \to B$ has a function $g : B \to A$ with $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$ if and only if f is injective and surjective.*

Proof. If $f(a_1) = f(a_2)$, applying g gives $a_1 = a_2$, so f is injective. For each $b \in B$, $b = f(g(b))$, so it is surjective. Conversely, bijectivity supplies exactly one a for each $b = f(a)$; assigning $g(b) = a$ defines both inverse identities. □

Example. A many-to-one mapping from a continuous waveform to its power spectrum is not invertible: phase information has been discarded.

7.6 Lean 4: two-sided inverses and bijections

A function $f : A \to B$ admits an inverse $g : B \to A$ satisfying $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$ if and only if f is both injective and surjective.

The forward direction uses the left inverse to show injectivity and the right inverse to exhibit a preimage for each $b \in B$. Conversely, surjectivity selects a preimage for every output; injectivity makes that choice a left inverse as well. This construction uses classical choice, a dependency recorded by Lean's axiom audit.

```
-- BRIDGE-007: a two-sided inverse exists exactly for bijections.
theorem inverse_iff_bijective {β : Type*} (f : α → β) :
    (∃ g : β → α, Function.LeftInverse g f ∧
      Function.RightInverse g f) ↔ Function.Bijective f := by
  constructor
  · rintro ⟨g, hleft, hright⟩
    constructor
    · intro a₁ a₂ heq
      calc
        a₁ = g (f a₁) := (hleft a₁).symm
        _ = g (f a₂) := congrArg g heq
        _ = a₂ := hleft a₂
    · intro b
      exact ⟨g b, hright b⟩
  · intro h
    let g : β → α := fun b => Classical.choose (h.2 b)
    have hright : Function.RightInverse g f := by
      intro b
      exact Classical.choose_spec (h.2 b)
    refine ⟨g, ?_, hright⟩
```

```
    intro a
    apply h.1
    exact hright (f a)

-- BRIDGE-008: pairs of natural indices form a countable type.
```

Bridge 8: Pairs of natural indices can be enumerated

Proposition 7.9. *The set* $\mathbb{N}_0 \times \mathbb{N}_0$ *is countable.*

Proof. Arrange the pairs along diagonals with fixed sum $s = m + n$: $(0, 0)$, then $(0, 1), (1, 0)$, then $(0, 2), (1, 1), (2, 0)$, and so on. Every pair appears on its finite-sum diagonal. Numbering entries as they are visited gives an enumeration by natural numbers. □

Example. A countable list of possible channel–time-bin indices need not contain every continuous-time event. There is a difference between indexing samples and describing physical time.

7.7 Lean 4: countability of natural-number pairs

A pair $(m, n) \in \mathbb{N} \times \mathbb{N}$ can be enumerated by visiting finite diagonals with fixed $m + n$. Lean expresses this conclusion as a `Countable` instance for the product type. The following proof asks Mathlib to supply the product instance from the countability of its factors. This certifies the countability claim, but does not formalize the manuscript's particular diagonal enumeration algorithm.

```
-- BRIDGE-008: pairs of natural indices form a countable type.
theorem natural_pairs_countable : Countable (ℕ × ℕ) := by
  infer_instance

#print axioms natural_pairs_countable
```

Exercises

1. Construct two sets with equal cardinality that are not equal.
2. Explain why "within 1 ms" is not generally transitive.
3. For a many-to-one f, exhibit E, F for which $f(E \cap F) \neq f(E) \cap f(F)$.

Part II

The construction of number systems

Chapter 8

Natural numbers, counting, and induction

Reading goal 8.1. See how counting can be built from basic rules, rather than accepted as an unexplained operation.

8.1 Counting precedes measurement

When we say there were seven action potentials, we count discrete events. Counting does not require time units, voltage calibration, or a model of firing probability. The natural numbers are the first formal setting for such statements. We adopt $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$, reserving $\mathbb{N}_+ = \{1, 2, \ldots\}$. Authors differ about including zero, so every technical treatment should make its convention explicit.

8.2 Successors and the Peano viewpoint

Imagine a set N with a distinguished element 0 and successor operation $S : N \to N$. The basic idea is that zero is not the successor of anything; distinct inputs have distinct successors; and any subset containing zero and closed under the successor operation contains all of N. The final clause is the induction principle. These structural properties establish a unique natural-number system up to isomorphism, under the usual axiomatic background [23].

Define addition recursively by $n + 0 = n$ and $n + S(m) = S(n + m)$. Define multiplication through $n \cdot 0 = 0$ and $n \cdot S(m) = n \cdot m + n$. These are *definitions*, while commutativity, associativity, and distributivity are derived properties. We ordinarily write $n + 1$ rather than $S(n)$ after connecting the symbols.

8.3 Induction as a way to prove infinite families of claims

Suppose we wish to prove $\sum_{k=1}^{n} k = n(n+1)/2$ for all positive integers n. The base case $n = 1$ holds. Assume it is true for n. Then

$$\sum_{k=1}^{n+1} k = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}.$$

Therefore the claim holds for $n + 1$; induction establishes every case. Induction is a proof principle, not an empirical inference from observing a finite number of examples.

8.4 Finite sets and combinatorics

Two finite sets have the same cardinality if there exists a bijection between them. The number of length-n binary vectors is 2^n; the number of ways to choose k distinct objects from n without order is $\binom{n}{k} = n!/[k!(n-k)!]$. If 20 channels are inspected in all unordered pairs, the number of pairs is $\binom{20}{2} = 190$. This count matters when reporting a family of connectivity tests.

Worked example 8.2 (Discrete time windows)**.** A recording sampled at $f_s = 1000$ samples/s for 10 seconds has $N = f_s T = 10000$ samples if we choose the index set $\{0, \dots, 9999\}$. The sample at index n corresponds to $t_n = n/f_s$, so the last recorded time is 9.999 s, not 10 s. Distinguish the *duration of acquisition* from the *timestamp of its final sample*.

8.5 Proving the arithmetic laws from the definitions

The recursive definitions of addition and multiplication say nothing explicit about commutativity or associativity. These must be proved by induction, and the proofs show how a small set of rules generates all of school arithmetic.

Proposition 8.3 (Associativity of addition)**.** *For all $k, m, n \in \mathbb{N}_0$, $(k + m) + n = k + (m + n)$.*

Proof. Fix k and m, and induct on n. For $n = 0$, both sides equal $k + m$ by the rule $x + 0 = x$. Suppose the identity holds for n. Then

$$(k+m)+S(n) = S\big((k+m)+n\big) = S\big(k+(m+n)\big) = k+S(m+n) = k+\big(m+S(n)\big),$$

where the first and third equalities use the recursive rule $x + S(y) = S(x + y)$, the second uses the induction hypothesis, and the last uses the recursive rule once more. □

Proposition 8.4 (Commutativity of addition)**.** *For all $m, n \in \mathbb{N}_0$, $m + n = n + m$.*

Proof. We first prove two lemmas by induction on n: (a) $0+n = n$, and (b) $S(m)+n = S(m+n)$. For (a), $0 + 0 = 0$, and $0 + S(n) = S(0 + n) = S(n)$. For (b), $S(m) + 0 = S(m) = S(m + 0)$, and $S(m) + S(n) = S(S(m) + n) = S(S(m + n)) = S(m + S(n))$. Now induct on n in the main claim. The case $n = 0$ is $m + 0 = m = 0 + m$ by (a). If $m + n = n + m$, then $m + S(n) = S(m + n) = S(n + m) = S(n) + m$ by (b). □

Analogous inductions establish associativity and commutativity of multiplication and the distributive law $k(m+n) = km+kn$. The order relation is then *defined* by $m \le n$ iff $n = m+p$ for some $p \in \mathbb{N}_0$, and one proves that it is a total order compatible with addition and multiplication.

8.6 Recursion as a way to define sequences

The *recursion theorem* states that, given a set X, an element $x_0 \in X$, and a function $g : X \to X$, there is exactly one sequence $x : \mathbb{N}_0 \to X$ with $x(0) = x_0$ and $x(S(n)) = g(x(n))$. It is the justification for every iterative definition: factorials, powers, and numerical algorithms such as Newton's method. Uniqueness follows from induction on n; existence requires a construction (the intersection of all relations with the right closure properties) which we omit.

Worked example 8.5 (A recursive filter is a recursive definition)**.** The exponential smoother $y_0 = x_0$, $y_n = \alpha y_{n-1} + (1-\alpha)x_n$ for $0 < \alpha < 1$ is defined by recursion. Unrolling by induction gives

$$y_n = \alpha^n x_0 + (1-\alpha)\sum_{k=1}^{n} \alpha^{n-k} x_k,$$

which can be verified by checking $n = 0$ and the step $n \to n+1$. The recursion is computationally cheap; the closed form shows the geometric weighting of past samples. This is the simplest infinite-impulse-response filter, revisited in Chapter 28.

8.7 Divisibility, remainders, and modular indices

Theorem 8.6 (Division with remainder)**.** *For $n \in \mathbb{N}_0$ and $d \in \mathbb{N}_+$ there are unique $q, r \in \mathbb{N}_0$ with $n = qd + r$ and $0 \le r < d$.*

Proof. Existence: the set $\{n - qd : q \in \mathbb{N}_0,\ qd \le n\}$ is a nonempty subset of $\mathbb{N}_0$ (it contains n), so by well ordering it has a least element $r = n - qd$. If $r \ge d$, then $n - (q+1)d = r - d \ge 0$ is a smaller element, a contradiction. Uniqueness: if $qd + r = q'd + r'$ with $0 \le r, r' < d$, then $|r - r'| < d$ is a multiple of d, hence 0, and then $q = q'$. □

Remainders appear constantly in indexing. Converting a linear sample index n into (trial, within-trial sample) for trials of length L is $n = qL + r$; circular buffers use $r = n \bmod L$; and the discrete Fourier transform treats its indices modulo N. Off-by-one errors are often confusions between the ranges $\{0, \dots, L-1\}$ and $\{1, \dots, L\}$.

8.8 Counting principles

Three counting rules recur:

1. *Sum rule*: if a finite set is a disjoint union $A \cup B$, then $|A \cup B| = |A| + |B|$.
2. *Product rule*: $|A \times B| = |A|\,|B|$.
3. *Ordered selections*: the number of injective functions from a k-set to an n-set is $n(n-1)\cdots(n-k+1) = n!/(n-k)!$.

Each unordered k-subset corresponds to exactly $k!$ ordered selections, which gives $\binom{n}{k} = n!/[k!(n-k)!]$. Pascal's rule $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ follows by splitting subsets according to whether they contain a fixed element, and induction with Pascal's rule proves the binomial theorem

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

With $a = p$ and $b = 1 - p$, the terms are the binomial probabilities of observing k spikes in n independent bins, and the identity says they sum to one.

Exercises

1. Prove by induction that $1 + 3 + \cdots + (2n-1) = n^2$.
2. Count unordered neuron pairs from $N = 80$ recorded units.
3. Show that a finite sequence of N entries is a function on $\{0, \dots, N-1\}$.
4. Explain why mathematical induction and statistical generalization from trials are different operations.
5. Prove by induction that $n \cdot 1 = n$ and $1 \cdot n = n$ from the recursive definition of multiplication.
6. Use the recursion of the exponential smoother to prove its closed form by induction.

7. Compute q and r for $n = 123457$ samples and trial length $L = 2500$; interpret them.
8. Prove Pascal's rule by a counting argument.
9. How many ordered sequences of 3 distinct stimuli can be chosen from 8? How many unordered sets?

Chapter 9

Integers: extending counting to differences

Reading goal 9.1. Understand negative quantities through equivalence classes and learn what a legitimate algebraic extension must preserve.

9.1 Why natural numbers are not enough

A count cannot be negative, but a *difference between counts* can be. We need numbers representing expressions like $3 - 5$ without insisting that subtraction remain inside the natural numbers. Define a relation on pairs $(a, b) \in \mathbb{N}_0^2$ by

$$(a, b) \sim (c, d) \quad \Longleftrightarrow \quad a + d = b + c.$$

This is an equivalence relation; its classes represent formal differences $a - b$. For example, $(3, 5) \sim (0, 2)$ because $3 + 2 = 5 + 0$. We call the set of equivalence classes $\mathbb{Z}$.

9.2 Operations on classes

Write $[(a, b)]$ for the class of (a, b). Addition can be defined by

$$[(a, b)] + [(c, d)] = [(a + c, b + d)],$$

and multiplication by

$$[(a, b)] \cdot [(c, d)] = [(ac + bd, ad + bc)].$$

A good definition must be *well defined*: choosing different representatives of the same classes must not change the resulting class. This issue appears again in quotient spaces and latent-variable models; an output must not depend on an arbitrary representation of the same underlying object.

9.3 Order and absolute value

Integers inherit addition, multiplication, and a total order. Negative numbers permit signed offsets and differences, but not all integers possess multiplicative inverses within $\mathbb{Z}$. The absolute value $|n|$ removes sign: $|n| = n$ when $n \geq 0$, and $|n| = -n$ otherwise. It satisfies the triangle inequality $|a + b| \leq |a| + |b|$.

9.4 A neuroscience example

Spike-count differences $D = N_{\text{post}} - N_{\text{pre}}$ may be negative although each count is nonnegative. However, raw subtraction is sensitive to the lengths of the counting intervals. If the intervals differ, compare normalized rates N/T rather than interpreting the count difference uncritically. Mathematical legitimacy does not guarantee scientific comparability.

9.5 Proving that the operations are well defined

Proposition 9.2. *Integer addition and multiplication defined on classes do not depend on the chosen representatives.*

Proof. Suppose $(a,b) \sim (a',b')$ and $(c,d) \sim (c',d')$, so $a+b' = b+a'$ and $c+d' = d+c'$. For addition we need $(a+c, b+d) \sim (a'+c', b'+d')$, i.e. $a+c+b'+d' = b+d+a'+c'$, which is the sum of the two hypotheses. For multiplication we need

$$ac + bd + a'd' + b'c' = ad + bc + a'c' + b'd'.$$

It suffices to change one representative at a time. Replacing (a,b) by (a',b') with c,d fixed, the required identity $ac+bd+a'd+b'c = ad+bc+a'c+b'd$ rearranges to $c(a+b') + d(b+a') = c(b+a') + d(a+b')$, which holds because $a+b' = b+a'$. The second representative is handled symmetrically. □

The map $n \mapsto [(n,0)]$ embeds $\mathbb{N}_0$ into $\mathbb{Z}$ injectively and preserves addition and multiplication. Every class $[(a,b)]$ equals either $[(n,0)]$ or $[(0,n)]$ for a unique n, which we write n or $-n$. In this sense $\mathbb{Z}$ *contains* $\mathbb{N}_0$, even though, strictly, the new numbers are sets of pairs.

9.6 The ring axioms

With these operations, $\mathbb{Z}$ is a *commutative ring with identity*: addition is associative and commutative with identity 0 and additive inverses; multiplication is associative and commutative with identity 1; and multiplication distributes over addition. The ring has no zero divisors: $ab = 0$ implies $a = 0$ or $b = 0$. Consequently cancellation holds: if $ac = bc$ and $c \neq 0$, then $a = b$.

Proposition 9.3. *In any ring,* $0 \cdot a = 0$ *and* $(-1)a = -a$. *Hence* $(-a)(-b) = ab$.

Proof. $0 \cdot a = (0+0)a = 0 \cdot a + 0 \cdot a$; adding $-(0 \cdot a)$ to both sides gives $0 = 0 \cdot a$. Next $a + (-1)a = (1 + (-1))a = 0 \cdot a = 0$, so $(-1)a$ is the additive inverse of a. Finally $(-a)(-b) = (-1)(-1)ab$, and $(-1)(-1) = -(-1) = 1$. □

The rule "a negative times a negative is positive" is therefore not a convention but a consequence of distributivity. Any arithmetic in which distributivity holds must satisfy it.

9.7 Divisibility and the greatest common divisor

For integers a, b not both zero, $\gcd(a,b)$ is the largest nonnegative common divisor (equivalently, the positive generator of the ideal a
$mathbbZ + b$
$mathbbZ$).

Theorem 9.4 (Bézout)**.** *For integers a, b not both zero there exist integers x, y with $ax + by = \gcd(a,b)$.*

Proof. Let $I = \{ax + by : x, y \in \mathbb{Z}\}$, and let g be its smallest positive element (by well ordering). Divide a by g: $a = qg + r$ with $0 \le r < g$. Then $r = a - qg \in I$, so minimality forces $r = 0$; thus $g \mid a$, and likewise $g \mid b$. Any common divisor of a and b divides every element of I, in particular g. Hence $g = \gcd(a, b)$. □

The Euclidean algorithm computes the gcd by repeated division, using $\gcd(a, b) = \gcd(b, a \bmod b)$. Gcds arise when combining signals with different sampling rates: resampling from 30 kHz to 1.25 kHz uses the rational factor $1250/30000 = 1/24$, and polyphase resampling routines reduce such ratios by their gcd.

Worked example 9.5 (Signed lags)**.** In a cross-correlogram between spike trains u and v, the lag $\tau = t_v - t_u$ takes integer values in units of bins. Negative lags mean v preceded u. The set of lags in a window $\{-K, \dots, K\}$ has $2K + 1$ elements, and reversing the roles of u and v maps lag τ to $-\tau$. A peak at negative lag is thus not a different phenomenon from a peak at positive lag with the neurons swapped; the sign records a convention of which train is the reference.

Exercises

1. Verify symmetry and transitivity of the equivalence relation on ordered pairs.
2. Show that $[(a, b)] + [(b, a)]$ is the integer zero.
3. Explain why subtraction of two nonnegative power estimates need not be nonnegative.
4. Prove that $\mathbb{Z}$ has no zero divisors, assuming the corresponding fact for $\mathbb{N}_0$.
5. Use the Euclidean algorithm to compute $\gcd(30000, 1250)$ and find Bézout coefficients.
6. Prove that if $c \neq 0$ and $ac = bc$ in $\mathbb{Z}$, then $a = b$.
7. Define $\le$ on $\mathbb{Z}$ by $[(a, b)] \le [(c, d)]$ iff $a + d \le b + c$, and prove that it is well defined.

Chapter 10

Rational numbers, ratios, and fields

Reading goal 10.1. Construct fractions and learn why units, denominators, and field axioms matter in quantitative science.

10.1 Fractions as equivalence classes

Take integer pairs (a, b) with $b \neq 0$. Say $(a, b) \sim (c, d)$ when $ad = bc$. The resulting equivalence class represents the fraction a/b. Ratios like $1/2 = 2/4$ are different representations of the same rational number. Addition and multiplication take their familiar forms

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \qquad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

Rationals form the field $\mathbb{Q}$: nonzero values possess multiplicative inverses, and the field operations obey associativity, commutativity, and distributivity. Division by zero is undefined: a rate with zero exposure duration is not an extreme number but an invalid operation.

10.2 What does a ratio mean physically?

Suppose a unit emits 75 spikes over 15 seconds. Its time-averaged rate is $75/15 = 5$ spikes/s. The numerator and denominator have different roles and units. When a software routine returns a mean across observations, ask what each observation represents and whether every observation has the same weight. The mean of within-trial rates need not equal total count divided by total exposure if trial durations differ.

10.3 Rationals are dense but incomplete

Between any two rational numbers $a < b$, their midpoint $(a + b)/2$ is another rational. Yet the equation $x^2 = 2$ has no rational solution. To show this, suppose $x = p/q$ is in lowest terms and $p^2 = 2q^2$. Then p must be even; substituting $p = 2k$ shows q also even, contradicting coprimality. Thus the rational numbers have gaps from the viewpoint of limits and geometry.

Worked example 10.2 (Units and normalization)**.** Compare spikes/bin with spikes/s. If the bin width is Δt seconds, the rate estimate is $r = n/\Delta t$ in spikes/s. Changing Δt changes the variance and resolution of the estimate even if the underlying experiment is unchanged. This anticipates the bias-variance trade-off in spike-rate estimation.

10.4 Well-definedness and the field axioms

As with integers, the operations on fractions must be checked to be independent of representatives. If $a/b = a'/b'$ (so $ab' = a'b$) then

$$\frac{a}{b}\cdot\frac{c}{d} = \frac{ac}{bd} \quad\text{and}\quad \frac{a'}{b'}\cdot\frac{c}{d} = \frac{a'c}{b'd}$$

are equal because $ac\cdot b'd = (ab')cd = (a'b)cd = a'c\cdot bd$. A similar computation handles addition.

A *field* is a commutative ring with identity $1 \neq 0$ in which every nonzero element has a multiplicative inverse. In $\mathbb{Q}$ the inverse of a/b with $a \neq 0$ is b/a. The field axioms imply all the familiar algebra of fractions; for example, $x \neq 0$ and $xy = xz$ give $y = z$ by multiplying with x^{-1}.

10.5 Order and the Archimedean property

Call a/b *positive* if $ab > 0$; this is independent of representative. Define $x < y$ iff $y - x$ is positive. Then $\mathbb{Q}$ is an *ordered field*: the order is total, and $x < y$ implies $x + z < y + z$, while $x < y$ and $z > 0$ imply $xz < yz$.

Proposition 10.3 (Archimedean property)**.** *For every rational $x > 0$ and every rational y, there is a natural number n with $nx > y$.*

Proof. Write $x = a/b$ and $y = c/d$ with $a, b, d > 0$. Take $n = |c|b + 1$. Then $nx = (|c|b+1)a/b \geq |c| + a/b > |c| \geq y$, since $a \geq 1$ and $d \geq 1$ give $|c| \geq |c|/d \geq c/d$. □

The Archimedean property says there are no infinitely large or infinitely small positive rationals. It justifies statements such as "choose $N > 1/\varepsilon$" in convergence proofs.

10.6 Decimal expansions

Proposition 10.4. *The decimal expansion of a rational number is eventually periodic, and every eventually periodic decimal represents a rational number.*

Proof. For p/q with $q > 0$, long division produces successive remainders in $\{0, \dots, q-1\}$. By the pigeonhole principle a remainder must repeat within $q+1$ steps, after which the digits repeat. Conversely, if $x = 0.\overline{d_1 \cdots d_k}$, then $10^k x - x = d_1 \cdots d_k$ is an integer, so $x = d_1 \cdots d_k/(10^k - 1)$; a finite preperiod only shifts and adds a rational. □

Numerical software does not store decimals: it stores binary floating-point numbers, which are rationals with power-of-two denominators. The decimal 0.1 has an infinite binary expansion and is therefore stored inexactly. This is the origin of the familiar fact that $0.1 + 0.2 \neq 0.3$ in double precision, discussed further in Chapter 39.

10.7 Weighted averages and ratio estimators

Given counts n_j and exposures $T_j > 0$, there are two natural rate summaries:

$$\bar{r}_{\text{pooled}} = \frac{\sum_j n_j}{\sum_j T_j}, \qquad \bar{r}_{\text{mean}} = \frac{1}{J}\sum_j \frac{n_j}{T_j}.$$

The pooled rate is the weighted average of the trial rates $r_j = n_j/T_j$ with weights $w_j = T_j / \sum_k T_k$:

$$\bar{r}_{\text{pooled}} = \sum_j w_j r_j.$$

The two summaries coincide when all T_j are equal. Otherwise they answer different questions: the pooled rate is "spikes per second of recording", and the mean of rates is "the typical trial's rate". Under a constant-rate Poisson model, the pooled rate is the maximum-likelihood estimator; the mean of rates gives more influence to short, noisy trials.

Worked example 10.5 (Simpson's paradox with rates)**.** Two conditions are recorded on two days. On day 1, condition A yields 20 spikes in 10 s (2.0 spikes/s) and B yields 210 spikes in 100 s (2.1 spikes/s). On day 2, A yields 900 spikes in 100 s (9.0 spikes/s) and B yields 95 spikes in 10 s (9.5 spikes/s). Condition B has the higher rate on *each* day. Pooled over days, however, A has $920/110 \approx 8.36$ spikes/s and B has $305/110 \approx 2.77$ spikes/s. The reversal occurs because A was recorded mostly on the high-rate day and B mostly on the low-rate day. Rational arithmetic is exact here; the paradox lies in which question each ratio answers.

Exercises

1. Explain what fails if $b = 0$ is allowed in a fraction.
2. Calculate the pooled firing rate across a 5-s trial with 20 spikes and a 15-s trial with 30 spikes; compare with the unweighted average of the two trial rates.
3. Prove that between two unequal rationals there is a rational number.
4. Show that the addition of fractions is well defined.
5. Prove that $\mathbb{Q}$ is not closed under taking square roots of positive elements.
6. Convert $0.1\overline{36}$ to a fraction in lowest terms.
7. Show that $\bar{r}_{\text{pooled}}$ lies between $\min_j r_j$ and $\max_j r_j$.
8. In the Simpson example, compute the mean of daily rates for each condition and explain why it agrees with the day-wise comparison.

Chapter 11

The real number system, order, and completeness

Reading goal 11.1. Understand why the real line is more than rational arithmetic, and why continuity and convergence depend on completeness.

11.1 The real line as an ordered field

We wish to represent distances, times, and voltages without rational-number gaps. The real numbers $\mathbb{R}$ are an *ordered field* satisfying a further completeness property. One characterization is the *least-upper-bound property*: every nonempty subset of $\mathbb{R}$ bounded above has a least upper bound. If $A = \{x \in \mathbb{R} : x^2 < 2,\ x \geq 0\}$, then $\sup A = \sqrt{2}$. This value cannot be supplied by rational numbers alone.

A real number may be developed rigorously using Dedekind cuts or equivalence classes of Cauchy sequences of rational numbers; either construction requires proving that the resulting operations and order are well defined. We will use the completeness axiom without adopting one construction as primary. For a fuller foundational treatment, consult [23].

11.2 Bounds, supremum, and infimum

A number M is an upper bound for A if $x \leq M$ for every $x \in A$. The supremum is the smallest such bound; it need not belong to the set. For the open interval $(0, 1)$, the supremum is 1, although 1 is not a member. The infimum is defined dually. A *maximum* must be a member of the set and greater than or equal to every other member; distinguish maximum from supremum.

11.3 Distances and the triangle inequality

The absolute value defines distance on $\mathbb{R}$ by $d(x, y) = |x - y|$. Distance is nonnegative, symmetric, zero precisely for equal points, and obeys $d(x, z) \leq d(x, y) + d(y, z)$. These four properties define a *metric* on a more general space. A tolerance criterion for a numerical solution is meaningful only with a specified distance, scale, and units.

11.4 Interval notation and physical uncertainty

The interval $[a, b]$ includes endpoints, whereas (a, b) excludes them. An experimental measurement can be modeled as an uncertain value lying in some interval, but the interval should not be mistaken for a statistical confidence interval without an inference procedure. A confidence interval concerns the long-run coverage of an estimator under a model; it is not simply an arbitrary range surrounding a measurement.

Worked example 11.2 (The value of completeness)**.** Take a sequence of rational approximations $1, 1.4, 1.41, 1.414, \ldots$ to $\sqrt{2}$. Each term can be rational while its limit is not. A space of numbers meant to support convergence of such sequences must contain the limiting value. The same idea underlies convergence of numerical algorithms for signal processing.

11.5 Consequences of completeness

The least-upper-bound property is a single axiom, but nearly every existence theorem in analysis rests on it. We record the first consequences.

Proposition 11.3 (Approximation property of the supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and $s = \sup A$. For every $\varepsilon > 0$ there is $a \in A$ with $s - \varepsilon < a \le s$.*

Proof. If no such a existed, then $a \le s - \varepsilon$ for every $a \in A$, so $s - \varepsilon$ would be an upper bound smaller than the least upper bound s. □

Theorem 11.4 (Archimedean property of $\mathbb{R}$)**.** *For every real x there is a natural number $n > x$. Equivalently, for every $\varepsilon > 0$ there is n with $1/n < \varepsilon$.*

Proof. If not, $\mathbb{N}_+$ is bounded above by x, so it has a supremum s. By the approximation property with $\varepsilon = 1$ there is n with $s - 1 < n$, whence $n + 1 > s$, contradicting that s bounds $\mathbb{N}_+$. □

Theorem 11.5 (Density of the rationals)**.** *Between any two reals $x < y$ there is a rational number.*

Proof. By the Archimedean property choose n with $1/n < y - x$. Let m be the smallest integer with $m > nx$ (it exists by well ordering, applied after shifting). Then $m - 1 \le nx$, so $m \le nx + 1 < ny$. Hence $x < m/n < y$. □

Density justifies approximating any real parameter by a finite-precision number to arbitrary accuracy. It does *not* imply that every real is rational, and indeed the irrationals are also dense: if $x < y$, there is a rational q with $x/\sqrt{2} < q < y/\sqrt{2}$ (for $q \neq 0$), and then $q\sqrt{2}$ is irrational and lies between x and y.

11.6 Existence of square roots

Theorem 11.6. *There is a unique positive real s with $s^2 = 2$.*

Proof. Let $A = \{x \ge 0 : x^2 < 2\}$, which is nonempty ($1 \in A$) and bounded above by 2. Let $s = \sup A$. Suppose $s^2 < 2$. For $0 < h < 1$, $(s + h)^2 \le s^2 + h(2s + 1)$, which is less than 2 once $h < (2 - s^2)/(2s + 1)$; then $s + h \in A$, contradicting that s is an upper bound. Suppose $s^2 > 2$. For $0 < h < s$, $(s - h)^2 \ge s^2 - 2sh > 2$ once $h < (s^2 - 2)/(2s)$; then $s - h$ is an upper bound of A smaller than s, a contradiction. Hence $s^2 = 2$. Uniqueness: if $s, t > 0$ and $s^2 = t^2$ then $(s - t)(s + t) = 0$ with $s + t > 0$, so $s = t$. □

The argument is a template: guess the object as a supremum, then rule out the two strict inequalities using small perturbations. The same template proves the intermediate value theorem in Chapter 15.

11.7 Nested intervals and uncountability

Theorem 11.7 (Nested interval property)**.** *If $[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$ is a decreasing sequence of nonempty closed bounded intervals, then $\bigcap_n [a_n, b_n] \neq \varnothing$. If moreover $b_n - a_n \to 0$, the intersection is a single point.*

Proof. The left endpoints are nondecreasing and bounded above by every b_m, so $a = \sup_n a_n$ exists and $a_n \le a \le b_m$ for all n, m. Thus a belongs to every interval. If two points $a < a'$ belonged to all intervals then $b_n - a_n \ge a' - a > 0$ for every n. □

Theorem 11.8 (Cantor)**.** *The interval $[0, 1]$ is uncountable.*

Proof. Let $x_1, x_2, \ldots$ be any sequence in $[0, 1]$. Divide $[0, 1]$ into three closed thirds; at least one does not contain x_1; call it I_1. Divide I_1 into thirds and pick one, I_2, not containing x_2. Continuing, we obtain nested closed intervals of length 3^{-n} with $x_n \notin I_n$. Their intersection contains a point x that differs from every x_n. Thus no sequence lists all of $[0, 1]$. □

11.8 Floating-point numbers are not real numbers

A double-precision floating-point number is a rational of the form $\pm m \cdot 2^e$ with a 53-bit integer significand m and a bounded exponent e. The set of such numbers is finite. It is not closed under addition (results are rounded), addition is not associative, and the distance between consecutive representable numbers grows with magnitude. The *machine epsilon* $\epsilon_{\text{mach}} = 2^{-52} \approx 2.2 \times 10^{-16}$ is the gap between 1 and the next larger double. Every real x in the normal range is rounded to a double $\mathrm{fl}(x)$ with relative error $|\mathrm{fl}(x) - x|/|x| \le \epsilon_{\text{mach}}/2$.

Worked example 11.9 (Timestamps in long recordings)**.** A 24-hour recording at 30 kHz has about 2.6×10^9 samples. Storing timestamps in seconds as doubles gives an absolute resolution near $t = 86400$ s of about $86400 \cdot 2.2 \times 10^{-16} \approx 1.9 \times 10^{-11}$ s, ample for spike timing. Storing them as single-precision floats ($\epsilon \approx 1.2 \times 10^{-7}$) gives about 10 ms resolution near the end of the day, which would destroy millisecond spike timing. Storing sample indices as 64-bit integers avoids the issue entirely. Completeness of $\mathbb{R}$ is a property of the mathematical model, not of the stored numbers.

Exercises

1. Find the supremum, infimum, minimum, and maximum, when defined, for $A = (0, 1]$.
2. Prove the reverse triangle inequality $\big||x| - |y|\big| \le |x - y|$.
3. Explain why a bounded measurement record does not imply its generating stochastic process has bounded support.
4. Prove that $\inf A = -\sup(-A)$ for a nonempty bounded set A.
5. Show that $\sup\{1 - 1/n : n \in \mathbb{N}_+\} = 1$ using the Archimedean property.
6. Show that the nested interval property fails for open intervals $(0, 1/n)$.

7. Adapt the square-root proof to show that every positive real has a unique positive cube root.
8. Find the largest timestamp, in seconds, at which single precision still resolves $1/30000\,\mathrm{s}$.

Chapter 12

Complex numbers: amplitude and phase

Reading goal 12.1. Build the complex plane from real coordinates and understand Euler's formula before meeting Fourier and Hilbert transforms.

12.1 Pairs with new arithmetic

A complex number $z = a+ib$ has real part a and imaginary part b, where $i^2 = -1$. We may regard z as the ordered pair (a, b) of reals, with addition $(a, b) + (c, d) = (a + c, b + d)$ and multiplication $(a, b)(c, d) = (ac - bd, ad + bc)$. The conjugate is $\bar{z} = a - ib$ and $|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$. For $z \neq 0$, $z^{-1} = \bar{z}/|z|^2$.

12.2 A plane with angles

Write $z = r(\cos\theta + i\sin\theta)$ for modulus $r \geq 0$ and argument θ. Angles differing by integer multiples of 2π represent the same direction. Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ gives the polar representation $z = re^{i\theta}$. Multiplication adds arguments and multiplies moduli:

$$(r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1+\theta_2)}.$$

The periodicity of phase will require circular, rather than ordinary linear, statistics.

12.3 Why complex numbers are indispensable for signals

A sine wave can be written $x(t) = A\cos(2\pi ft + \phi) = \Re\{Ae^{i(2\pi ft+\phi)}\}$. Multiplication by $e^{i\omega t}$ shifts the angular coordinate at a constant rate. The Fourier transform treats a signal as a superposition of such rotating complex exponentials. The complex coefficient encodes two quantities: magnitude and phase. Discarding one generally loses information about the waveform.

Worked example 12.2 (Phase wraps)**.** The angles 179° and -179° differ numerically by 358° but lie just 2° apart on a circle. Their arithmetic mean is 0°, which points in the opposite direction. A circular mean instead uses the argument of $e^{i179^\circ} + e^{-i179^\circ}$ and correctly points near 180°.

12.4 The complex numbers form a field

With the multiplication rule $(a,b)(c,d) = (ac - bd, ad + bc)$, the set $\mathbb{C} = \mathbb{R}^2$ satisfies all field axioms: the operations are associative and commutative, multiplication distributes over addition, $(0,0)$ and $(1,0)$ are identities, and every nonzero (a,b) has inverse $(a,-b)/(a^2+b^2)$. The map $a \mapsto (a,0)$ embeds $\mathbb{R}$ in $\mathbb{C}$, and $i = (0,1)$ satisfies $i^2 = (-1,0) = -1$. Unlike $\mathbb{R}$, the complex field admits *no* order compatible with its arithmetic: in an ordered field every nonzero square is positive, but $i^2 = -1$ and $1^2 = 1$ cannot both be positive.

Proposition 12.3 (Properties of conjugation and modulus)**.** *For $z, w \in \mathbb{C}$: $\overline{z+w} = \bar z + \bar w$, $\overline{zw} = \bar z \bar w$, $|zw| = |z||w|$, $\Re z = (z+\bar z)/2$, $\Im z = (z-\bar z)/(2i)$, and $|z+w| \le |z| + |w|$.*

Proof. The first two follow by expanding both sides. Then $|zw|^2 = zw\overline{zw} = z\bar z w \bar w = |z|^2|w|^2$. For the triangle inequality,

$$|z+w|^2 = |z|^2 + |w|^2 + 2\Re(z\bar w) \le |z|^2 + |w|^2 + 2|z||w| = (|z| + |w|)^2,$$

using $\Re u \le |u|$ and $|z\bar w| = |z||w|$. □

12.5 Euler's formula and de Moivre

For now we take $e^{i\theta}$ to be defined by $\cos\theta + i\sin\theta$; Chapter 15 justifies this through power series. The addition formulas for sine and cosine are then equivalent to the exponential law

$$e^{i(\alpha+\beta)} = e^{i\alpha} e^{i\beta}.$$

Iterating gives de Moivre's formula $(\cos\theta + i\sin\theta)^n = \cos n\theta + i \sin n\theta$, and the trigonometric identities

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

These two formulas convert any product of sines and cosines into a sum of exponentials, which is how frequency mixing is analysed: $\cos(\omega_1 t)\cos(\omega_2 t) = \tfrac{1}{2}[\cos((\omega_1-\omega_2)t) + \cos((\omega_1+\omega_2)t)]$. A nonlinearity applied to a sum of two rhythms creates power at sum and difference frequencies, a source of spurious cross-frequency coupling.

12.6 Roots of unity and the discrete Fourier basis

The equation $z^N = 1$ has exactly N complex solutions, the *Nth roots of unity* ω^k for $k = 0, \dots, N-1$, where $\omega = e^{2\pi i/N}$. They are equally spaced on the unit circle.

Proposition 12.4 (Orthogonality of roots of unity)**.** *For integers k,*

$$\sum_{n=0}^{N-1} \omega^{kn} = \begin{cases} N, & k \equiv 0 \pmod N, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. If $k \equiv 0$ then every term is 1. Otherwise $q = \omega^k \neq 1$ and the geometric sum is $(1-q^N)/(1-q)$; since $q^N = (\omega^N)^k = 1$, the numerator vanishes. □

This identity is the entire algebraic content of the discrete Fourier transform's invertibility. It also explains why a sinusoid whose frequency is not an integer multiple of f_s/N "leaks" into many DFT bins: the corresponding geometric sum does not cancel.

12.7 Complex numbers as rotations and scalings

Multiplication by a fixed $c = re^{i\phi}$ maps the plane to itself by rotating through ϕ and scaling by r. As a map of $\mathbb{R}^2$, it is the matrix

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad c = a + ib.$$

This identifies complex numbers with a special class of 2×2 real matrices. A linear filter acting on a single sinusoid at frequency f does exactly this: it multiplies the complex amplitude by the frequency response $H(f)$, rescaling amplitude by $|H(f)|$ and shifting phase by $\arg H(f)$. Composing two filters multiplies their responses, which adds their phase shifts.

Worked example 12.5 (Averaging phases correctly)**.** Suppose spike phases relative to theta are $\theta_1, \dots, \theta_n$. The *mean resultant vector* is $\bar{z} = \frac{1}{n}\sum_j e^{i\theta_j}$. Its argument is the circular mean phase and its modulus $R = |\bar{z}| \in [0, 1]$ measures concentration; $R = 1$ iff all phases coincide, by the equality case of the triangle inequality. For phases $\{170°, -170°, 180°\}$, $\bar{z} \approx \frac{1}{3}(-0.985 + 0.174i - 0.985 - 0.174i - 1) = -0.990$, so the mean phase is $180°$ and $R \approx 0.99$, as intuition demands.

Exercises

1. Compute $(1 + i)^2$ and $|1 + i|$.
2. Show $|zw| = |z||w|$ using complex conjugation.
3. Express $\cos\theta \cos\phi$ as a combination of complex exponentials and simplify.
4. Explain why phase averages need not be meaningful when oscillatory amplitude is close to zero.
5. Show that $\mathbb{C}$ cannot be ordered: assume $i > 0$ or $i < 0$ and derive a contradiction with the ordered-field axioms.
6. Derive the identities for $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$ from $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$.
7. List the 6th roots of unity in Cartesian form and verify that their sum is zero.
8. Show that $|z + w| = |z| + |w|$ if and only if one of z, w is a nonnegative real multiple of the other.
9. Expand $\cos^3(\omega t)$ into a sum of cosines and identify which frequencies a cubic nonlinearity creates.

Chapter 13

Bridges between counting and number systems

Reading goal 13.1. Understand why integers, rational numbers, real numbers, and complex numbers have the algebraic properties used later in measurement and numerical analysis.

Bridge 9: Associativity of addition from recursion

Proposition 13.2. *If addition on $\mathbb{N}_0$ is defined by $a+0=a$ and $a+S(b)=S(a+b)$, then $(a+b)+c=a+(b+c)$.*

Proof. Induct on c. For $c=0$ both sides equal $a+b$. Assume the identity for c. Then $(a+b)+S(c)=S((a+b)+c)=S(a+(b+c))=a+S(b+c)=a+(b+S(c))$. □

Example. Accumulating three disjoint spike-count windows is independent of how we group the additions. It is not independent of whether the windows overlap.

13.1 Lean 4: addition associativity by induction

The recursive equations $a+0=a$ and $a+S(c)=S(a+c)$ lead to associativity by induction on the third argument. The base case simplifies by addition with zero. The successor step rewrites addition at a successor, then invokes the induction hypothesis. The formal proof mirrors the English argument in Bridge 9 rather than merely citing a library theorem of associativity.

```
import Mathlib

namespace Con2physFormal.BridgesNumbers

-- BRIDGE-009: associativity of addition, proved by induction
-- on the third summand using the recursive successor equation.
theorem add_assoc_induction (a b c : ℕ) :
    (a + b) + c = a + (b + c) := by
  induction c with
  | zero =>
      simp only [Nat.add_zero]
  | succ c ih =>
      simp only [Nat.add_succ, ih]

-- BRIDGE-010: integer representatives as pairs of naturals.
-- (a,b) represents the formal difference a-b.
```

```
def integerPairRel (p q : ℕ × ℕ) : Prop :=
  p.1 + q.2 = p.2 + q.1

theorem integerPairRel_refl (p : ℕ × ℕ) :
    integerPairRel p p := by
  simp [integerPairRel, Nat.add_comm]

theorem integerPairRel_symm (p q : ℕ × ℕ) :
    integerPairRel p q → integerPairRel q p := by
  intro h
  exact h.symm

theorem integerPairRel_trans (p q r : ℕ × ℕ) :
    integerPairRel p q → integerPairRel q r → integerPairRel p r := by
  intro hpq hqr
  dsimp [integerPairRel] at *
  omega

theorem integerPairRel_equivalence :
    Equivalence integerPairRel := by
  constructor
  · exact integerPairRel_refl
  · exact integerPairRel_symm
  · exact integerPairRel_trans

def integerPairAdd (p q : ℕ × ℕ) : ℕ × ℕ :=
  (p.1 + q.1, p.2 + q.2)

theorem integerPairAdd_respects (p p' q q' : ℕ × ℕ)
    (hpp : integerPairRel p p') (hqq : integerPairRel q q') :
    integerPairRel (integerPairAdd p q) (integerPairAdd p' q') := by
  dsimp [integerPairRel, integerPairAdd] at *
  omega

#print axioms integerPairRel_equivalence
#print axioms integerPairAdd_respects

#print axioms add_assoc_induction

end Con2physFormal.BridgesNumbers
```

Bridge 10: Constructing integers from pairs of naturals

Proposition 13.3. *For $(a,b),(c,d) \in \mathbb{N}_0^2$, define $(a,b) \sim (c,d)$ when $a+d = b+c$. This is an equivalence relation. Addition $[(a,b)]+[(c,d)] = [(a+c,b+d)]$ is independent of representatives.*

Proof. Reflexivity and symmetry follow from equality. If $a+d=b+c$ and $c+f=d+e$, adding equations and canceling $c+d$ yields $a+f=b+e$, so transitivity holds. If $a+b'=b+a'$ and $c+d'=d+c'$, adding gives $(a+c)+(b'+d')=(b+d)+(a'+c')$, proving representative-independent addition. □

Example. The signed time difference between events is modeled by an integer when timestamps lie on a sample grid; negative offsets have a meaning not available in the natural numbers.

13.2 Lean 4: integers as equivalence classes of pairs

Represent the formal difference $a - b$ by $(a, b) \in \mathbb{N}^2$. Two pairs represent the same integer when $a + d = b + c$. Reflexivity, symmetry, and transitivity are proved separately. Addition of representatives is componentwise, $(a, b) + (c, d) = (a + c, b + d)$; the last theorem shows that equivalent inputs give equivalent outputs, making this operation well-defined on equivalence classes.

```
-- BRIDGE-010: integer representatives as pairs of naturals.
-- (a,b) represents the formal difference a-b.
def integerPairRel (p q : ℕ × ℕ) : Prop :=
  p.1 + q.2 = p.2 + q.1

theorem integerPairRel_refl (p : ℕ × ℕ) :
    integerPairRel p p := by
  simp [integerPairRel, Nat.add_comm]

theorem integerPairRel_symm (p q : ℕ × ℕ) :
    integerPairRel p q → integerPairRel q p := by
  intro h
  exact h.symm

theorem integerPairRel_trans (p q r : ℕ × ℕ) :
    integerPairRel p q → integerPairRel q r → integerPairRel p r := by
  intro hpq hqr
  dsimp [integerPairRel] at *
  omega

theorem integerPairRel_equivalence :
    Equivalence integerPairRel := by
  constructor
  · exact integerPairRel_refl
  · exact integerPairRel_symm
  · exact integerPairRel_trans

def integerPairAdd (p q : ℕ × ℕ) : ℕ × ℕ :=
  (p.1 + q.1, p.2 + q.2)

theorem integerPairAdd_respects (p p' q q' : ℕ × ℕ)
    (hpp : integerPairRel p p') (hqq : integerPairRel q q') :
    integerPairRel (integerPairAdd p q) (integerPairAdd p' q') := by
  dsimp [integerPairRel, integerPairAdd] at *
  omega

#print axioms integerPairRel_equivalence
#print axioms integerPairAdd_respects
```

Scope: this code establishes an equivalence relation and congruence of representative addition. It does not yet define the quotient type or establish an explicit equivalence with `Int`.

Bridge 11: Rational arithmetic is independent of representation

Proposition 13.4. *For integers a, c and nonzero integers b, d, set $(a, b) \sim (c, d)$ when $ad = bc$. Rational addition $[(a, b)] + [(c, d)] = [(ad + bc, bd)]$ is well-defined.*

Proof. If $ab' = a'b$ and $cd' = c'd$, then after multiplying by dd' and bb', respectively, one obtains $(ad + bc)b'd' = (a'd' + b'c')bd$. Thus equivalent input fractions yield equivalent output fractions. The nonzero-denominator condition ensures the construction remains in its domain. □

Example. 1/2 and 500/1000 represent the same rational sampling interval; changing the fraction's written form should not change computed sampling times in exact arithmetic.

Bridge 12: There is a rational between distinct reals

Proposition 13.5. *If $a < b$ are real, there exists $q \in \mathbb{Q}$ with $a < q < b$.*

Proof. Choose $n \in \mathbb{N}$ with $n(b-a) > 1$ using the Archimedean property (proved next). Choose an integer m with $na < m \leq na + 1$. Since $m \leq na + 1 < nb$, the rational $q = m/n$ is strictly between a and b. □

Example. Any ideal real-valued threshold can be approximated by rational thresholds, but a recorded voltage has finite instrument resolution and may not distinguish them.

Bridge 13: Archimedean property from completeness

Proposition 13.6. *For every real x there is a natural number $n > x$.*

Proof. Otherwise $\mathbb{N}_0$ would be nonempty and bounded above. Let $s = \sup \mathbb{N}_0$. Then $s - 1$ is not an upper bound, so some $n > s - 1$ belongs to $\mathbb{N}_0$. Its successor $n + 1 > s$ also belongs to $\mathbb{N}_0$, contradicting that s is an upper bound. □

Example. Given any finite desired time resolution $\Delta > 0$, a sufficiently large integer sampling-rate multiple can make $1/n < \Delta$ in an ideal mathematical model. Hardware bandwidth is a separate question.

Bridge 14: A supremum can be approached from below

Proposition 13.7. *If a nonempty set $A \subset \mathbb{R}$ is bounded above and $s = \sup A$, then for every $\varepsilon > 0$ some $a \in A$ satisfies $s - \varepsilon < a \leq s$.*

Proof. If no such a existed, every element of A would be at most $s - \varepsilon$. Then $s - \varepsilon$ would be an upper bound smaller than the least upper bound s, a contradiction. □

Example. The maximal attainable value of an optimization criterion can be a supremum without being reached by any tested finite set of hyperparameters.

Bridge 15: Floor partitions nonnegative time into half-open bins

Proposition 13.8. *For $\Delta > 0$ and $t \geq 0$, there is a unique $k \in \mathbb{N}_0$ with $k\Delta \leq t < (k+1)\Delta$, namely $k = \lfloor t/\Delta \rfloor$.*

Proof. The integer-part property gives $k \leq t/\Delta < k + 1$. Multiplying by positive Δ yields the interval condition. Two distinct integers cannot satisfy the inequalities because their half-open intervals do not overlap. □

Example. The timestamp $t = 0.100$ s belongs to the second 0.1-s bin $[0.1, 0.2)$ if timestamps and boundaries are exact. Floating-point computations require explicit endpoint handling.

Bridge 16: Conjugation makes complex magnitude multiplicative

Proposition 13.9. *For $z, w \in \mathbb{C}$, $\overline{zw} = \bar{z}\bar{w}$ and $|zw| = |z||w|$.*

Proof. Expanding $(a + ib)(c + id)$ and reversing the sign of the imaginary part proves the conjugation identity. Since $|z|^2 = z\bar{z}$, we have $|zw|^2 = zw\overline{zw} = (z\bar{z})(w\bar{w}) = |z|^2|w|^2$. Taking nonnegative square roots establishes the result. □

Example. Complex Fourier coefficients encode both amplitude and phase; replacing every coefficient by its modulus loses phase information even though magnitudes multiply neatly.

Exercises

1. Prove cancellation in the natural numbers using induction.
2. Exhibit two distinct integer pairs representing the same integer.
3. Show that the union of two adjacent half-open time bins is again a half-open interval.

Part III

Calculus and analysis

Chapter 14

Sequences, limits, and continuity

Reading goal 14.1. Understand the precision behind "approaches", "converges", and "continuous" using explicit tolerances.

14.1 An infinite list is a function

The sequence $x_n = 1/n$ approaches zero as n grows. But what should "approaches" mean? We say $x_n \to L$ if, for every tolerance $\varepsilon > 0$, there exists an index N_ε such that $|x_n - L| < \varepsilon$ whenever $n \geq N_\varepsilon$. The order of the quantifiers matters. We must succeed for *every* tolerance, although the required index may depend on the chosen tolerance.

For $x_n = 1/n$, choose $N_\varepsilon > 1/\varepsilon$. Then $n \geq N_\varepsilon$ implies $|1/n| < \varepsilon$. Convergence is not the same as attaining the limit; no term in this sequence equals zero. Nor is convergence established by drawing the first few points. An infinite claim requires an argument that controls all sufficiently late terms.

14.2 Cauchy sequences and completeness

A sequence is *Cauchy* if its sufficiently late terms become arbitrarily close to one another: $\forall \varepsilon > 0$ there exists N such that $|x_m - x_n| < \varepsilon$ for all $m, n \geq N$. In a complete metric space every Cauchy sequence converges to a point of the space. The rational approximations to $\sqrt{2}$ form a Cauchy sequence of rationals but fail to converge *within* the rationals. This shows why completeness is essential in analysis.

14.3 Functions of a real variable

We say $f(x)$ tends to L as $x \to a$ if for every $\varepsilon > 0$ we can find $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - L| < \varepsilon$. A function is continuous at a when its limit equals its value at that point. Uniform continuity strengthens the statement by demanding one δ for a given ε that works across the entire domain.

14.4 Why this matters for continuous-time recordings

A continuous-time model $x(t)$ does not mean its actual neuronal source changes smoothly; it means the mathematical function has a value for each real-valued time in its domain. A sampled waveform records only certain time points. Interpolating between them requires

assumptions about smoothness or bandwidth. Similarly, increasing the sampling frequency is not a mathematical proof that a measurement device approximates the underlying field arbitrarily well: instrument bandwidth and noise also matter.

Worked example 14.2 (Finite-difference derivatives)**.** For a smooth $x(t)$, the quantity $[x(t+h)-x(t)]/h$ may approach a derivative as $h \to 0$. In recorded data, however, taking ever smaller h may amplify measurement noise. Convergence of an ideal mathematical expression and consistency of a noisy numerical estimator are separate questions.

14.5 Algebra of limits

Proofs from the ε–N definition become routine once a few general rules are established.

Proposition 14.3 (Uniqueness and boundedness)**.** *A convergent sequence has exactly one limit and is bounded.*

Proof. If $x_n \to L$ and $x_n \to L'$ with $L \neq L'$, take $\varepsilon = |L-L'|/2$. For large n both $|x_n - L| < \varepsilon$ and $|x_n - L'| < \varepsilon$, so $|L-L'| < 2\varepsilon = |L-L'|$, a contradiction. For boundedness, choose N with $|x_n - L| < 1$ for $n \geq N$; then $|x_n| \leq \max\{|x_1|, \dots, |x_{N-1}|, |L|+1\}$ for all n. □

Theorem 14.4 (Limit laws)**.** *If $x_n \to a$ and $y_n \to b$, then $x_n + y_n \to a+b$, $x_n y_n \to ab$, and, if $b \neq 0$, $x_n/y_n \to a/b$ (for n large enough that $y_n \neq 0$).*

Proof. Sum: given ε, choose N so that $|x_n - a| < \varepsilon/2$ and $|y_n - b| < \varepsilon/2$ for $n \geq N$, and use the triangle inequality. Product: write

$$x_n y_n - ab = x_n(y_n - b) + b(x_n - a).$$

Since (x_n) is bounded by some M, $|x_n y_n - ab| \leq M|y_n - b| + |b||x_n - a|$, which can be made smaller than ε. Quotient: it suffices to show $1/y_n \to 1/b$. For large n, $|y_n| > |b|/2$, so $|1/y_n - 1/b| = |b - y_n|/(|y_n||b|) \leq 2|y_n - b|/|b|^2 \to 0$. □

Proposition 14.5 (Squeeze)**.** *If $x_n \leq y_n \leq z_n$ for all large n and $x_n \to L$, $z_n \to L$, then $y_n \to L$.*

Proof. For large n, $L - \varepsilon < x_n \leq y_n \leq z_n < L + \varepsilon$. □

14.6 Monotone sequences and the Bolzano–Weierstrass theorem

Theorem 14.6 (Monotone convergence)**.** *A nondecreasing sequence that is bounded above converges to its supremum.*

Proof. Let $s = \sup_n x_n$, which exists by completeness. Given $\varepsilon > 0$, the approximation property gives N with $x_N > s - \varepsilon$. For $n \geq N$, monotonicity gives $s - \varepsilon < x_N \leq x_n \leq s$. □

Theorem 14.7 (Bolzano–Weierstrass)**.** *Every bounded real sequence has a convergent subsequence.*

Proof. Call an index m a *peak* if $x_m \geq x_n$ for all $n > m$. If there are infinitely many peaks $m_1 < m_2 < \cdots$, then $x_{m_1} \geq x_{m_2} \geq \cdots$ is a bounded nonincreasing subsequence. Otherwise, after the last peak every index n_1 has a later n_2 with $x_{n_2} > x_{n_1}$, and repeating yields an increasing subsequence. In both cases, monotone convergence applies. □

Theorem 14.8 (Cauchy criterion)**.** *A real sequence converges if and only if it is Cauchy.*

Proof. If $x_n \to L$, then $|x_m - x_n| \le |x_m - L| + |L - x_n| < \varepsilon$ for large m, n. Conversely, a Cauchy sequence is bounded (as in the boundedness proof), so by Bolzano–Weierstrass a subsequence converges to some L. Given ε, choose N with $|x_m - x_n| < \varepsilon/2$ for $m, n \ge N$ and a subsequence index $n_k \ge N$ with $|x_{n_k} - L| < \varepsilon/2$; then $|x_n - L| < \varepsilon$ for $n \ge N$. □

The Cauchy criterion is what iterative numerical methods actually check: a stopping rule such as "stop when successive iterates differ by less than 10^{-8}" is a finite surrogate of the Cauchy condition. It is only a surrogate. The harmonic partial sums $H_n = \sum_{k \le n} 1/k$ have successive differences $1/n \to 0$ but diverge, so small steps alone do not prove convergence.

14.7 Series

A series $\sum_{k=1}^{\infty} a_k$ converges if its partial sums $S_n = \sum_{k \le n} a_k$ converge. The geometric series $\sum_{k \ge 0} r^k$ converges to $1/(1-r)$ iff $|r| < 1$, since $S_n = (1 - r^{n+1})/(1 - r)$. A series converges *absolutely* if $\sum |a_k| < \infty$; absolute convergence implies convergence (by the Cauchy criterion and the triangle inequality) and allows rearrangement without changing the sum. The alternating harmonic series $\sum (-1)^{k+1}/k$ converges, but not absolutely, and can be rearranged to converge to any prescribed real number. The comparison test states that $0 \le a_k \le b_k$ and $\sum b_k < \infty$ imply $\sum a_k < \infty$.

Absolute summability is the stability condition for discrete filters (Chapter 28): if $\sum_k |h_k| < \infty$ and $|x_n| \le M$, then $|\sum_k h_k x_{n-k}| \le M \sum_k |h_k|$.

14.8 Continuity on closed intervals

Theorem 14.9 (Intermediate value theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < c < f(b)$, there is $x \in (a, b)$ with $f(x) = c$.*

Proof. Let $x = \sup\{t \in [a, b] : f(t) < c\}$. If $f(x) < c$, continuity keeps $f < c$ slightly to the right of x, contradicting the supremum; $x < b$ because $f(b) > c$. If $f(x) > c$, continuity keeps $f > c$ on an interval $(x - \delta, x]$, so $x - \delta$ is a smaller upper bound, a contradiction. Hence $f(x) = c$. □

Theorem 14.10 (Extreme value theorem)**.** *A continuous function on a closed bounded interval is bounded and attains its maximum and minimum.*

Proof. If f were unbounded, there would be x_n with $|f(x_n)| > n$. A subsequence converges to some $x \in [a, b]$, and continuity forces $f(x_{n_k}) \to f(x)$, contradicting unboundedness. Let $M = \sup f$; choose x_n with $f(x_n) > M - 1/n$, extract a convergent subsequence with limit x^*, and conclude $f(x^*) = M$ by continuity. □

Both theorems fail on open intervals or for discontinuous functions: $f(x) = 1/x$ on $(0, 1]$ is unbounded, and a step function skips intermediate values. A threshold-crossing detector implicitly relies on the intermediate value theorem for the underlying continuous signal; on samples, the crossing time must be interpolated.

Worked example 14.11 (Bisection as a constructive proof)**.** To find a zero of a continuous f with $f(a) < 0 < f(b)$, evaluate f at the midpoint and keep the half-interval on which the sign changes. After n steps the interval has length $(b - a)2^{-n}$ and, by the nested interval property, the endpoints converge to a zero. To locate a threshold crossing between two samples to $1\,\mu$s when the samples are $33\,\mu$s apart requires $\lceil \log_2 33 \rceil = 6$ bisection steps on an interpolated signal. The accuracy refers to the interpolant, not to the unobserved voltage.

Exercises

1. Prove $2/n \to 0$ from the definition.
2. Give a sequence that is bounded but does not converge.
3. Determine whether the step function $f(x) = 0$ for $x < 0$, $f(x) = 1$ for $x \geq 0$ is continuous at zero.
4. Explain the conceptual difference between a limit as $n \to \infty$ and a sample size n that remains finite.
5. Prove from the definition that $(n+1)/(2n+3) \to 1/2$.
6. Show that $x_n = (1 + 1/n)^n$ is increasing and bounded above by 3, and conclude that it converges.
7. Prove that $\sum_{k \geq 1} 1/k^2$ converges by comparison with $\sum 1/(k(k-1))$ for $k \geq 2$.
8. Show that a polynomial of odd degree has a real root.
9. Give a continuous function on $(0, 1)$ that is bounded but does not attain its supremum.

Chapter 15

Derivatives, integrals, and differential equations

Reading goal 15.1. Build rate of change and accumulation from limits and relate them to state variables and observed neural signals.

15.1 A derivative is a limit of ratios

For a differentiable function f, the derivative at t is

$$f'(t) = \lim_{h \to 0} \frac{f(t+h) - f(t)}{h}.$$

Its units are output units divided by input units: differentiating voltage with respect to time gives volts per second. A derivative expresses an instantaneous rate of change under a continuous model; a finite difference merely estimates this rate using discrete observations. The distinction becomes serious for noisy measurements.

15.2 Integration as accumulation

A Riemann integral $\int_a^b f(t)\,dt$ is a limit of sums of function values multiplied by small interval widths. If $r(t)$ is a firing intensity in spikes/s, its integral over $[a, b]$ has units of expected spikes, and under a conditional-intensity point-process model it represents integrated intensity, not the realized count. If $p(t)$ is power in an appropriate density scale, integrating it over time yields an accumulated quantity with corresponding units.

The fundamental theorem of calculus connects these operations: if $F(t) = \int_a^t f(s)\,ds$ and f is continuous, then $F'(t) = f(t)$. Thus differentiation and integration are linked through limiting constructions.

15.3 Differential equations: rules for evolving states

A differential equation specifies how a variable changes, not necessarily its value at every time. Consider

$$\tau \frac{dx}{dt} = -x + u(t).$$

With $u(t) = u_0$ constant, the solution is $x(t) = u_0 + [x(0) - u_0]e^{-t/\tau}$. The time constant $\tau > 0$ controls relaxation. Without an initial condition, many trajectories satisfy the same

differential equation. This is a model for first-order filtering and a building block for membrane and firing-rate dynamics.

15.4 A distinction between models and data

Recorded spikes are events and need not be represented by a differentiable function. The firing-rate function is an inferred or modeled quantity. Writing $\dot{r}(t)$ for the derivative of an estimated rate assumes a degree of smoothness that the raw point process does not possess. We will state these choices explicitly when we develop state-space models.

15.5 Rules of differentiation

Proposition 15.2 (Differentiability implies continuity)**.** *If f is differentiable at t, it is continuous at t.*

Proof. $f(t+h) - f(t) = h \cdot \frac{f(t+h)-f(t)}{h} \to 0 \cdot f'(t) = 0$ as $h \to 0$. □

The converse is false: $|t|$ is continuous but not differentiable at 0.

Theorem 15.3 (Product and chain rules)**.** *If f, g are differentiable at t, then $(fg)'(t) = f'(t)g(t) + f(t)g'(t)$. If g is differentiable at t and f at $g(t)$, then $(f \circ g)'(t) = f'(g(t))g'(t)$.*

Proof. For the product,

$$\frac{f(t+h)g(t+h) - f(t)g(t)}{h} = \frac{f(t+h) - f(t)}{h} g(t+h) + f(t) \frac{g(t+h) - g(t)}{h},$$

and $g(t+h) \to g(t)$ by continuity. For the chain rule, define $\phi(k) = [f(g(t)+k) - f(g(t))]/k$ for $k \neq 0$ and $\phi(0) = f'(g(t))$; ϕ is continuous at 0. With $k = g(t+h) - g(t)$, we have $f(g(t+h)) - f(g(t)) = \phi(k)k$ in all cases, so the difference quotient equals $\phi(k)\,[g(t+h)-g(t)]/h \to f'(g(t))g'(t)$. □

15.6 The mean value theorem

Theorem 15.4 (Rolle and mean value)**.** *Let f be continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$, there is $c \in (a, b)$ with $f'(c) = 0$. In general there is $c \in (a, b)$ with*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. By the extreme value theorem f attains a maximum and a minimum on $[a, b]$. If both occur at endpoints, f is constant and $f' \equiv 0$. Otherwise an extremum occurs at an interior c, where the one-sided difference quotients have opposite signs, forcing $f'(c) = 0$. For the general case apply Rolle's theorem to $g(t) = f(t) - \frac{f(b)-f(a)}{b-a}(t - a)$. □

Consequences: a function with $f' \equiv 0$ on an interval is constant; $f' > 0$ implies f is strictly increasing; and if $|f'| \leq K$ then $|f(x) - f(y)| \leq K|x - y|$ (a *Lipschitz* bound). The last statement bounds how fast a signal can change given a bound on its slope, which links bandwidth to sample-to-sample variation.

15.7 Taylor's theorem and the error of finite differences

Theorem 15.5 (Taylor with Lagrange remainder)**.** *If f is $n+1$ times differentiable on an interval containing t and $t+h$, then*

$$f(t+h) = \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!} h^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} h^{n+1}$$

for some ξ between t and $t+h$.

The proof applies Rolle's theorem repeatedly to the difference between f and its Taylor polynomial. Two consequences matter for sampled signals. The forward difference satisfies

$$\frac{f(t+h)-f(t)}{h} = f'(t) + \tfrac{h}{2} f''(\xi),$$

an $O(h)$ error, whereas the central difference $[f(t+h)-f(t-h)]/(2h) = f'(t) + O(h^2)$. If each sample carries independent noise of standard deviation σ, the forward difference adds noise of standard deviation $\sqrt{2}\sigma/h$. The total error $\approx \tfrac{h}{2}|f''| + \sqrt{2}\sigma/h$ is minimized at $h^* = \sqrt{2\sqrt{2}\sigma/|f''|}$: shrinking h below this value makes the derivative estimate *worse*.

15.8 The exponential function

Define $\exp(x) = \sum_{k\ge 0} x^k/k!$, which converges absolutely for every real or complex x by comparison with a geometric series once $k > 2|x|$. Termwise multiplication of absolutely convergent series and the binomial theorem give $\exp(x+y) = \exp(x)\exp(y)$, and termwise differentiation (justified by uniform convergence on bounded sets) gives $\exp' = \exp$. Writing $e^x = \exp(x)$ and splitting the series for $e^{i\theta}$ into even and odd powers yields

$$e^{i\theta} = \sum_{m\ge 0} \frac{(-1)^m \theta^{2m}}{(2m)!} + i \sum_{m\ge 0} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!} = \cos\theta + i\sin\theta,$$

which proves Euler's formula once cosine and sine are defined by these series.

15.9 Linear first-order equations in general

Theorem 15.6. *Let $\tau > 0$ and u be continuous. The unique solution of $\tau \dot{x} = -x + u(t)$ with $x(0) = x_0$ is*

$$x(t) = x_0 e^{-t/\tau} + \frac{1}{\tau}\int_0^t e^{-(t-s)/\tau} u(s)\, ds.$$

Proof. Multiply by the integrating factor $e^{t/\tau}/\tau$: $\frac{d}{dt}(e^{t/\tau}x) = e^{t/\tau}u(t)/\tau$. Integrate from 0 to t and multiply by $e^{-t/\tau}$. If x, y are two solutions, $w = x - y$ satisfies $\tau\dot{w} = -w$, $w(0) = 0$, so $\frac{d}{dt}(e^{t/\tau}w) = 0$ and $w \equiv 0$. □

The solution is a *convolution* of the input with the exponential kernel $k(t) = \tau^{-1} e^{-t/\tau}\mathbf{1}_{t\ge 0}$. A leaky integrator, an RC membrane, and an exponential synaptic filter are the same mathematical object. Its step response reaches $1 - e^{-1} \approx 63\%$ of its final value after one time constant.

Worked example 15.7 (Membrane time constant from a current step)**.** A passive membrane obeys $C\dot{V} = -(V - E_L)/R + I(t)$, i.e. $\tau\dot{V} = -(V-E_L) + RI$ with $\tau = RC$. For a step $I(t) = I_0$ at $t = 0$ from rest, $V(t) = E_L + RI_0(1 - e^{-t/\tau})$. With $R = 100\,\mathrm{M\Omega}$ and $C = 200\,\mathrm{pF}$, $\tau = 20\,\mathrm{ms}$ and a $50\,\mathrm{pA}$ step produces a $5\,\mathrm{mV}$ steady depolarization. Fitting τ from data presumes the passive model; active conductances, space clamp errors, and series resistance change the curve.

Exercises

1. Differentiate $f(t) = e^{-t/\tau}$ and confirm the result has units of inverse time.
2. Solve $dx/dt = -ax$ for $a > 0$ and a specified initial condition.
3. Explain why a spike-count cumulative function is not ordinarily differentiable at spike times.
4. Compute $\int_0^T 3\,dt$ and interpret it for a rate of 3 spikes/s.
5. Use the mean value theorem to prove $|\sin x - \sin y| \le |x - y|$.
6. Show that the central difference has error $O(h^2)$ using Taylor's theorem to third order.
7. Solve $\tau \dot{x} = -x + \sin(\omega t)$ with $x(0) = 0$ and find the steady-state amplitude and phase lag.
8. Show that $f(t) = t^2 \sin(1/t)$ (with $f(0) = 0$) is differentiable at 0 but f' is not continuous there.
9. Derive the optimal step h^* for the noisy forward difference.

Chapter 16

Several variables, optimization, and geometry

Reading goal 16.1. Generalize derivatives to gradients and Jacobians and understand curvature, critical points, and constrained optimization.

16.1 The gradient is a local direction of change

For $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is $\nabla f = (\partial f/\partial x_1, \dots, \partial f/\partial x_n)^\top$. Its dot product with a direction v gives the directional derivative $D_v f = \nabla f \cdot v$ for a differentiable function. In a neighborhood of x, $f(x+h) \approx f(x) + \nabla f(x)^\top h$. The gradient depends on the units and coordinate scaling of the input variables.

A vector-valued map $F : \mathbb{R}^n \to \mathbb{R}^m$ has Jacobian $J_{ij} = \partial F_i/\partial x_j$. Near a point, $F(x+h) \approx F(x) + J_F(x)h$. The chain rule for $G \circ F$ is $J_{G\circ F}(x) = J_G(F(x))J_F(x)$, provided the dimensions match.

16.2 Second derivatives and the Hessian

The Hessian of a twice differentiable scalar function is the matrix of second partial derivatives $H_{ij} = \partial^2 f/(\partial x_i \partial x_j)$. At a stationary point $\nabla f = 0$, a positive-definite Hessian gives a strict local minimum; an indefinite Hessian indicates a saddle. The quadratic approximation

$$f(x+h) \approx f(x) + \nabla f(x)^\top h + \tfrac{1}{2} h^\top H_f(x) h$$

provides a local account of curvature and motivates Newton-type optimization.

16.3 Optimization as fitting a model

Least squares seeks $\hat{\theta} = \arg\min_\theta \sum_i [y_i - f_\theta(x_i)]^2$. The objective is a chosen measure of mismatch, not a guarantee that the parameter has a physiological meaning. Regularization changes the optimization problem, e.g.

$$\hat{\theta} = \arg\min_\theta \left\{ \|y - X\theta\|_2^2 + \lambda \|\theta\|_2^2 \right\}.$$

The penalty limits parameter magnitude and can improve stability, but also introduces bias. Constraints such as $\theta \geq 0$ are meaningful only if supported by the model and units.

16.4 The manifold idea starts locally

A smooth curve in $\mathbb{R}^3$ may be one-dimensional although it occupies a three-dimensional ambient space. Near a regular point, its variation can be approximated by a tangent direction. General smooth manifolds are spaces locally modeled on $\mathbb{R}^d$; the ambient dimension and intrinsic dimension need not coincide. PCA identifies a linear subspace, but whether the data genuinely concentrate around a nonlinear manifold is an additional question.

16.5 Differentiability in several variables

Partial derivatives alone do not make a function well behaved. The function $f(x,y) = xy/(x^2+y^2)$ with $f(0,0) = 0$ has both partial derivatives equal to zero at the origin, yet is not even continuous there (along $y = x$ it equals $1/2$). The correct notion is *(Fréchet) differentiability*: $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at x if there is a vector g with

$$\lim_{h \to 0} \frac{|f(x+h) - f(x) - g^{\top} h|}{\|h\|} = 0,$$

in which case $g = \nabla f(x)$. A useful sufficient condition is that all partial derivatives exist near x and are continuous at x. When second partials are continuous, mixed partials commute (Schwarz–Clairaut), so the Hessian is symmetric.

Proposition 16.2 (Steepest ascent)**.** *If f is differentiable at x and $\nabla f(x) \neq 0$, then among unit vectors v the directional derivative $D_v f(x)$ is maximized by $v = \nabla f(x)/\|\nabla f(x)\|$, with maximal value $\|\nabla f(x)\|$.*

Proof. By the Cauchy–Schwarz inequality, $D_v f = \nabla f^{\top} v \le \|\nabla f\|\|v\| = \|\nabla f\|$, with equality iff v is a positive multiple of ∇f. □

16.6 Convexity

A set C is convex if it contains the segment between any two of its points. A function f on a convex set is *convex* if

$$f(\lambda x + (1-\lambda) y) \le \lambda f(x) + (1-\lambda) f(y) \qquad (0 \le \lambda \le 1).$$

Theorem 16.3. *Let f be twice continuously differentiable on an open convex set. Then f is convex iff its Hessian is positive semidefinite everywhere. If f is convex and differentiable, every stationary point is a global minimizer.*

Proof of the second statement. For differentiable convex f, the convexity inequality rearranges to $[f(x + \lambda(y-x)) - f(x)]/\lambda \le f(y) - f(x)$; letting $\lambda \downarrow 0$ gives the *supporting hyperplane inequality* $f(y) \ge f(x) + \nabla f(x)^{\top}(y-x)$. If $\nabla f(x) = 0$, then $f(y) \ge f(x)$ for every y. □

Convexity is why some fitting problems are reliable and others are not. Least squares, ridge regression, and the Poisson GLM with canonical log link (Chapter 35) have convex negative log-likelihoods, so any local optimizer that converges finds the global optimum. Fitting a mixture model, a neural network, or a tensor decomposition is nonconvex; different initializations may give different answers, and agreement across restarts is evidence, not proof, of a global optimum.

16.7 Constrained optimization and Lagrange multipliers

Theorem 16.4 (Lagrange multipliers, one constraint)**.** *Let f, g be continuously differentiable and suppose x^* is a local extremum of f subject to $g(x) = 0$, with $\nabla g(x^*) \neq 0$. Then $\nabla f(x^*) = \lambda \nabla g(x^*)$ for some scalar λ.*

The geometric idea: on the constraint surface, f cannot increase to first order in any tangent direction, so ∇f has no component tangent to the surface and must be parallel to its normal ∇g. The PCA derivation in Chapter 20 is exactly this theorem with $f(v) = v^\top S v$ and $g(v) = v^\top v - 1$. The multiplier has a meaning: it is the rate at which the optimal value changes as the constraint level changes. In PCA it equals the captured variance.

16.8 Gradient descent and Newton's method

To minimize f, *gradient descent* iterates $x_{k+1} = x_k - \eta \nabla f(x_k)$. For the quadratic $f(x) = \frac{1}{2} x^\top A x$ with A symmetric positive definite and eigenvalues $0 < \mu \leq \cdots \leq L$, the iteration is $x_{k+1} = (I - \eta A) x_k$, which converges iff $0 < \eta < 2/L$; the best fixed step $\eta = 2/(\mu + L)$ contracts errors by the factor $(\kappa - 1)/(\kappa + 1)$ per step, with condition number $\kappa = L/\mu$. Badly scaled variables (e.g. one regressor in volts and another in microvolts) produce large κ and slow convergence; standardizing regressors is therefore a numerical as well as a statistical decision.

Newton's method uses the quadratic model: $x_{k+1} = x_k - H_f(x_k)^{-1} \nabla f(x_k)$. On a quadratic it converges in one step; near a nondegenerate minimum it converges quadratically (the number of correct digits roughly doubles each step). Its cost is forming and solving with the Hessian. Iteratively reweighted least squares for GLMs is Newton's method applied to the log-likelihood.

Worked example 16.5 (A constrained variance problem)**.** Maximize $f(x, y) = 3x^2 + 2xy + 3y^2$ on the unit circle $x^2 + y^2 = 1$. Lagrange's condition gives $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$, an eigenvalue problem with eigenvalues 4 (direction $(1, 1)/\sqrt{2}$) and 2 (direction $(1, -1)/\sqrt{2}$). The maximum is 4, the minimum is 2. If f is the variance of a projected two-channel signal, the first direction is the common-mode component and the second is the differential (bipolar) component.

Exercises

1. Compute the gradient and Hessian of $f(x, y) = x^2 + 2y^2$.
2. Use the chain rule to differentiate $g(t) = \sin(x(t)^2)$.
3. Compare L2 and L1 penalties conceptually in regression.
4. Give an example of a curved one-dimensional set in $\mathbb{R}^2$.
5. Show that $f(x, y) = xy/(x^2 + y^2)$, $f(0, 0) = 0$, is discontinuous at the origin.
6. Prove that a sum of convex functions is convex, and that $\|x\|_2^2$ is convex.
7. Use Lagrange multipliers to minimize $x^2 + y^2$ subject to $x + 2y = 5$.
8. For $f(x) = \frac{1}{2}(x_1^2 + 100 x_2^2)$, compute the condition number and the best fixed-step gradient-descent contraction factor.
9. Perform two Newton steps for $f(x) = x^4/4 - x$ starting at $x_0 = 2$.

Chapter 17

Integration, function spaces, and transforms

Reading goal 17.1. Prepare the mathematical ground for Hilbert spaces, Fourier coefficients, energy, and stochastic-process covariance.

17.1 From vectors of numbers to functions

The Euclidean inner product $\sum_{i=1}^{n} x_i y_i$ measures agreement between two finite vectors. For sufficiently integrable real functions on an interval, the analogous quantity is

$$\langle f, g \rangle = \int_a^b f(t) g(t)\, dt.$$

For complex functions we conjugate the second or first argument according to convention; here we use $\langle f, g \rangle = \int_a^b f(t)\overline{g(t)}\, dt$, linear in the first argument. The induced norm is $\|f\|_2 = (\int |f|^2)^{1/2}$. Signals with finite squared integral belong to the space L^2, after identifying functions equal almost everywhere.

17.2 Orthogonality and approximation

Two functions are orthogonal when their inner product is zero. If $e_1, \ldots, e_k$ are orthonormal, the projection of f onto their span is $\sum_j \langle f, e_j \rangle e_j$. The error is orthogonal to the span: this is the same geometry later used in least squares, PCA, and finite-basis spectral expansions.

17.3 Measure and probability: what does "almost everywhere" mean?

A measure assigns nonnegative sizes to suitable sets. In the familiar one-dimensional setting, interval length is the basic example. The Lebesgue integral extends Riemann integration and provides a stable theory for limits of integrable functions. "Almost everywhere" means except on a set of measure zero. Two functions that differ at finitely many time points can represent the same L^2 element, even though pointwise spike events at those instants could be scientifically significant. This reminds us that the choice of mathematical object must fit the observation type.

17.4 Generalized signals and impulses

The ideal Dirac delta $\delta(t - t_0)$ is not an ordinary L^2 function. In generalized-function notation its defining property is $\int f(t)\delta(t - t_0)\,dt = f(t_0)$ for suitable test functions. A spike train can be represented formally as $s(t) = \sum_j \delta(t - t_j)$, but practical recordings contain finite-time waveforms and discrete timestamps. Both descriptions are useful, and neither should be confused with the other.

Worked example 17.2 (Projection onto one sinusoid)**.** On $[0, T]$, let $e(t) = \sqrt{2/T}\cos(2\pi kt/T)$ for a nonzero integer k. Its squared integral is one. The coefficient $a = \int_0^T x(t)e(t)\,dt$ determines the best approximation $ae(t)$ to $x(t)$ within this one-dimensional subspace under squared-integral error. This is the geometric core of Fourier analysis.

17.5 The Riemann integral and its limits

For a bounded f on $[a, b]$ and a partition $a = t_0 < \cdots < t_n = b$, the upper and lower sums are $U = \sum_k \sup_{[t_{k-1},t_k]} f\,\Delta t_k$ and $L = \sum_k \inf_{[t_{k-1},t_k]} f\,\Delta t_k$. The function is Riemann integrable when $\inf U = \sup L$ over all partitions. Every continuous function on $[a, b]$ is integrable (because it is uniformly continuous there), and so is every monotone function.

The indicator of the rationals on $[0, 1]$ (the Dirichlet function) is not Riemann integrable: every subinterval contains rationals and irrationals, so $U = 1$ and $L = 0$ for every partition. In Lebesgue's theory, the rationals form a set of measure zero and the integral is 0. This example is artificial, but the deficiency it reveals is serious: Riemann integrals do not behave well under limits.

17.6 When can limits and integrals be exchanged?

Let $f_n(t) = n\,\mathbf{1}_{(0,1/n)}(t)$ on $[0, 1]$. Then $f_n(t) \to 0$ for every t, but $\int_0^1 f_n = 1$ for every n. Pointwise convergence does not imply convergence of integrals; the mass escapes into a spike that becomes infinitely narrow, which is precisely how the delta function is approximated.

Theorem 17.3 (Uniform convergence)**.** *If $f_n \to f$ uniformly on $[a, b]$ (that is, $\sup_t |f_n(t) - f(t)| \to 0$) and each f_n is integrable, then f is integrable and $\int f_n \to \int f$.*

Proof of the convergence statement. $\left|\int f_n - \int f\right| \le \int |f_n - f| \le (b-a)\sup_t |f_n(t) - f(t)| \to 0$. □

Lebesgue's *dominated convergence theorem* gives a far more flexible criterion: if $f_n \to f$ almost everywhere and $|f_n| \le g$ for one integrable g, then $\int f_n \to \int f$. In the example above no integrable dominating function exists, since $\sup_n f_n(t) \approx 1/t$ near zero. Dominated convergence justifies differentiating under the integral sign, exchanging expectation and limits, and many steps in spectral estimation.

17.7 Inequalities in inner-product spaces

Theorem 17.4 (Cauchy–Schwarz)**.** *In any inner-product space, $|\langle f, g\rangle| \le \|f\|\,\|g\|$, with equality iff f and g are linearly dependent.*

Proof. If $g = 0$ there is nothing to prove. Otherwise let $c = \langle f, g\rangle / \|g\|^2$ and expand $0 \le \|f - cg\|^2 = \|f\|^2 - |\langle f, g\rangle|^2/\|g\|^2$. Equality holds iff $f = cg$. □

The triangle inequality $\|f+g\| \le \|f\| + \|g\|$ follows by expanding $\|f+g\|^2$ and applying Cauchy–Schwarz. The normalized quantity $\langle f, g\rangle/(\|f\|\|g\|)$ lies in $[-1, 1]$ for real functions; applied to centered data vectors it is the Pearson correlation, and applied to cross-spectra across trials it underlies coherence. Cauchy–Schwarz is the reason both are bounded by one.

Theorem 17.5 (Pythagoras and Bessel)**.** *If $e_1, \dots, e_k$ are orthonormal, then for every f,*

$$\Big\| f - \sum_j \langle f, e_j\rangle e_j \Big\|^2 = \|f\|^2 - \sum_j |\langle f, e_j\rangle|^2, \qquad \textit{hence} \qquad \sum_j |\langle f, e_j\rangle|^2 \le \|f\|^2.$$

Proof. Let $p = \sum_j \langle f, e_j\rangle e_j$. Then $\langle f - p, e_j\rangle = 0$ for every j, so $f - p \perp p$ and $\|f\|^2 = \|f-p\|^2 + \|p\|^2$, where $\|p\|^2 = \sum_j |\langle f, e_j\rangle|^2$ by orthonormality. □

Bessel's inequality says that the energy captured by any orthonormal set of components cannot exceed the total energy. When the orthonormal set is *complete* (its span is dense), equality holds for an infinite sum: this is Parseval's identity, and the Fourier basis is the central example.

17.8 Completeness of function spaces

A normed space is *complete* (a Banach space; a Hilbert space if the norm comes from an inner product) if every Cauchy sequence converges in the space. The continuous functions on $[0, 1]$ with the L^2 norm are not complete: continuous ramps approximating a step function form a Cauchy sequence whose limit is discontinuous. The space L^2 is the completion of this space, which is why Fourier series, least-squares approximation, and stochastic integrals are formulated there. As with $\mathbb{Q} \subset \mathbb{R}$, completeness is the property that guarantees that approximation procedures have something to converge to.

17.9 Convolution

For integrable f, g on $\mathbb{R}$, the convolution is $(f * g)(t) = \int f(s) g(t - s)\, ds$. It is commutative and associative, and $\|f * g\|_1 \le \|f\|_1 \|g\|_1$. Convolution with a nonnegative kernel of unit integral is a weighted local average; convolving a spike train $\sum_j \delta(t - t_j)$ with a Gaussian kernel k_σ gives the familiar smoothed rate $\sum_j k_\sigma(t - t_j)$. The bandwidth σ trades temporal resolution for variance, and no choice of σ recovers the "true" rate without a model of what the rate is.

Worked example 17.6 (Energy of a windowed sinusoid)**.** Let $x(t) = A\cos(2\pi f_0 t)$ on $[0, T]$ with $f_0 T$ a positive integer. Then $\|x\|_2^2 = A^2 T/2$. In the orthonormal basis of the previous example, only the coefficient with $k = f_0 T$ is nonzero, and it equals $A\sqrt{T/2}$; Bessel's inequality holds with equality. If $f_0 T$ is not an integer, energy spreads across many coefficients, but by Parseval their squared sum is still $\|x\|_2^2$. Leakage redistributes energy; it does not create or destroy it.

Exercises

1. Verify that $\int_0^T \sin(2\pi n t/T)\cos(2\pi m t/T)\, dt = 0$ for positive integers m, n.
2. Explain why delta functions require care in ordinary function spaces.
3. State the difference between orthogonality and statistical independence.
4. Show that $f_n(t) = t^n$ converges pointwise on $[0, 1]$ to a discontinuous limit, but not uniformly.

5. Prove the triangle inequality in an inner-product space from Cauchy–Schwarz.
6. Compute $\mathbf{1}_{[0,1]} * \mathbf{1}_{[0,1]}$ and sketch it.
7. Show that a Gaussian smoothing kernel of standard deviation σ applied to a single spike produces a function with integral one.
8. Exhibit a Cauchy sequence of continuous functions in the $L^2[0,1]$ norm whose limit is a step function.

Chapter 18

Bridges from number systems to calculus

Reading goal 18.1. Learn to justify limits, continuity, differentiation, and integration rather than treating calculus rules as unexplained computational recipes.

Bridge 17: A convergent sequence has only one limit

Proposition 18.2. *If $x_n \to a$ and $x_n \to b$ in $\mathbb{R}$, then $a = b$.*

Proof. Suppose $a \neq b$. Set $\varepsilon = |a-b|/3$. For all sufficiently large n, both $|x_n - a| < \varepsilon$ and $|x_n - b| < \varepsilon$. The triangle inequality yields $|a-b| \leq |a-x_n| + |x_n - b| < 2|a-b|/3$, a contradiction. □

Example. Two numerical routines approximating the same well-defined limit may have different finite-sample outputs; a proven convergent sequence cannot have two distinct limiting values.

Bridge 18: Limits respect addition and multiplication

Proposition 18.3. *If real sequences $x_n \to a$ and $y_n \to b$, then $x_n + y_n \to a+b$ and $x_n y_n \to ab$.*

Proof. For sums, $|(x_n+y_n)-(a+b)| \leq |x_n-a| + |y_n-b| \to 0$. For products, write $x_n y_n - ab = (x_n - a)y_n + a(y_n - b)$. A convergent sequence is bounded, so the first term tends to zero and the second does too. □

Example. If a mean and a calibration gain both converge in a controlled limit, their product converges. It does not follow that multiplying two biased finite-sample estimators produces an unbiased estimate.

Bridge 19: Bounded monotone real sequences converge

Proposition 18.4. *Every nondecreasing real sequence bounded above converges to the supremum of its values.*

Proof. Let $s = \sup\{x_n : n \geq 0\}$. For any $\varepsilon > 0$, the supremum approximation property provides N with $s - \varepsilon < x_N \leq s$. Monotonicity then gives $s - \varepsilon < x_n \leq s$ for all $n \geq N$, which is convergence to s. □

Example. A running maximum of a bounded score converges mathematically, but the limit may depend on the candidate set and need not estimate true predictive accuracy.

Bridge 20: Composition preserves continuity

Proposition 18.5. *If f is continuous at a and g is continuous at $f(a)$, then $g \circ f$ is continuous at a.*

Proof. Given an output tolerance ε, continuity of g supplies a tolerance η on its input. Continuity of f supplies δ such that $|x - a| < \delta$ implies $|f(x) - f(a)| < \eta$. The latter implies $|g(f(x)) - g(f(a))| < \varepsilon$. □

Example. A continuously varying model amplitude followed by a continuous nonlinear transfer function remains continuous. Thresholding is discontinuous and can violate this conclusion.

Bridge 21: A continuous curve cannot jump over an intermediate value

Theorem 18.6 (Intermediate value theorem)**.** *If f is continuous on $[a, b]$ and $f(a) < c < f(b)$, then some $t \in (a, b)$ satisfies $f(t) = c$.*

Proof. Let $S = \{x \in [a, b] : f(x) \leq c\}$ and $s = \sup S$. Continuity at s rules out $f(s) < c$ (then a nearby point to the right would still lie in S), and rules out $f(s) > c$ (then nearby points to the left could not approach s from S). Hence $f(s) = c$. □

Example. If a truly continuous filtered trace changes sign, it must cross zero. A pair of digital samples with opposite signs only implies a crossing if a continuous interpolation model is justified.

Bridge 22: A continuous function has extrema on a compact interval

Theorem 18.7 (Extreme value theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous, it attains both a minimum and a maximum.*

Proof. Continuity on the compact interval makes $f([a, b])$ compact: for any convergent sequence of images, extract a convergent subsequence of preimages and use continuity to identify its limit. The image is thus bounded and closed, so its supremum and infimum belong to it. □

Example. A continuous fit score on a closed, bounded parameter interval has an optimum. Open intervals, discontinuities, or unconstrained coefficients require separate existence arguments.

Bridge 23: Sum and product differentiation rules

Proposition 18.8. *For functions differentiable at t, $(f+g)'(t) = f'(t) + g'(t)$ and $(fg)'(t) = f'(t)g(t) + f(t)g'(t)$.*

Proof. Linearity follows by splitting the difference quotient. For the product, add and subtract $f(t+h)g(t)$ in the numerator:

$$\frac{f(t+h)g(t+h) - f(t)g(t)}{h} = f(t+h)\frac{g(t+h)-g(t)}{h} + g(t)\frac{f(t+h)-f(t)}{h}.$$

Differentiability implies continuity of f, so the limit yields the formula. □

Example. Differentiating an ideal amplitude-modulated oscillation contains both envelope change and carrier change, explaining why temporal derivatives mix these mechanisms.

Bridge 24: The chain rule follows from local linear approximation

Theorem 18.9 (Chain rule)**.** *If f is differentiable at t and g at $f(t)$, then $(g \circ f)'(t) = g'(f(t))f'(t)$.*

Proof. Write $g(y+u) = g(y)+g'(y)u+r(u)u$ where $r(u) \to 0$ as $u \to 0$, with $r(0) = 0$. Substitute $y = f(t)$ and $u = f(t+h) - f(t)$; divide by h. The first term converges to $g'(f(t))f'(t)$ and the remainder to zero because u/h stays bounded. □

Example. For a fitted intensity $\lambda(t) = \exp(\eta(t))$, we have $\lambda'(t) = \lambda(t)\eta'(t)$, provided η is differentiable.

Bridge 25: Differentiability provides a first-order remainder

Proposition 18.10. *A function f differentiable at a satisfies $f(a+h) = f(a) + f'(a)h + o(h)$ as $h \to 0$.*

Proof. Set $r(h) = [f(a+h) - f(a)]/h - f'(a)$ for nonzero h and $r(0) = 0$. Differentiability says $r(h) \to 0$. Rearranging gives $f(a+h) = f(a) + f'(a)h + hr(h)$, and $hr(h) = o(h)$. □

Example. A local linear approximation of a nonlinear observation model is accurate only near its expansion point; the remainder is not globally negligible.

Bridge 26: Integrating a rate recovers accumulated change

Theorem 18.11 (Fundamental theorem of calculus)**.** *If f is continuous on $[a,b]$ and $F(t) = \int_a^t f(s)\,ds$, then $F'(t) = f(t)$ at interior points.*

Proof. For $h \neq 0$ with $t+h \in [a,b]$, $[F(t+h)-F(t)]/h$ is the average of f over the short interval from t to $t+h$. By continuity, this average differs from $f(t)$ by at most $\sup_{s \text{ between } t,t+h} |f(s) - f(t)|$, which tends to zero. □

Example. Integrating a conditional spike intensity gives an expected count only under an appropriate point-process model; the calculus theorem alone does not establish that the fitted intensity is correct.

Exercises

1. Prove continuity of $x \mapsto x^2$ using limit arithmetic.
2. Explain which hypothesis fails for the Heaviside step function in the intermediate value theorem.
3. Simulate first-difference derivatives of a noisy sine wave at increasing sample rates.

Part IV

Linear algebra and numerical models

Chapter 19

Vectors, matrices, and linear mappings

Reading goal 19.1. Learn the grammar of population matrices before using covariance, singular values, or latent factors.

19.1 Arrays are not merely spreadsheets

A vector is an element of a vector space: an object that can be added to another vector and scaled while obeying specific algebraic rules. A matrix $A \in \mathbb{R}^{m\times n}$ represents a linear mapping $A : \mathbb{R}^n \to \mathbb{R}^m$. Its jth column is the image of the jth standard basis vector. The product Ax forms linear combinations of these columns. Dimensions are part of the meaning: AB is defined only when the number of columns of A equals the number of rows of B.

For a population of N neurons sampled at T time points, set $X \in \mathbb{R}^{T\times N}$, with rows indexing time and columns indexing neurons. Then Xv is a time series obtained by weighting neurons with $v \in \mathbb{R}^N$. If a recording is stored in channel-by-time orientation, as often happens in MATLAB, it must be transposed before blindly applying a formula whose convention is time-by-neuron.

19.2 Rank, kernels, and invertibility

The rank of A is the dimension of its image or column space. Its kernel $\ker A = \{x : Ax = 0\}$ contains directions invisible to the mapping. Rank-nullity says $\dim \ker A + \operatorname{rank} A = n$. A square matrix is invertible precisely when its kernel contains only zero. A measurement operator with a nontrivial kernel cannot uniquely reconstruct arbitrary inputs: different sources differ by kernel vectors while producing identical observations.

19.3 Inner products and projections

The Euclidean inner product is $\langle x, y\rangle = x^\top y$, with length $\|x\|_2 = \sqrt{x^\top x}$. For a matrix U whose columns are orthonormal, $U^\top U = I$, the orthogonal projection onto their span is $P = UU^\top$. Check $P^2 = P$ and $P^\top = P$. The residual $(I - P)x$ is orthogonal to the selected subspace.

19.4 Least squares as orthogonal projection

If X has full column rank, minimizing $\|y - X\beta\|_2^2$ yields the normal equations $X^\top X\hat{\beta} = X^\top y$ and solution $\hat{\beta} = (X^\top X)^{-1}X^\top y$. In practice, QR or SVD is usually numerically safer than explicitly forming the inverse of $X^\top X$; squaring the condition number can magnify numerical problems.

Worked example 19.2 (Which axis contains the data?)**.** A matrix of 30 channels and 1,000,000 samples has shape 30×10^6. An operation computing covariances among channels should treat the 30 channels as variables, not the million sample positions. A mistaken transpose can yield a million-by-million covariance matrix or an out-of-memory failure. Mathematical notation doubles as a memory-design contract.

19.5 Vector spaces, subspaces, and bases

A real *vector space* is a set V with addition and scalar multiplication satisfying the usual axioms (associativity, commutativity, a zero vector, additive inverses, and distributive laws). Examples include $\mathbb{R}^n$, the $m \times n$ matrices, polynomials of degree at most d, and the continuous functions on an interval. A *subspace* is a nonempty subset closed under addition and scalar multiplication; it contains 0.

Vectors $v_1, \dots, v_k$ are *linearly independent* if $\sum_j c_j v_j = 0$ forces all $c_j = 0$. They *span* V if every vector is a linear combination of them, and they form a *basis* if both hold.

Theorem 19.3 (Steinitz exchange; invariance of dimension)**.** *If $v_1, \dots, v_k$ are linearly independent in V and $w_1, \dots, w_m$ span V, then $k \le m$. Consequently any two finite bases of V have the same number of elements, the* dimension *of V.*

Sketch. Write $v_1 = \sum_i a_i w_i$; some $a_i \neq 0$ since $v_1 \neq 0$, so w_i can be replaced by v_1 without changing the span. Repeat with v_2, which must involve some remaining w (otherwise v_2 would be a combination of v_1, contradicting independence). After k steps all v_j have entered, so at least k of the w's existed. □

Coordinates are only defined relative to a basis. A recording vector in "channel coordinates" becomes a different list of numbers after re-referencing (a change of basis), although it represents the same physical potential distribution up to the reference convention.

19.6 Linear maps and their matrices

A map $T : V \to W$ is *linear* if $T(au + bv) = aT(u) + bT(v)$. Once bases are chosen, T is represented by the matrix whose jth column holds the coordinates of T applied to the jth basis vector. Composition of linear maps corresponds to matrix multiplication, which is why matrix multiplication is associative but generally not commutative.

Theorem 19.4 (Rank–nullity)**.** *For a linear map $T : V \to W$ with* $\dim V = n$, $\dim \ker T + \dim \operatorname{im} T = n$.

Proof. Let $u_1, \dots, u_k$ be a basis of $\ker T$ and extend it to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of V, so $n = k + r$. The vectors $Tv_1, \dots, Tv_r$ span the image, since $Tu_i = 0$. They are independent: if $\sum c_j T v_j = 0$, then $\sum c_j v_j \in \ker T$, so it is a combination of the u_i, and independence of the full basis forces all $c_j = 0$. Thus $\dim \operatorname{im} T = r$. □

Proposition 19.5. *For a square matrix $A \in \mathbb{R}^{n\times n}$ the following are equivalent: (i) A is invertible; (ii) $\ker A = \{0\}$; (iii) $\operatorname{rank} A = n$; (iv) the columns of A form a basis; (v) $\det A \neq 0$; (vi) $Ax = b$ has exactly one solution for every b.*

The equivalence of (ii) and (iii) is rank–nullity; the others follow from it and the definitions. For non-square matrices, injectivity and surjectivity separate: a tall matrix ($m > n$) with full column rank is injective but not surjective, so $Ax = b$ has at most one solution and usually none, which is the setting of least squares.

19.7 Row and column spaces; the four fundamental subspaces

For $A \in \mathbb{R}^{m\times n}$, the column space $\operatorname{col}(A) \subseteq \mathbb{R}^m$ and the null space $\ker A \subseteq \mathbb{R}^n$ are accompanied by the row space $\operatorname{col}(A^\top) \subseteq \mathbb{R}^n$ and the left null space $\ker A^\top \subseteq \mathbb{R}^m$.

Theorem 19.6. $\ker A$ *is the orthogonal complement of* $\operatorname{col}(A^\top)$ *in* $\mathbb{R}^n$*, and* $\ker A^\top$ *is the orthogonal complement of* $\operatorname{col}(A)$ *in* $\mathbb{R}^m$*. In particular,* $\operatorname{rank} A = \operatorname{rank} A^\top$.

Proof. $Ax = 0$ iff every row of A has zero inner product with x, iff x is orthogonal to the row space. Applying this to $A^\top$ gives the second statement. By rank–nullity, $\operatorname{rank} A = n - \dim\ker A = n - (n - \operatorname{rank} A^\top) = \operatorname{rank} A^\top$. □

The left null space has a direct reading for least squares: the residual $y - X\hat\beta$ lies in $\ker X^\top$, i.e. it is orthogonal to every regressor. If a nuisance regressor (e.g. running speed) is included, the residual is uncorrelated with it in-sample by construction, which is not evidence that the nuisance was "removed" from the underlying process.

19.8 Norms and condition numbers

The operator norm of A is $\|A\|_2 = \max_{\|x\|=1} \|Ax\|_2$. For invertible A the *condition number* is $\kappa(A) = \|A\|_2\|A^{-1}\|_2 \geq 1$.

Proposition 19.7 (Sensitivity of linear systems)**.** *If $Ax = b$ and $A(x + \delta x) = b + \delta b$, then*

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa(A)\frac{\|\delta b\|}{\|b\|}.$$

Proof. $\delta x = A^{-1}\delta b$ gives $\|\delta x\| \leq \|A^{-1}\|\|\delta b\|$, and $b = Ax$ gives $\|b\| \leq \|A\|\|x\|$. Multiply the two inequalities. □

A condition number of 10^8 means that relative errors of 10^{-16} in the data can become relative errors of 10^{-8} in the solution, and relative measurement noise of 10^{-3} can be amplified to make the solution meaningless. Since $\kappa(X^\top X) = \kappa(X)^2$, forming the normal equations explicitly squares the problem's sensitivity.

Worked example 19.8 (Common-average referencing has a kernel)**.** For C channels, common-average referencing applies $R = I - \frac{1}{C}\mathbf{1}\mathbf{1}^\top$. Then $R\mathbf{1} = 0$, so the constant vector spans $\ker R$; by rank–nullity $\operatorname{rank} R = C - 1$. R is the orthogonal projection onto the subspace of zero-sum vectors. Any signal identical on all channels is annihilated, and the referenced data cannot be "un-referenced" without knowing that common component. Downstream covariance matrices of referenced data are singular, which must be handled before inverting them (e.g. in Mahalanobis distances or beamformers).

Exercises

1. Verify the projection identities for a unit vector u and $P = uu^\top$.
2. Find the kernel of $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$.
3. Derive the normal equations by differentiating the squared residual norm.
4. Explain why rank deficiency compromises parameter uniqueness.
5. Show that the set of $n \times n$ symmetric matrices is a subspace of dimension $n(n+1)/2$.
6. Verify $R^2 = R$ and $R^\top = R$ for the common-average reference matrix.
7. Compute $\kappa(A)$ for $A = \operatorname{diag}(1, 10^{-4})$ and interpret it.
8. Prove that $\ker(X^\top X) = \ker X$, and conclude that $X^\top X$ is invertible iff X has full column rank.
9. Show that the residual of least squares is orthogonal to the fitted values.

Chapter 20

Eigenvalues, singular values, and principal components

Reading goal 20.1. Understand the spectrum of a matrix, its geometry, and the exact optimization solved by PCA.

20.1 Eigenvectors as preserved directions

For a square matrix A, an eigenvector $v \neq 0$ satisfies $Av = \lambda v$. The scalar λ describes the action on that direction. A real symmetric matrix admits an orthonormal basis of real eigenvectors and a factorization $A = Q\Lambda Q^\top$. A covariance matrix is symmetric positive semidefinite, so its eigenvalues are nonnegative.

20.2 Singular value decomposition

Every real matrix $X \in \mathbb{R}^{T\times N}$ admits an SVD $X = U\Sigma V^\top$ with nonnegative singular values in Σ and orthonormal singular vectors. This construction exists even for rectangular or rank-deficient matrices. The right singular vectors describe directions in variable space; the left singular vectors describe directions in observation space. Truncating to the first k components gives the best rank-k approximation in Frobenius norm (Eckart–Young theorem), provided singular values are ordered descending.

20.3 Deriving PCA

Let X_c denote the column-centered observations and $S = X_c^\top X_c/(T-1)$. A projected observation onto a unit vector v has sample variance $v^\top Sv$. The maximum-variance direction solves

$$v_1 = \arg\max_{\|v\|=1} v^\top Sv.$$

Using the Lagrangian $\mathcal{L}(v, \lambda) = v^\top Sv - \lambda(v^\top v - 1)$ gives $Sv = \lambda v$. The maximizing direction is an eigenvector of the largest eigenvalue. Subsequent components maximize variance subject to orthogonality. If $X_c = U\Sigma V^\top$, then $\lambda_j = \sigma_j^2/(T-1)$.

20.4 What PCA does not promise

PCA maximizes sample variance, not class separation, behavioral relevance, temporal predictability, or a causal mechanism. Large variance can come from artifacts, gain fluctuations, or unmodeled experimental factors. Cross-validation, null models, and careful centering are essential. As [21, 10] discuss, structured population dynamics may require objectives different from variance maximization.

Worked example 20.2 (An effective dimension)**.** If $\lambda_1, \dots, \lambda_N$ are nonnegative variances, the participation ratio $d_{\mathrm{PR}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ equals N when all eigenvalues are equal and equals one when only one is positive. It is a continuous summary of spectral concentration, not an integer count of "real" sources.

20.5 The spectral theorem

Theorem 20.3 (Spectral theorem for real symmetric matrices)**.** *If $A \in \mathbb{R}^{n\times n}$ is symmetric, then all eigenvalues of A are real, eigenvectors for distinct eigenvalues are orthogonal, and there is an orthogonal Q with $A = Q\Lambda Q^\top$, Λ diagonal.*

Proof of the first two statements. Let $Av = \lambda v$ with $v \in \mathbb{C}^n$ nonzero. Then $\bar{v}^\top A v = \lambda \|v\|^2$, and taking conjugate transposes (using $A = A^\top$ real) gives $\bar{v}^\top A v = \bar{\lambda}\|v\|^2$. Hence $\lambda = \bar{\lambda}$. If $Au = \lambda u$ and $Av = \mu v$ with $\lambda \neq \mu$, then $\lambda u^\top v = (Au)^\top v = u^\top A v = \mu u^\top v$, so $u^\top v = 0$. The full diagonalization follows by induction on n: the orthogonal complement of an eigenvector is invariant under A. □

Definition 20.4. *A symmetric A is* positive semidefinite *(PSD) if $x^\top A x \geq 0$ for all x, and* positive definite *(PD) if $x^\top A x > 0$ for $x \neq 0$.*

By the spectral theorem, A is PSD iff all eigenvalues are nonnegative, and PD iff all are positive. Every sample covariance $S = X_c^\top X_c/(T-1)$ is PSD because $v^\top S v = \|X_c v\|^2/(T-1) \geq 0$. If $T - 1 < N$ (fewer centered observations than variables), $\operatorname{rank} S \leq T - 1 < N$ and S is singular: at least $N - T + 1$ eigenvalues are exactly zero, regardless of the true covariance.

20.6 The Rayleigh quotient and Courant–Fischer

For symmetric A with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$ and orthonormal eigenvectors q_j,

$$\lambda_1 = \max_{x \neq 0} \frac{x^\top A x}{x^\top x}, \qquad \lambda_n = \min_{x \neq 0} \frac{x^\top A x}{x^\top x}.$$

Proof. Write $x = \sum_j c_j q_j$. Then $x^\top A x / x^\top x = \sum_j \lambda_j c_j^2 / \sum_j c_j^2$, a weighted average of the eigenvalues, which lies between λ_n and λ_1 and attains them at q_n and q_1. □

More generally, the Courant–Fischer theorem states $\lambda_k = \max_{\dim W = k} \min_{x \in W \setminus \{0\}} x^\top A x / x^\top x$. A consequence is *Weyl's inequality*: for symmetric A, E, $|\lambda_k(A + E) - \lambda_k(A)| \leq \|E\|_2$. Eigenvalues are stable under perturbation. Eigenvectors are not: if two eigenvalues are separated by a gap δ, a perturbation of size $\|E\|$ can rotate the corresponding eigenvectors by an angle of order $\|E\|/\delta$ (Davis–Kahan). Principal components with nearly equal variances are therefore poorly determined and should not be interpreted individually.

20.7 Proof of the SVD and its geometry

Theorem 20.5. *Every $X \in \mathbb{R}^{T\times N}$ can be written $X = U\Sigma V^\top$ with $U \in \mathbb{R}^{T\times T}$ and $V \in \mathbb{R}^{N\times N}$ orthogonal and Σ diagonal (rectangular) with entries $\sigma_1 \ge \sigma_2 \ge \cdots \ge 0$.*

Proof. $X^\top X$ is symmetric PSD; diagonalize it as $V\Lambda V^\top$ with $\lambda_1 \ge \cdots \ge \lambda_N \ge 0$. Put $\sigma_j = \sqrt{\lambda_j}$ and, for $\sigma_j > 0$, $u_j = Xv_j/\sigma_j$. These u_j are orthonormal, since $u_i^\top u_j = v_i^\top X^\top X v_j/(\sigma_i\sigma_j) = \lambda_j\delta_{ij}/(\sigma_i\sigma_j) = \delta_{ij}$. Extend them to an orthonormal basis of $\mathbb{R}^T$. Then $Xv_j = \sigma_j u_j$ for every j (both sides vanish when $\sigma_j = 0$ because $\|Xv_j\|^2 = \lambda_j$), which is $XV = U\Sigma$. □

Geometrically, X maps the unit sphere in $\mathbb{R}^N$ to an ellipsoid whose semi-axes have lengths σ_j along directions u_j. The rank is the number of nonzero singular values, $\|X\|_2 = \sigma_1$, $\|X\|_F^2 = \sum_j \sigma_j^2$, and $\kappa(X) = \sigma_1/\sigma_{\min}$.

Theorem 20.6 (Eckart–Young)**.** *Let $X_k = \sum_{j\le k} \sigma_j u_j v_j^\top$. For every matrix Y of rank at most k, $\|X - Y\|_F^2 \ge \|X - X_k\|_F^2 = \sum_{j>k} \sigma_j^2$.*

The theorem gives PCA a second interpretation: the first k components provide the best rank-k reconstruction of the centered data in squared error. The *fraction of variance explained* by k components is $\sum_{j\le k}\sigma_j^2 / \sum_j \sigma_j^2$.

20.8 Non-symmetric matrices and dynamics

Matrices that describe dynamics, such as $x_{t+1} = Ax_t$, are usually not symmetric. Their eigenvalues may be complex, appearing in conjugate pairs for real A, and their eigenvectors need not be orthogonal or even form a basis. A complex pair $re^{\pm i\theta}$ corresponds to rotation by angle θ per step with growth or decay factor r. If $A = V\Lambda V^{-1}$ is diagonalizable then $A^t = V\Lambda^t V^{-1}$ and the solution decays iff all $|\lambda_j| < 1$. When V is badly conditioned ("non-normal" dynamics), $\|A^t\|$ can grow transiently by a large factor before decaying, even though every eigenvalue is inside the unit circle. Transient amplification of this kind has been proposed for motor cortex and is invisible in the eigenvalues alone.

Worked example 20.7 (Spurious structure in pure noise)**.** Let X be $T \times N$ with independent standard normal entries, $T = 200$, $N = 100$. The true covariance is I, so every direction has variance 1. Yet the eigenvalues of $S = X^\top X/T$ spread approximately over $[(1 - \sqrt{N/T})^2, (1 + \sqrt{N/T})^2] \approx [0.09, 2.91]$ (the Marchenko–Pastur law). The leading "component" explains about 2.9 times the average variance, purely by sampling. Comparing an empirical spectrum against such a noise baseline, or against shuffled data, is required before calling a component a population signal.

Exercises

1. Derive $\lambda_j = \sigma_j^2/(T-1)$ for centered observations.
2. Compare participation ratios for $(10, 0, 0)$ and $(4, 3, 3)$.
3. Explain why replacing trials by a trial average can remove single-trial fluctuations.
4. Give a case where the leading principal component tracks recording artifact rather than neural activity.
5. Show that a symmetric matrix with all eigenvalues positive is positive definite.
6. Compute the SVD of $\begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$ via the eigen-decomposition of $X^\top X$.

7. Prove $\|X\|_F^2 = \sum_j \sigma_j^2$ using $\|X\|_F^2 = \operatorname{tr}(X^\top X)$.

8. Find the eigenvalues of $\begin{pmatrix} 0.9 & -0.3 \\ 0.3 & 0.9 \end{pmatrix}$ and describe the trajectory of $x_{t+1} = Ax_t$.

9. With $T = 50$ and $N = 200$, how many eigenvalues of the sample covariance are exactly zero?

Chapter 21

Tensors, reduced-rank models, and inverse problems

Reading goal 21.1. Move from matrix factorization to multivariate measurements, and learn how identifiable models differ from underdetermined ones.

21.1 More than two indices

An experiment with neurons n, time t, and trial r may be stored as a third-order tensor $\mathcal{X}_{ntr}$. Its modes represent different axes of variation. A CANDECOMP/PARAFAC (CP) approximation is

$$\mathcal{X}_{ntr} \approx \sum_{k=1}^{K} a_{nk} b_{tk} c_{rk}.$$

Interpretation requires confronting scaling and permutation ambiguities: multiplying one factor by α and dividing another by α leaves the product unchanged. Additional uniqueness conditions depend on the factors and rank. The mathematical pattern is broader than one physiological explanation.

21.2 Reduced-rank regression

For predictor matrix $X \in \mathbb{R}^{T \times p}$ and response matrix $Y \in \mathbb{R}^{T \times q}$, ordinary multivariate regression fits $Y \approx XB$. Reduced-rank regression requires $\operatorname{rank}(B) \leq r$, controlling the dimension of predictive communication. Calling the resulting dimensions "communication subspaces" is an interpretation built on the predictive model, not proof that specific axons mediate the influence.

21.3 The forward model is not the inverse model

Let measurements obey $y = Hx + \varepsilon$, where x is a source, H a measurement operator, and ε noise. The *forward problem* computes y from a chosen source and measurement physics. The *inverse problem* estimates x from y. If H has a nontrivial kernel, x and $x + z$ generate the same noiseless measurement whenever $Hz = 0$. This is an identifiability failure, not a limitation remedied merely by better numerical precision.

For extracellular electric potentials, source geometry and volume conduction affect H, and several current patterns may explain the same electrode measurements. Biophysical treatments emphasize the interpretation of forward models and the assumptions behind inverse methods

[8, 16]. Regularization can choose one plausible solution, but it cannot magically restore missing independent information.

21.4 Regularized reconstruction

One possible estimator is $\hat{x} = \arg\min_x \{\|y - Hx\|_2^2 + \lambda \|Lx\|_2^2\}$. If the normal equations yield an invertible matrix, $\hat{x} = (H^\top H + \lambda L^\top L)^{-1} H^\top y$. The operator L expresses a prior preference (e.g., smoothness), while λ sets its influence. The result answers a conditional question: which sources fit the observations well while obeying the chosen regularity assumption?

21.5 The Moore–Penrose pseudoinverse

Let $X = U\Sigma V^\top$ with r nonzero singular values. The *pseudoinverse* is $X^+ = V\Sigma^+ U^\top$, where Σ^+ inverts the nonzero singular values and transposes the shape.

Theorem 21.2. *For any y, the vector $\hat{\beta} = X^+ y$ is the minimum-norm solution among all minimizers of $\|y - X\beta\|_2$.*

Proof. In the coordinates $\gamma = V^\top \beta$ and $c = U^\top y$, orthogonality gives $\|y - X\beta\|^2 = \sum_{j \le r} (c_j - \sigma_j \gamma_j)^2 + \sum_{j > r} c_j^2$. The minimizers are exactly the γ with $\gamma_j = c_j / \sigma_j$ for $j \le r$ and arbitrary γ_j for $j > r$. Since $\|\beta\| = \|\gamma\|$, the smallest-norm choice sets $\gamma_j = 0$ for $j > r$, which is $\hat{\beta} = V\Sigma^+ U^\top y$. □

The theorem identifies the arbitrary choice made by a pseudoinverse: components of β in $\ker X$ are invisible to the data and are set to zero. In source localization, this "minimum-norm estimate" favors superficial sources, because deep sources need larger currents to produce the same scalp or electrode field; depth weighting changes the norm and therefore the answer.

21.6 Ill-posedness and the Picard condition

Expand the noiseless data $y = Xx$ in the singular basis: $c_j = u_j^\top y = \sigma_j (v_j^\top x)$. The pseudoinverse estimate is

$$\hat{x} = \sum_{j \le r} \frac{u_j^\top y}{\sigma_j} v_j.$$

With noise $y = Xx + \varepsilon$, the term for component j contains $u_j^\top \varepsilon / \sigma_j$. When singular values decay towards zero, as they do for smoothing operators such as volume conduction, the noise is divided by tiny numbers and dominates. A problem is *well posed* in Hadamard's sense if a solution exists, is unique, and depends continuously on the data; inverse problems in electrophysiology typically fail the second or third condition. The discrete *Picard condition* asks that $|u_j^\top y|$ decay faster than σ_j, so that the true signal is recoverable at least in the leading components.

21.7 Tikhonov regularization through the SVD

With $L = I$, the Tikhonov estimate $\hat{x}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ becomes

$$\hat{x}_\lambda = \sum_j \frac{\sigma_j^2}{\sigma_j^2 + \lambda} \frac{u_j^\top y}{\sigma_j} v_j.$$

The *filter factors* $\phi_j = \sigma_j^2/(\sigma_j^2+\lambda)$ are near 1 for $\sigma_j^2 \gg \lambda$ and near 0 for $\sigma_j^2 \ll \lambda$. Regularization is therefore a soft truncation of poorly determined components. *Truncated SVD* uses $\phi_j \in \{0,1\}$ instead. Choosing λ (by cross-validation, generalized cross-validation, the L-curve, or a noise-level-based discrepancy principle) is a modeling decision with consequences for spatial resolution.

Proposition 21.3 (Bias of ridge regression)**.** *If $y = X\beta + \varepsilon$ with $\mathbb{E}\varepsilon = 0$, then $\mathbb{E}\hat\beta_\lambda = \sum_j \phi_j (v_j^\top \beta) v_j$, so the ridge estimator is biased towards zero in every direction with $\lambda > 0$, most strongly along small singular values.*

Proof. Substitute $u_j^\top y = \sigma_j v_j^\top \beta + u_j^\top \varepsilon$ into the filter-factor formula and take expectations. $\square$

21.8 Tensor rank and uniqueness

For a third-order tensor, the *CP rank* is the smallest K for which an exact decomposition $\sum_{k\le K} a_k \circ b_k \circ c_k$ exists. Unlike matrix rank, tensor rank is hard to compute, can depend on whether the field is $\mathbb{R}$ or $\mathbb{C}$, and the set of tensors of rank at most K need not be closed, so a "best rank-K approximation" may fail to exist (factor norms diverge while the fit improves). In exchange, CP decompositions are often unique up to scaling and permutation, which matrix factorizations never are.

Theorem 21.4 (Kruskal, stated)**.** *Let k_A denote the largest number k such that every k columns of A are linearly independent (the Kruskal rank), and similarly k_B, k_C. If $k_A + k_B + k_C \ge 2K + 2$, the rank-$K$ CP decomposition is unique up to permutation and scaling of the components.*

Uniqueness is the mathematical reason tensor methods are popular for neuron $\times$ time $\times$ trial data: a matrix factorization $X \approx AB^\top$ can be replaced by $(AM)(BM^{-\top})^\top$ for any invertible M, so its factors are not individually interpretable without further constraints, whereas a CP decomposition satisfying Kruskal's condition is.

Worked example 21.5 (A two-electrode inverse problem)**.** Two electrodes record $y = Hx$ from three candidate sources with

$$H = \begin{pmatrix} 1 & 0.5 & 0.2 \\ 0.2 & 0.5 & 1 \end{pmatrix}.$$

The rows are independent, so $\operatorname{rank} H = 2$ and, by rank–nullity, $\ker H$ is one-dimensional. A spanning vector is the cross product of the two rows, $(0.4, -0.96, 0.4)^\top$, i.e. $z \propto (5, -12, 5)^\top$; indeed $Hz = (5 - 6 + 1,\ 1 - 6 + 5)^\top = 0$. Any source pattern can be altered by a multiple of z without changing the recordings: a strong middle source can be traded for two outer sources of opposite sign. A minimum-norm estimate removes the z component; a sparsity prior may instead drive one source to zero. The data alone cannot decide between them.

Exercises

1. Exhibit two distinct $x \in \mathbb{R}^2$ satisfying $[1\ 1]x = 2$.
2. Explain why a CP decomposition's factors may be arbitrarily rescaled.
3. Describe what you would need to claim that a reduced-rank relationship reflects direct anatomical connectivity.
4. Show that $XX^+X = X$ and $X^+XX^+ = X^+$.
5. Show that the filter factors ϕ_j are decreasing in λ and compute λ for which $\phi_j = 1/2$.
6. Verify that $(a\alpha)\circ(b/\alpha)\circ c = a \circ b \circ c$ for any $\alpha \neq 0$.
7. For $X = \operatorname{diag}(1, 10^{-3})$ and noise of size 10^{-2} in each coordinate, compare the errors of the pseudoinverse and of truncated SVD with one component.

Chapter 22

Bridges from vectors to neural population geometry

Reading goal 22.1. Derive basic linear-algebra tools from definitions and see exactly how they lead to PCA, projections, and regression.

Bridge 27: Linear independence makes coordinate representations unique

Proposition 22.2. *If $v_1, \dots, v_k$ are linearly independent vectors, every vector in their span has exactly one representation $x = \sum_{j=1}^{k} a_j v_j$.*

Proof. Existence follows from the definition of span. If $x = \sum a_j v_j = \sum b_j v_j$, subtracting gives $\sum (a_j - b_j) v_j = 0$. Linear independence forces $a_j - b_j = 0$ for each j, so coefficients are unique. □

Example. A latent subspace can have unique coordinates after fixing a basis, but another rotated basis describes the same subspace. "Component 1" need not be a unique biological source.

Bridge 28: Dimensions split into visible and invisible directions

Theorem 22.3 (Rank–nullity)**.** *For a linear map $A : V \to W$ with $\dim V = n < \infty$, $\dim \ker A + \dim \operatorname{im} A = n$.*

Proof. Choose a basis $u_1, \dots, u_r$ of $\ker A$ and extend it to a basis of V with $v_1, \dots, v_{n-r}$. Their images $Av_1, \dots, Av_{n-r}$ span $\operatorname{im} A$. If a linear combination of these images vanishes, the corresponding combination of v_j lies in the kernel and must have zero coefficients by independence of the full basis. Thus the images form a basis of dimension $n - r$. □

Example. A lead-field map with nonzero nullspace cannot uniquely recover all neural current distributions from extracellular electrodes.

Bridge 29: Cauchy–Schwarz bounds the inner product

Theorem 22.4 (Cauchy–Schwarz)**.** *For a real inner-product space, $|\langle x, y \rangle| \le \|x\| \|y\|$.*

Proof. For $y \neq 0$, nonnegativity of $\|x - \alpha y\|^2$ with $\alpha = \langle x, y\rangle / \|y\|^2$ gives $\|x\|^2 - \langle x, y\rangle^2 / \|y\|^2 \geq 0$. Rearranging and taking square roots proves the claim. If $y = 0$ both sides vanish. □

Example. Applying this inequality to centered random variables implies $|\rho| \leq 1$ when their variances are positive.

Bridge 30: The triangle inequality follows from Cauchy–Schwarz

Proposition 22.5. *For the norm $\|x\| = \sqrt{\langle x, x\rangle}$, $\|x + y\| \leq \|x\| + \|y\|$.*

Proof. Expand $\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2$. The previous bridge bounds the middle term by $2\|x\|\|y\|$, making the sum no larger than $(\|x\| + \|y\|)^2$. Nonnegative square roots preserve the inequality. □

Example. Reconstruction error from two perturbations is at most the sum of their norm bounds; that inequality says nothing about whether the model is physiologically meaningful.

Bridge 31: Orthogonal components add energy, not amplitude

Proposition 22.6. *If $\langle x, y\rangle = 0$, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.*

Proof. The general squared-norm expansion has the cross term $2\langle x, y\rangle$. Orthogonality makes this term zero. □

Example. Orthogonal Fourier components add squared energy over a complete ideal window. Raw amplitudes are not generally additive as magnitudes.

Bridge 32: Independent vectors can be orthogonalized

Theorem 22.7 (Gram–Schmidt procedure)**.** *Every finite linearly independent family $v_1, \ldots, v_k$ in an inner-product space spans an orthonormal family $q_1, \ldots, q_k$ with matching partial spans.*

Proof. Set $u_1 = v_1$ and $q_1 = u_1 / \|u_1\|$. Recursively let $u_j = v_j - \sum_{i<j} \langle v_j, q_i\rangle q_i$. This vector is perpendicular to preceding q_i by cancellation of inner products, and it cannot be zero without making v_j a combination of preceding vectors. Normalize $q_j = u_j / \|u_j\|$ and proceed inductively. □

Example. Orthonormal bases permit stable projections, but finite-precision implementations often prefer Householder QR over classical Gram–Schmidt.

Bridge 33: Projection solves constrained least squares

Proposition 22.8. *If columns of $Q \in \mathbb{R}^{n \times k}$ are orthonormal, the closest vector to y in $\operatorname{im} Q$ is $QQ^\top y$.*

Proof. Write $p = QQ^\top y$ and $r = y - p$. Then $Q^\top r = Q^\top y - Q^\top QQ^\top y = 0$. For any $z \in \operatorname{im} Q$, $y - z = r + (p - z)$ is a sum of orthogonal vectors, so $\|y - z\|^2 = \|r\|^2 + \|p - z\|^2 \geq \|r\|^2$. The minimum occurs at $z = p$. □

Example. Linear neural decoding fits a projection onto a predictor subspace. Validation must still separate training and held-out trials.

Bridge 34: Covariance matrices are positive semidefinite

Proposition 22.9. *For a random vector $X \in \mathbb{R}^n$ with finite second moments, its covariance matrix C is symmetric positive semidefinite: $v^\top C v \geq 0$ for all v.*

Proof. Symmetry follows from $\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$. Expanding the variance of the scalar $v^\top X$ yields $v^\top C v = \mathrm{Var}(v^\top X) \geq 0$. □

Example. PCA uses the eigendecomposition of C and nonnegative eigenvalues. A numerical covariance estimate with materially negative eigenvalues calls for an implementation or modeling audit.

Exercises

1. Find a basis and nullspace for $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$.
2. Derive the ordinary least-squares normal equations by orthogonality.
3. Explain why PCA axes can change substantially when leading eigenvalues are nearly equal.

Part V

Probability, statistics, and time

Chapter 23

Probability begins with sets of possible outcomes

Reading goal 23.1. Translate uncertainty into an explicit sample space, events, conditional probabilities, and independence assumptions.

23.1 Experiments and events

A random experiment has possible outcomes in a *sample space* Ω. An *event* is a subset $A \subseteq \Omega$. A probability measure P assigns numbers to events with $P(A) \geq 0$, $P(\Omega) = 1$, and countable additivity across disjoint events. This connects the set operations learned in Chapter 4 to the calculus of uncertainty. A model of stochastic spike counts has a different sample space from a model of electrode voltage traces; neither should be silently substituted for the other.

23.2 Conditional probability

When $P(B) > 0$, define $P(A \mid B) = P(A \cap B)/P(B)$. Rearranging gives the multiplication rule $P(A \cap B) = P(A \mid B)P(B)$. If A and B are independent, $P(A \cap B) = P(A)P(B)$. Independence is a model property, not merely a statement that the observed sample correlation vanishes.

The law of total probability for a partition (B_j) is $P(A) = \sum_j P(A \mid B_j)P(B_j)$. Bayes' theorem follows:

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_k P(A \mid B_k)P(B_k)}.$$

In Bayesian analysis, a prior over model parameters and a likelihood for observed data yield a posterior. The posterior is conditional on both the likelihood and the prior; a credible interval and a frequentist confidence interval do not have interchangeable interpretations.

23.3 Conditioning changes the question

Two neurons can appear correlated because each varies with movement speed. Conditioning on behavior may reduce this apparent association. But conditioning can also introduce new dependencies, as in collider bias. A scientific claim must specify what was conditioned on, why, and whether the variables were measured accurately. The conditional modeling framework for spike jitter makes this especially explicit [1].

Worked example 23.2 (A base-rate effect)**.** Suppose only 1 percent of candidate events are genuine. A detector identifies 90 percent of genuine events and falsely flags 5 percent of nongenuine

events. The probability that a flagged event is genuine is $0.90(0.01)/[0.90(0.01)+0.05(0.99)] \approx 0.154$. High sensitivity does not by itself imply high positive predictive value.

23.4 Consequences of the axioms

Every rule of elementary probability follows from three axioms: $P(A) \geq 0$, $P(\Omega) = 1$, and countable additivity, $P(\bigcup_j A_j) = \sum_j P(A_j)$ for pairwise disjoint events.

Proposition 23.3. *For events A, B: (i) $P(\varnothing) = 0$; (ii) $P(A^c) = 1 - P(A)$; (iii) if $A \subseteq B$ then $P(A) \leq P(B)$ and $P(B \setminus A) = P(B) - P(A)$; (iv) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; (v) $P(\bigcup_j A_j) \leq \sum_j P(A_j)$ for any countable family (the* union bound*).*

Proof. (i) Apply additivity to $\Omega, \varnothing, \varnothing, \ldots$: $1 = 1 + \sum P(\varnothing)$. (ii) $\Omega = A \cup A^c$ disjointly. (iii) $B = A \cup (B \setminus A)$ disjointly and $P(B \setminus A) \geq 0$. (iv) $A \cup B = A \cup (B \setminus A)$ disjointly and $P(B \setminus A) = P(B) - P(A \cap B)$ by (iii). (v) Set $B_1 = A_1$, $B_j = A_j \setminus \bigcup_{i<j} A_i$; the B_j are disjoint with the same union and $B_j \subseteq A_j$. □

The union bound is the mathematical basis of the Bonferroni correction: if each of m tests has false-positive probability at most α/m under its null, the probability of at least one false positive is at most α, *whatever* the dependence between tests. Its generality is the source of its conservatism.

23.5 Continuity of probability and sigma-algebras

Proposition 23.4 (Continuity from below and above)**.** *If $A_1 \subseteq A_2 \subseteq \cdots$ then $P(\bigcup_n A_n) = \lim_n P(A_n)$. If $A_1 \supseteq A_2 \supseteq \cdots$ then $P(\bigcap_n A_n) = \lim_n P(A_n)$.*

Proof. For increasing events, write the union as the disjoint union of $A_1, A_2 \setminus A_1, A_3 \setminus A_2, \ldots$; countable additivity gives $P(\bigcup A_n) = \lim_N \sum_{n \leq N} P(A_n \setminus A_{n-1}) = \lim_N P(A_N)$. The decreasing case follows by complements. □

On uncountable sample spaces, such as the set of all voltage traces, one cannot in general assign probabilities to *every* subset consistently. Probabilities are assigned to a *σ-algebra* $\mathcal{F}$: a collection of subsets containing Ω and closed under complements and countable unions. The triple $(\Omega, \mathcal{F}, P)$ is a *probability space*. The Borel σ-algebra on $\mathbb{R}$, generated by intervals, contains every set one meets in practice. The restriction is technical but explains the word "measurable" attached to random variables.

23.6 Independence of several events

Events $A_1, \ldots, A_n$ are *mutually independent* if $P(\bigcap_{i \in J} A_i) = \prod_{i \in J} P(A_i)$ for every subset J of indices. Pairwise independence is weaker.

Worked example 23.5 (Pairwise but not mutually independent)**.** Toss two fair coins. Let A = "first is heads", B = "second is heads", C = "the coins agree". Each has probability $1/2$, and each pair intersects in a single outcome of probability $1/4$, so the events are pairwise independent. But $A \cap B \subseteq C$, so $P(A \cap B \cap C) = 1/4 \neq 1/8$. In neural terms: three binary units can be pairwise uncorrelated while their joint activity is fully constrained, which is why pairwise statistics cannot certify the absence of higher-order structure.

23.7 Conditional probability as a new probability measure

For fixed B with $P(B) > 0$, the function $A \mapsto P(A \mid B)$ satisfies all three axioms; it is a probability measure on the same σ-algebra, concentrated on B. Every theorem of probability therefore holds conditionally. In particular, *conditional independence* of A and C given B means $P(A \cap C \mid B) = P(A \mid B)P(C \mid B)$, and it neither implies nor is implied by unconditional independence.

Proposition 23.6 (Chain rule)**.** *If $P(A_1 \cap \cdots \cap A_{n-1}) > 0$, then*

$$P(A_1 \cap \cdots \cap A_n) = P(A_1)\, P(A_2 \mid A_1) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

The proof is induction on the multiplication rule. The chain rule is the probability of a sequence written as a product of conditional probabilities of each step given the past, the form taken by every likelihood for time series and spike trains (Chapter 34).

Worked example 23.7 (Conditioning creates dependence)**.** Two neurons fire independently with probability 0.1 each in a bin. Suppose a recording is retained only if at least one of them fires (a collider). Given retention, $P(\text{both} \mid \text{retained}) = 0.01/0.19 \approx 0.053$, while $P(\text{first} \mid \text{retained}) = P(\text{second} \mid \text{retained}) = 0.1/0.19 \approx 0.526$, whose product is 0.277. The neurons are now strongly *negatively* dependent: knowing one did not fire guarantees that the other did. Selection on an outcome manufactures structure.

Exercises

1. Derive $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
2. Give independent events and dependent events from a six-sided die experiment.
3. Explain why a detector's posterior event probability depends on prevalence.
4. Prove the union bound for finitely many events by induction from (iv).
5. With $m = 190$ pairwise tests, what per-test threshold does Bonferroni give for $\alpha = 0.05$?
6. Show that if A and B are independent then so are A and B^c.
7. Verify the numbers in the collider example and compute the conditional correlation of the two indicator variables.
8. Show that $P(\cdot \mid B)$ satisfies countable additivity.

Chapter 24

Random variables, expectation, and statistical moments

Reading goal 24.1. Define distributions and calculate moments before discussing firing-rate variability or moving-window envelope statistics.

24.1 A random variable is a function

A random variable $X : \Omega \to \mathbb{R}$ assigns a numerical value to each possible outcome, subject to a measurability requirement that ensures events such as $\{\omega : X(\omega) \leq x\}$ have defined probabilities. Its cumulative distribution function is $F_X(x) = P(X \leq x)$. Discrete random variables have probability mass functions; sufficiently regular continuous variables have probability densities. A density is not itself a probability; probabilities arise through integration over regions.

24.2 Means, variance, skewness, kurtosis

The mean is $\mu = E[X]$; variance is $\sigma^2 = E[(X - \mu)^2]$; standardized skewness is $\gamma_1 = E[(X - \mu)^3]/\sigma^3$; and standardized kurtosis is $\gamma_2 = E[(X - \mu)^4]/\sigma^4$. Some authors report excess kurtosis $\gamma_2 - 3$, which equals zero for a Gaussian distribution. If $\sigma = 0$, standardized higher moments are undefined. Higher moments are sensitive to rare large amplitudes and may require large samples for stable estimation.

For a time-indexed envelope $A(t)$, a sliding-window empirical mean is $\hat{\mu}(t) = \frac{1}{M} \sum_{j=0}^{M-1} A(t + j\Delta t)$ with analogous central moments. If consecutive windows overlap heavily, the resulting statistics are strongly dependent: sampling an estimate every 100 ms does not create a new independent 10-s experiment at every step.

24.3 Covariance and correlation

Covariance is $\mathrm{Cov}(X, Y) = E[(X - EX)(Y - EY)]$. Its scale depends on units, while Pearson correlation normalizes by standard deviations: $\rho = \mathrm{Cov}(X, Y)/(\sigma_X \sigma_Y)$ for nonzero variances. A zero correlation does not guarantee independence in general. Let X be symmetric around zero and $Y = X^2$; then $\mathrm{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0$, although Y is determined by X.

24.4 The law of total variance

For random variables X, Z with finite variance,

$$\operatorname{Var}(X) = E[\operatorname{Var}(X \mid Z)] + \operatorname{Var}(E[X \mid Z]).$$

The first term measures variation remaining after conditioning; the second measures variation in conditional means. This identity is an essential bridge to decomposing neuronal trial-to-trial variability into point-process and gain contributions [20].

Worked example 24.2 (Poisson versus overdispersed counts)**.** For $N \sim \text{Poisson}(\lambda)$, $E[N] = \operatorname{Var}(N) = \lambda$. If $N \mid \Lambda \sim \text{Poisson}(\Lambda)$ and Λ varies, then $\operatorname{Var}(N) = E[\Lambda] + \operatorname{Var}(\Lambda)$. Extra variance can arise from latent fluctuations even if spikes are conditionally Poisson.

24.5 Expectation as an integral and its linearity

For a discrete random variable, $\mathbb{E}[X] = \sum_x x\, P(X = x)$; for one with density f, $\mathbb{E}[X] = \int x f(x)\, dx$, provided the sum or integral converges absolutely. In general $\mathbb{E}[X] = \int_\Omega X\, dP$ is the Lebesgue integral with respect to the probability measure. The most used property is linearity, $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$, which holds *without any independence assumption.*

Worked example 24.3 (Expected coincidences by linearity)**.** Two spike trains are binned into n bins with firing probabilities p and q per bin, and coincidences are counted as $C = \sum_{k=1}^{n} X_k Y_k$. If the trains are independent in each bin, $\mathbb{E}[X_k Y_k] = pq$, so $\mathbb{E}[C] = npq$ regardless of any dependence *across* bins. The variance of C, by contrast, does depend on the temporal dependence structure, so "expected coincidences" is easy and "significance of coincidences" is not.

Proposition 24.4 (Law of the unconscious statistician)**.** *For a function g, $\mathbb{E}[g(X)] = \sum_x g(x) P(X = x)$ or $\int g(x) f(x)\, dx$, without computing the distribution of $g(X)$.*

In particular $\mathbb{E}[X^2]$ and $\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$ can be computed directly from the distribution of X.

24.6 Fundamental inequalities

Theorem 24.5 (Markov and Chebyshev)**.** *If $X \geq 0$ and $a > 0$, then $P(X \geq a) \leq \mathbb{E}[X]/a$. For any X with finite variance and $k > 0$, $P(|X - \mathbb{E}X| \geq k) \leq \operatorname{Var}(X)/k^2$.*

Proof. $a\mathbf{1}_{\{X \geq a\}} \leq X$ pointwise; take expectations. Chebyshev is Markov applied to $(X - \mathbb{E}X)^2$ and $a = k^2$. □

Theorem 24.6 (Jensen)**.** *If φ is convex and X, $\varphi(X)$ are integrable, then $\varphi(\mathbb{E}X) \leq \mathbb{E}[\varphi(X)]$.*

Proof. By convexity there is a supporting line at $m = \mathbb{E}X$: $\varphi(x) \geq \varphi(m) + c(x - m)$ for all x. Substitute X and take expectations; the linear term has mean zero. □

Jensen's inequality shows that $\mathbb{E}[X^2] \geq (\mathbb{E}X)^2$, that the mean of log-power is at most the log of mean power (so averaging in decibels and in linear units give different answers), and that the KL divergence is nonnegative.

24.7 Common distributions

Distribution	Support	Mean	Variance
Bernoulli(p)	$\{0,1\}$	p	$p(1-p)$
Binomial(n,p)	$\{0,\dots,n\}$	np	$np(1-p)$
Poisson(λ)	$\mathbb{N}_0$	λ	λ
Geometric(p) (trials to success)	$\mathbb{N}_+$	$1/p$	$(1-p)/p^2$
Uniform(a,b)	$[a,b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	$[0,\infty)$	$1/\lambda$	$1/\lambda^2$
Gamma(k,λ)	$[0,\infty)$	k/λ	k/λ^2
Normal(μ,σ^2)	$\mathbb{R}$	μ	σ^2

These are connected. A binomial count with $n\to\infty$, $p\to 0$ and $np\to\lambda$ converges to Poisson(λ), which is why counts of rare events in many small bins are approximately Poisson. Waiting times between events of a homogeneous Poisson process are exponential, and the waiting time to the kth event is gamma. The exponential distribution is the only continuous distribution on $[0,\infty)$ with the *memoryless* property $P(X>s+t\mid X>s)=P(X>t)$, which is the property a refractory neuron violates.

24.8 Independence of random variables and sums

Random variables X and Y are independent if $P(X\in A, Y\in B)=P(X\in A)P(Y\in B)$ for all Borel sets A,B. Then $\mathbb{E}[g(X)h(Y)]=\mathbb{E}[g(X)]\mathbb{E}[h(Y)]$ for integrable g,h, and in particular $\operatorname{Cov}(X,Y)=0$. For any random variables,

$$\operatorname{Var}\Big(\sum_i X_i\Big)=\sum_i \operatorname{Var}(X_i)+2\sum_{i<j}\operatorname{Cov}(X_i,X_j).$$

For uncorrelated summands the variances add. For a population of N neurons with equal variances σ^2 and equal pairwise correlation ρ, the variance of the summed activity is $N\sigma^2[1+(N-1)\rho]$. Even a small $\rho=0.05$ dominates when $N=100$: the summed variance is about six times the independent value. Weak pairwise correlations have large population consequences.

24.9 Moment generating functions

The moment generating function $M_X(s)=\mathbb{E}[e^{sX}]$, when finite near $s=0$, determines the distribution and generates moments by $M_X^{(k)}(0)=\mathbb{E}[X^k]$. For independent X,Y, $M_{X+Y}=M_XM_Y$. For Poisson(λ), $M(s)=\exp(\lambda(e^s-1))$, so the sum of independent Poisson variables with means λ_1,λ_2 is Poisson($\lambda_1+\lambda_2$): pooling spike counts from independent Poisson neurons gives a Poisson count.

Exercises

1. Calculate the population mean, variance, skewness, and kurtosis of the discrete values $0,1,2$ under equal probabilities.
2. Prove $\operatorname{Var}(aX+b)=a^2\operatorname{Var}(X)$.
3. Explain the difference between window overlap and an actual temporal correlation in the source.
4. Derive the overdispersion identity from total variance.

5. Use Chebyshev's inequality to bound the probability that a Poisson(100) count deviates from 100 by at least 30.
6. Show that the exponential distribution is memoryless.
7. Show by Jensen that $\mathbb{E}[\log P] \le \log \mathbb{E}[P]$ for a positive random power P, and give a numerical example.
8. Compute the variance of a population sum with $N = 50$, $\sigma^2 = 4$, $\rho = 0.1$.
9. Derive the Poisson moment generating function and use it to compute the variance.

Chapter 25

Inference, likelihood, and stochastic processes

Reading goal 25.1. Separate population parameters from estimators, and connect time-dependent random variables to autocorrelation and stationarity.

25.1 Data, models, estimators

An unknown parameter θ describes a proposed data-generating distribution $p(x \mid \theta)$. An *estimator* $\hat{\theta}(X_1, \ldots, X_n)$ is a function of random data; its realized value is an estimate. An estimator can be biased, have high variance, or be inconsistent. Mean squared error decomposes as $\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + [E\hat{\theta} - \theta]^2$ when these quantities exist. Estimation is therefore not reduced to maximizing an observed fit score.

The likelihood $L(\theta; x) = p(x \mid \theta)$ views the observed data as fixed and the parameter as variable. For independent Poisson counts n_i with intensities $\lambda_i(\theta)$, the log-likelihood is

$$\ell(\theta) = \sum_i \{n_i \log \lambda_i(\theta) - \lambda_i(\theta) - \log(n_i!)\}.$$

The assumption of independent counts must be justified or modified. A large likelihood does not prove the model is correct, and a small p-value does not give the probability that the null hypothesis is true.

25.2 Time indexed random variables

A stochastic process is a family $(X_t)_{t \in T}$ of random variables on a common probability space. For weak stationarity, the mean is constant and the covariance between X_t and X_s depends only on their lag $t - s$ (assuming finite second moments). Define autocovariance $C_X(\tau) = \mathrm{Cov}(X_t, X_{t+\tau})$. Its Fourier transform, under appropriate conditions, gives a power spectral density. Stationarity is a property of the model, not something one can prove from one short recording.

25.3 Conditional structure and hidden state

A state-dependent process may have time-varying conditional mean or variance despite an apparently stable recording. A latent state Z_t might represent arousal, behavior, or an abstract computational state. The distinction between $p(X_t \mid Z_t)$ and the marginal $p(X_t)$ explains why pooling states can create spurious correlations and overdispersion. This is a key insight in interpreting correlations [11, 20].

Worked example 25.2 (A stochastic oscillator is not just a sine wave)**.** A pure sinusoid has a well-defined frequency and deterministic phase evolution. A noisy oscillatory process may have a broad spectral peak and time-varying phase. Conversely, a stochastic process with $1/f$-like spectrum can place substantial power within the theta band without containing an independent theta oscillator. The distinction motivates explicit spectral modeling [15, 19].

25.4 Laws of large numbers and the central limit theorem

Theorem 25.3 (Weak law of large numbers)**.** *If $X_1, X_2, \dots$ are uncorrelated with common mean μ and variance $\sigma^2 < \infty$, then for every $\varepsilon > 0$, $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \to 0$.*

Proof. $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$ by the variance-of-sums formula; apply Chebyshev. □

Theorem 25.4 (Central limit theorem, iid case)**.** *If $X_1, X_2, \dots$ are iid with mean μ and finite variance $\sigma^2 > 0$, then $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ converges in distribution to $\mathcal{N}(0, 1)$.*

The proof uses characteristic functions and is given in standard texts. Its hypotheses matter: with dependence, the variance of $\bar{X}_n$ is $\frac{\sigma^2}{n}\big[1 + 2\sum_{k=1}^{n-1}(1 - k/n)\rho_k\big]$, where ρ_k is the lag-k autocorrelation. For positively autocorrelated data the naive standard error $\sigma/\sqrt{n}$ is too small, sometimes by a large factor. The *effective sample size* $n_{\mathrm{eff}} = n/[1 + 2\sum_k \rho_k]$ (for long series) quantifies how many independent observations the data are worth. For an AR(1) process with $\rho_k = \phi^k$, $n_{\mathrm{eff}} \approx n(1 - \phi)/(1 + \phi)$: with $\phi = 0.9$, $n_{\mathrm{eff}} \approx n/19$.

25.5 Properties of estimators

An estimator $\hat{\theta}_n$ is *unbiased* if $\mathbb{E}\hat{\theta}_n = \theta$ and *consistent* if $\hat{\theta}_n \to \theta$ in probability. The sample variance with divisor $n - 1$, $s^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$, is unbiased for iid data.

Proposition 25.5. *For iid X_i with variance σ^2, $\mathbb{E}[\sum_i (X_i - \bar{X})^2] = (n - 1)\sigma^2$.*

Proof. $\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2$. Taking expectations gives $n\sigma^2 - n \cdot \sigma^2/n$. □

Unbiasedness is not sacred. The divisor n gives a biased but lower-variance estimator, and the plug-in estimators of entropy or mutual information are biased in ways that depend strongly on sample size, which can create apparent condition differences when conditions have different numbers of trials.

25.6 Maximum likelihood

The *maximum-likelihood estimator* (MLE) maximizes $\ell(\theta) = \log L(\theta; x)$. For n iid Poisson counts with common mean λ, $\ell(\lambda) = \sum_i n_i \log\lambda - n\lambda + \text{const}$, so $\ell'(\lambda) = \sum_i n_i/\lambda - n = 0$ gives $\hat{\lambda} = \bar{n}$. For counts n_i in windows of length T_i with rate r, $\hat{r} = \sum n_i / \sum T_i$, the pooled rate.

Under regularity conditions, the MLE is consistent and asymptotically normal with variance given by the inverse *Fisher information* $I(\theta) = -\mathbb{E}[\ell''(\theta)]$. For the Poisson example, $I(\lambda) = n/\lambda$, so $\mathrm{se}(\hat{\lambda}) \approx \sqrt{\hat{\lambda}/n}$. The *Cramér–Rao bound* states that, under regularity, every unbiased estimator has variance at least $1/I(\theta)$. These asymptotic results presuppose that the model is correct; under misspecification the MLE converges to the parameter of the model closest to the truth in Kullback–Leibler divergence, and the correct variance is the "sandwich" $I^{-1}JI^{-1}$ with J the variance of the score.

25.7 Stationary processes: examples

- *White noise*: uncorrelated, mean zero, variance σ^2; autocovariance $\sigma^2 \mathbf{1}_{\tau=0}$; flat spectrum.
- *AR(1)*: $X_t = \phi X_{t-1} + \varepsilon_t$ with $|\phi| < 1$. Iterating gives $X_t = \sum_{k \geq 0} \phi^k \varepsilon_{t-k}$, so $\operatorname{Var} X_t = \sigma^2/(1-\phi^2)$ and $C_X(\tau) = \phi^{|\tau|}\sigma^2/(1-\phi^2)$. The autocorrelation decays geometrically with time constant $-1/\log|\phi|$ samples. If $|\phi| \geq 1$ no stationary solution of this form exists.
- *Random phase sinusoid*: $X_t = A\cos(\omega t + \Theta)$ with Θ uniform on $[0, 2\pi)$. Its mean is zero and $C_X(\tau) = \frac{A^2}{2}\cos(\omega\tau)$, which does not decay. If the amplitude A is itself random and independent of Θ, the process is still stationary, with $C_X(\tau) = \frac{1}{2}\mathbb{E}[A^2]\cos(\omega\tau)$, but a single realization reveals only its own amplitude.
- *Random walk*: $X_t = \sum_{s \leq t} \varepsilon_s$ has $\operatorname{Var} X_t = t\sigma^2$, so it is not stationary; its increments are.

A process is *ergodic* (for the mean) if time averages along one realization converge to the ensemble mean. Ergodicity is what licenses estimating expectations from a single long recording. It is an assumption, and it fails whenever a slowly varying latent state (arousal, drift) makes one session unrepresentative of the ensemble.

Worked example 25.6 (How many independent seconds?)**.** An LFP envelope sampled at 100 Hz has lag-one autocorrelation $\phi = 0.98$ at that sampling rate, and decays approximately geometrically. A 60-s segment contains $n = 6000$ samples but $n_{\text{eff}} \approx 6000 \cdot 0.02/1.98 \approx 61$. A t-test that treats the 6000 samples as independent overstates the evidence by a factor of about $\sqrt{6000/61} \approx 10$ in the test statistic.

Exercises

1. Explain what changes when observations are serially correlated but an iid likelihood is used.
2. Derive the mean and variance of an average of n independent variables with equal finite variance.
3. Define weak stationarity in your own words and give a counterexample.
4. Explain why trial averaging can suppress evidence of state switches.
5. Show that $\frac{1}{n}\sum(X_i - \bar{X})^2$ is biased, and compute its bias.
6. Derive the MLE of the rate of an exponential distribution from iid inter-spike intervals, and its Fisher information.
7. Derive the autocovariance of an AR(1) process from its moving-average representation.
8. Show that the random walk is not weakly stationary.
9. Compute n_{eff} for $n = 1000$ samples from an AR(1) with $\phi = 0.5$.

Chapter 26

Bridges from sets to random variables and spike counts

Reading goal 26.1. Understand random events, independence, averages, covariance, and the law of total variance from elementary probability definitions.

Bridge 35: Events form a sigma-algebra rather than an arbitrary list

Proposition 26.2. *If $\mathcal{F}$ is a sigma-algebra on Ω, it contains $\varnothing$, and it is closed under finite intersections and countable intersections.*

Proof. By definition $\Omega \in \mathcal{F}$, complements of members belong to $\mathcal{F}$, and countable unions of members belong to $\mathcal{F}$. Thus $\varnothing = \Omega^c$ is in $\mathcal{F}$. De Morgan's laws turn intersections into complements of unions of complements, so the intersections also belong to $\mathcal{F}$. □

Example. A predicate such as "the spike count exceeds five" must refer to a measurable event before assigning it a probability. In finite discrete experiments all subsets of the sample space can be used.

Bridge 36: Independence survives taking complements

Proposition 26.3. *If events A, B are independent, then A^c, B and A^c, B^c are independent.*

Proof. Since $B = (A \cap B) \sqcup (A^c \cap B)$, finite additivity gives $\mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B) = (1 - \mathbb{P}(A))\mathbb{P}(B)$. Hence A^c and B are independent. Repeating the same calculation with complement of B yields independence of both complements. □

Example. Independence of independent-trial events includes their non-occurrences. Recorded spike counts from adjacent bins generally do not automatically satisfy this condition.

Bridge 37: Expectation is linear even for dependent variables

Proposition 26.4. *If X, Y are integrable and $a, b \in \mathbb{R}$, then $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.*

Proof. In a finite experiment, expand the probability-weighted sum $\sum_\omega (aX(\omega) + bY(\omega))\mathbb{P}(\{\omega\})$ and distribute multiplication and addition. For integrable variables the corresponding property follows from linearity of the Lebesgue integral. No independence assumption is used. □

Example. The expected combined spike count of two units equals the sum of their expected individual counts even if their activities are strongly correlated.

Bridge 38: Indicator variables encode event counts

Proposition 26.5. *For measurable A, $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. For events $A_1, \dots, A_n$, the count $N = \sum_{j=1}^n \mathbf{1}_{A_j}$ satisfies $\mathbb{E}N = \sum_{j=1}^n \mathbb{P}(A_j)$.*

Proof. The indicator takes value one on A and zero on A^c, so its expectation is $1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c)$. Apply expectation linearity to the finite sum. □

Example. A per-bin count is a sum of event indicators. Its variance depends on cross-covariances, so its expectation alone cannot establish Poisson statistics.

Bridge 39: Conditional expectation obeys the tower property

Theorem 26.6 (Law of total expectation)**.** *If X is integrable and G a conditioning variable, then $\mathbb{E}[\mathbb{E}[X \mid G]] = \mathbb{E}[X]$.*

Proof. For a finite-valued G, write $\mathbb{E}[X \mid G = g]$ as a conditional weighted average. Multiplying by $\mathbb{P}(G = g)$ and summing over g recovers the sum over every outcome, hence $\mathbb{E}[X]$. The general statement follows from the defining integral identity of conditional expectation with the conditioning event chosen as the entire sample space. □

Example. A latent behavioral state changes conditional firing rate, but averaging over states recovers the unconditional mean if the weighting distribution is correct.

Bridge 40: Total variance separates within- and between-state effects

Theorem 26.7 (Law of total variance)**.** *If X is square-integrable, then* $\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid G)] + \mathrm{Var}(\mathbb{E}[X \mid G])$.

Proof. Use $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$ and $\mathrm{Var}(X \mid G) = \mathbb{E}[X^2 \mid G] - (\mathbb{E}[X \mid G])^2$. Take expectations of the latter, apply the tower property to X^2, and add $\mathrm{Var}(\mathbb{E}[X \mid G]) = \mathbb{E}[(\mathbb{E}[X \mid G])^2] - (\mathbb{E}X)^2$. The intermediate squared-conditional-mean terms cancel. □

Example. Trial-to-trial latent gain can produce spike-count variance above its conditional Poisson component. The decomposition does not say which physiological variable the gain represents.

Bridge 41: Covariance is bilinear under finite moments

Proposition 26.8. *For square-integrable X, Y, Z and scalars a, b, covariance is linear in its first argument:*

$$\operatorname{Cov}(aX + bY, Z) = a \operatorname{Cov}(X, Z) + b \operatorname{Cov}(Y, Z).$$

Proof. By definition, $\operatorname{Cov}(W, Z) = \mathbb{E}[(W - \mathbb{E}W)(Z - \mathbb{E}Z)]$. Insert $W = aX + bY$, use expectation linearity to center it as $a(X - \mathbb{E}X) + b(Y - \mathbb{E}Y)$, distribute, and use expectation linearity once more. □

Example. The variance of a pooled population count includes covariance between neurons. Ignoring correlated noise can misstate the precision of a population statistic.

Bridge 42: Random Poisson intensity creates overdispersion

Proposition 26.9. *If $N \mid G \sim \text{Poisson}(\mu G)$ with $\mu > 0$, $G > 0$, $\mathbb{E}G = 1$, and finite $\operatorname{Var}(G)$, then $\mathbb{E}N = \mu$ and $\operatorname{Var}(N) = \mu + \mu^2 \operatorname{Var}(G)$.*

Proof. Conditional Poisson mean and variance are both μG. Total expectation gives $\mathbb{E}N = \mu \mathbb{E}G = \mu$. Total variance gives $\operatorname{Var}(N) = \mathbb{E}[\mu G] + \operatorname{Var}(\mu G) = \mu + \mu^2 \operatorname{Var}(G)$. □

Example. The Fano factor is $\operatorname{Var}(N)/\mathbb{E}N = 1 + \mu \operatorname{Var}(G)$. It differs from Fano's information-theoretic inequality. An estimator such as `spike_count_fano.m` measures a sample statistic; it does not identify G without a latent-variable model.

Exercises

1. Produce two dependent Bernoulli variables and verify expectation linearity numerically.
2. Derive $\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2 \operatorname{Cov}(X, Y)$.
3. Compare Poisson and lognormal-gain mixed-Poisson simulations at equal mean counts.

Part VI

Mathematics of electrophysiological signals

Chapter 27

From a physical voltage to a sampled sequence

Reading goal 27.1. Understand the measurement chain, units, sample rates, aliasing, and the epistemic difference between source and signal.

27.1 What electrodes actually measure

Extracellular electrodes record potential differences relative to a reference. Their outputs reflect contributions of multiple transmembrane currents, electrode geometry, and conductive tissue. The recorded LFP is not a direct readout of a particular neuron or synapse. A physical forward model connects currents to voltages [8, 16]. Analog amplification, referencing, and filtering occur before the digitized values are stored.

27.2 Sampled time is a lattice

For sampling frequency f_s, write $\Delta t = 1/f_s$ and $x[n] = x(n\Delta t)$ for integer n in the acquisition index set. The Nyquist frequency is $f_s/2$. A sinusoid at a frequency above Nyquist can produce the same discrete samples as a lower-frequency sinusoid; this is *aliasing*. A pre-sampling low-pass filter is therefore required to suppress frequencies that would fold into the band of interest. Merely applying a digital low-pass filter after sampling cannot undo aliasing already introduced.

27.3 The ideal sampling theorem and its assumptions

If a continuous signal is strictly bandlimited to frequencies $|f| < f_s/2$, its samples determine the original signal through ideal sinc interpolation. Actual brain recordings are neither exactly bandlimited nor free of instrumentation artifacts. The theorem establishes a mathematical possibility under stated conditions, not a promise that every sampled waveform reproduces all physical current sources.

27.4 Storage, quantization, and memory mapping

A discrete recording is represented in finite precision. With C channels, N samples, and b bytes per stored sample, uncompressed size is CNb bytes (ignoring metadata and headers). Memory-mapped storage can support access to selected time blocks without allocating the entire

dataset. It does not guarantee fast noncontiguous access: array layout, strides, page faults, and disk bandwidth still matter. Always document sample type, endian convention, orientation, sampling rate, units, and channel metadata.

Worked example 27.2 (A large acquisition)**.** For 128 channels at 30,000 samples/s over one hour, stored as signed 16-bit integers, the raw size is $128 \cdot 30000 \cdot 3600 \cdot 2 = 27{,}648{,}000{,}000$ bytes (about 27.65 GB in decimal units). Converting the entire array to 64-bit floats multiplies memory by four. A streaming analysis therefore need not sacrifice mathematical precision in its intermediate accumulators, but should avoid careless full-array expansion.

27.5 Aliasing as an equivalence relation

Fix a sampling rate f_s. Two frequencies f and f' produce identical samples of the complex exponential $e^{2\pi i f t}$ at times n/f_s iff $e^{2\pi i (f-f')n/f_s} = 1$ for every integer n, i.e. iff $f - f'$ is an integer multiple of f_s. "Same samples" is thus the equivalence relation $f \sim f' \iff f - f' \in f_s\mathbb{Z}$ of Chapter 5, and each class has exactly one representative in $(-f_s/2, f_s/2]$. For *real* sinusoids, $\cos(2\pi f t)$ and $\cos(-2\pi f t)$ coincide, so each class of $|f|$ has a representative in $[0, f_s/2]$: the folded or *alias* frequency

$$f_{\text{alias}} = \left| f - f_s \cdot \operatorname{round}(f/f_s) \right|.$$

A 60 Hz line-noise harmonic at 540 Hz sampled at 1000 Hz appears at $|540 - 1000| = 460$ Hz; sampled at 500 Hz it appears at $|540 - 500| = 40$ Hz, inside the gamma band.

27.6 Quantization

An analog-to-digital converter with b bits over an input range of width V has step $q = V/2^b$. Rounding to the nearest level introduces an error $e \in [-q/2, q/2]$. If the signal varies over many levels, a standard model treats e as uniform on that interval and independent of the signal, giving error variance $q^2/12$. For a full-scale sinusoid the resulting signal-to-quantization-noise ratio is approximately $6.02b + 1.76$ dB.

Worked example 27.3 (Resolution of a 16-bit headstage)**.** A 16-bit converter spanning ± 5 mV has $q = 10\,\text{mV}/65536 \approx 0.15\,\mu\text{V}$ and quantization noise standard deviation $q/\sqrt{12} \approx 0.044\,\mu\text{V}$, far below typical thermal and biological noise of several microvolts. In this regime the uniform-error model is harmless. When gain is set so low that a signal spans only a few levels, the model fails: the error becomes deterministic and signal-dependent, and it creates harmonics rather than white noise.

27.7 Resampling

Changing the sampling rate by a rational factor L/M (in lowest terms) consists of three steps: insert $L - 1$ zeros between samples (upsampling), low-pass filter at the lower of the two Nyquist frequencies, and keep every Mth sample (downsampling). Omitting the filter before downsampling folds high frequencies into the band exactly as in analog sampling: decimating a 30 kHz wideband trace to 1 kHz by taking every 30th sample aliases spike energy into the LFP band. Library routines that "decimate" include the filter; plain indexing does not.

27.8 Timing and synchronization

Two acquisition systems running at nominally equal rates drift apart because their clocks differ by a few parts per million. A 10 ppm discrepancy accumulates to 36 ms per hour, larger than many

spike–LFP or spike–behavior timing effects. Alignment therefore requires shared synchronization pulses, and the mapping between clocks is estimated by regression of pulse times, typically linear ($t_B = a + bt_A$) over a session. Residuals of that regression measure the timing uncertainty that should be propagated into any latency claim.

27.9 Sample-and-hold and the zero-order hold

Physical converters do not take instantaneous point samples; they integrate over a short aperture, and digital-to-analog converters hold each value until the next sample. Holding for one sample period multiplies the spectrum by $\mathrm{sinc}(f/f_s)$, an attenuation of about 3.9 dB at Nyquist. This is a reminder that the "ideal sampling" of the previous sections is a model whose deviations are small but quantifiable.

Exercises

1. Determine the Nyquist frequency for an LFP sampled at 1250 Hz.
2. Calculate the storage size of 80 channels sampled at 2 kHz for 2 hours using float32.
3. Explain why the reference electrode can affect estimated coherence.
4. Distinguish a physical source model, an amplifier signal, and a floating-point array.
5. Compute the alias frequency of a 1.1 kHz component sampled at 1250 Hz.
6. What is the quantization step of a 24-bit converter spanning ± 2.5 V?
7. Explain why taking every 30th sample of a 30 kHz recording is not a valid way to obtain a 1 kHz LFP.
8. Two clocks differ by 20 ppm. How long until they disagree by 1 ms?

Chapter 28

Fourier analysis, convolution, and digital filtering

Reading goal 28.1. Derive spectral representations and see why filtering is a temporal operation with potential artifacts.

28.1 The discrete Fourier transform

For N discrete samples, define

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-i2\pi kn/N}, \qquad k = 0, \ldots, N-1.$$

The inverse is $x[n] = N^{-1} \sum_{k=0}^{N-1} X[k] e^{i2\pi kn/N}$. The frequency index represents $f_k = kf_s/N$ modulo f_s. Frequency resolution in the ordinary DFT grid is $\Delta f = f_s/N = 1/T$ where $T = N/f_s$. Zero-padding interpolates the frequency grid but does not create new resolving power for nearby components.

Parseval's identity for this convention is $\sum_n |x[n]|^2 = N^{-1} \sum_k |X[k]|^2$. One-sided and two-sided PSD conventions differ by factors of two away from DC and Nyquist; implementations must state normalization and units. The PSD estimates power per hertz, not simply squared FFT amplitude.

28.2 Convolution and linear time-invariant systems

A discrete linear time-invariant filter has output $y[n] = \sum_m h[m]x[n-m]$, where h is its impulse response. Under appropriate conditions, $Y(e^{i\omega}) = H(e^{i\omega})X(e^{i\omega})$. The transfer function has both magnitude and phase. Its impulse and step responses show how brief transients can be smeared in time or transformed into artificial-looking oscillations [14].

28.3 FIR, IIR, and zero-phase filtering

A finite impulse-response (FIR) filter has a finite set of coefficients; an infinite impulse-response (IIR) filter includes feedback. Applying a forward-backward filter can cancel net phase shift in the interior of a record but is noncausal and changes the effective magnitude response. Boundaries require special handling; filtering a short isolated segment can produce misleading edge effects. In an online decoder, a filter that uses future samples is not available without added latency.

28.4 Cross-correlation and spectral duality

For a weakly stationary process, the autocovariance $C_{xx}[\ell] = E[(X_n - \mu)(X_{n+\ell} - \mu)]$ is related to the PSD through a Fourier transform (Wiener–Khinchin). This relationship creates a bridge between lag-domain dependencies and frequency-domain power. Oscillatory-looking autocovariance and a spectral peak are related descriptions; neither alone proves the existence of a distinct biological oscillator.

Worked example 28.2 (Moving-average low-pass filter)**.** Take $y[n] = (x[n] + x[n-1])/2$. Its frequency response is $H(e^{i\omega}) = \frac{1}{2}(1 + e^{-i\omega}) = e^{-i\omega/2}\cos(\omega/2)$. Thus magnitude attenuation and phase delay both emerge from an apparently simple average. At Nyquist ($\omega = \pi$), the response is zero.

28.5 Proof of DFT inversion and Parseval

Theorem 28.3. *With $X[k] = \sum_{n=0}^{N-1} x[n]\omega^{-kn}$ and $\omega = e^{2\pi i/N}$, we have $x[n] = \frac{1}{N}\sum_k X[k]\omega^{kn}$ and $\sum_n |x[n]|^2 = \frac{1}{N}\sum_k |X[k]|^2$.*

Proof. Substitute the definition:

$$\frac{1}{N}\sum_k X[k]\omega^{kn} = \frac{1}{N}\sum_m x[m]\sum_k \omega^{k(n-m)} = x[n],$$

by the orthogonality of roots of unity (Chapter 12): the inner sum is N when $m = n$ and 0 otherwise. For Parseval, expand $\sum_k |X[k]|^2 = \sum_k \sum_{m,n} x[m]\overline{x[n]}\omega^{-k(m-n)}$ and apply the same identity. □

Equivalently, the vectors $e_k[n] = \omega^{kn}/\sqrt{N}$ form an orthonormal basis of $\mathbb{C}^N$, and the DFT, suitably scaled, is a unitary change of basis. Energy is preserved; it is merely redistributed among coordinates.

28.6 Symmetry and the meaning of negative frequencies

For real x, $X[N-k] = \overline{X[k]}$. Hence $|X[k]|$ is symmetric about $N/2$ and the bins $k > N/2$ carry no new information; they represent negative frequencies $-(N-k)f_s/N$. A real cosine at frequency f_0 appears as two lines of equal height at $\pm f_0$, each with half of its amplitude. A one-sided spectrum folds the negative-frequency energy onto the positive side, which is the origin of the factor of two for non-DC, non-Nyquist bins.

28.7 Circular convolution

Theorem 28.4 (DFT convolution theorem)**.** *If $y[n] = \sum_{m=0}^{N-1} h[m]x[(n-m) \bmod N]$, then $Y[k] = H[k]X[k]$.*

Proof. $Y[k] = \sum_n \sum_m h[m]x[(n-m) \bmod N]\omega^{-kn}$. Substitute $n' = (n-m) \bmod N$ and use $\omega^{-kn} = \omega^{-km}\omega^{-kn'}$ (valid modulo N because $\omega^N = 1$). The double sum factors into $H[k]X[k]$. □

The theorem concerns *circular* convolution: the signal is treated as periodic, and the end of the record wraps around to its start. Filtering by multiplying FFTs therefore contaminates the first samples with the last ones unless the signal is zero-padded to at least $N + L - 1$ samples for a filter of length L, after which circular and linear convolution agree on the samples of interest.

28.8 Leakage and windows

A finite record is the product of an infinite signal and a rectangular window $w[n] = \mathbf{1}_{0\le n<N}$. Multiplication in time is (circular) convolution in frequency, so the observed spectrum is the true spectrum smeared by the window's transform, the *Dirichlet kernel*

$$W(\theta) = \sum_{n=0}^{N-1} e^{-i\theta n} = e^{-i\theta(N-1)/2}\frac{\sin(N\theta/2)}{\sin(\theta/2)}.$$

Its main lobe has width $4\pi/N$ (in angular frequency) and its highest sidelobe is only about 13 dB below the peak. A sinusoid between DFT bins therefore leaks power into distant bins, which can hide a weak component near a strong one. Tapered windows (Hann, Hamming, Slepian) trade a wider main lobe for much lower sidelobes; the Hann window's first sidelobe is about 31 dB down. No window removes the trade-off, which is a finite-N instance of the uncertainty principle.

28.9 The uncertainty principle

Theorem 28.5 (Heisenberg–Gabor inequality)**.** *For $x \in L^2(\mathbb{R})$ with $\|x\|_2 = 1$ and sufficient decay, define $\sigma_t^2 = \int t^2|x(t)|^2dt$ and $\sigma_f^2 = \int f^2|\hat{x}(f)|^2df$ (centered at zero). Then $\sigma_t\sigma_f \ge 1/(4\pi)$, with equality exactly for Gaussians.*

Sketch. Integrate by parts: $1 = \int |x|^2dt = -2\Re\int t\,x(t)\overline{x'(t)}\,dt$. By Cauchy–Schwarz, $1 \le 2\|tx\|_2\|x'\|_2$, and by Plancherel $\|x'\|_2 = 2\pi\|f\hat{x}\|_2$. Equality in Cauchy–Schwarz requires $x' \propto tx$, which forces a Gaussian. □

No analysis method can produce a representation that is simultaneously arbitrarily sharp in time and in frequency. Wavelets and multitapers choose different compromises; they do not escape the bound.

28.10 Filter design specifications

A practical band-pass filter is specified by passband edges, stopband edges, passband ripple, and stopband attenuation. The transition width Δf largely determines the required FIR length: for a windowed-sinc design, $L \approx Af_s/(22\Delta f)$ for A decibels of attenuation (a rule of thumb due to Kaiser and Harris). A theta filter (4–12 Hz) at $f_s = 1000$ Hz with 2 Hz transitions and 60 dB attenuation thus needs roughly 1400 taps, 1.4 s of impulse response. Narrow bands at low frequencies imply long filters, and long filters smear events in time: a filter's impulse response is the shortest event it can resolve.

Worked example 28.6 (A filter that rings)**.** Pass a single brief transient, approximately an impulse, through a narrow 8 Hz band-pass filter. The output is the filter's impulse response: a windowed 8 Hz oscillation lasting several cycles. A Hilbert phase estimate of this output is perfectly "rhythmic" although the input contained no rhythm at all. Any event-triggered analysis built on narrow-band filtering must therefore compare against the response to non-oscillatory transients, or examine the unfiltered waveform.

Exercises

1. Calculate the DFT of a constant sequence and identify its DC term.
2. Derive the moving-average frequency response for three samples.

3. Explain the difference between a Fourier coefficient and a PSD estimate.
4. Design a simulation exposing a filter-induced false oscillatory event.
5. Prove that $X[N-k] = \overline{X[k]}$ for real x.
6. Compute the DFT of $x[n] = \cos(2\pi k_0 n/N)$ for integer $0 < k_0 < N/2$.
7. Show that zero-padding a length-N signal to length $2N$ produces DFT values that interpolate the original ones.
8. Estimate the FIR length for a 150–250 Hz ripple filter at $f_s = 1250$ Hz with 20 Hz transitions and 50 dB attenuation.
9. Verify the uncertainty bound with equality for $x(t) = (2a)^{1/4} e^{-\pi a t^2}$.

Chapter 29

Spectrograms, multitapers, and periodic-versus-aperiodic spectra

Reading goal 29.1. Distinguish a spectral estimate from the generating spectrum and avoid inferring oscillations from arbitrary band power.

29.1 Why spectral estimates fluctuate

A finite realization contains sampling variability and boundary effects. The periodogram's frequency resolution changes with recording duration, and its variance does not simply vanish by taking a long record without appropriate averaging. Windowing reduces leakage at the price of altered resolution. Welch estimates average periodograms across segments; the segments may overlap, but this does not make them independent replicates.

29.2 Spectrograms and the time-frequency trade-off

The short-time Fourier transform is

$$X(\tau, f) = \int_{-\infty}^{\infty} x(t) w(t - \tau) e^{-i2\pi f t}\, dt,$$

where w is a localized window. Narrow time windows localize fast transients while broadening spectral resolution. Long windows sharpen frequency information but smear event timing. A spectrogram $|X(\tau, f)|^2$ is not an instantaneous power measurement at a mathematical point: it summarizes a neighborhood determined by the window.

29.3 Multitaper estimation

The multitaper method forms spectral estimates using several mutually orthogonal tapers with controlled spectral concentration. Discrete prolate spheroidal sequences are commonly selected for a time-bandwidth product NW; the number of useful tapers is often on the order of $2NW - 1$. The smoothing bandwidth and record length must be reported. Multitaper methods are central to the Chronux approach [5].

29.4 Spectral peaks over the background

Brain spectra often contain aperiodic structure approximated locally by a $1/f^{\beta}$ law, along with putative narrowband components. A rise in band power may result from an oscillatory peak, a change in the aperiodic offset, a change in the exponent, or a frequency shift. Models such as FOOOF/specparam parameterize periodic and aperiodic components; IRASA uses irregular resampling to estimate the aperiodic part. These methods have distinct failure modes, especially when peaks are wide or overlap [15, 19].

Worked example 29.2 (Two indistinguishable band powers)**.** Imagine condition A has a 10-Hz Gaussian peak over a low background while condition B has no distinct peak but a sufficiently elevated broadband background. The integrals of their spectral power from 8 to 12 Hz may be equal. The same band-power summary corresponds to different generative accounts; it is therefore not a sufficient statistic for the existence of an oscillator.

29.5 The periodogram is inconsistent

For a stationary process sampled at f_s with N samples, the periodogram is

$$\hat{S}(f_k) = \frac{1}{N f_s}\Big|\sum_{n=0}^{N-1} x[n] e^{-2\pi i k n/N}\Big|^2.$$

Proposition 29.3. *If $x[n]$ is Gaussian white noise with variance σ^2, then for $0 < k < N/2$ the values $\hat{S}(f_k)$ are independent, each distributed as $\frac{\sigma^2}{f_s} \cdot \frac{\chi^2_2}{2}$, i.e. exponential with mean equal to the true spectral density σ^2/f_s.*

Sketch. The real and imaginary parts of $X[k]$ are linear combinations of independent Gaussians with orthogonal coefficient vectors $\cos(2\pi kn/N)$ and $\sin(2\pi kn/N)$, each of squared norm $N/2$. They are therefore independent $\mathcal{N}(0, N\sigma^2/2)$ variables, and $|X[k]|^2$ is $N\sigma^2/2$ times a χ^2_2 variable. Distinct bins use orthogonal coefficient vectors and are independent. □

The standard deviation of an exponential variable equals its mean. Doubling the record length doubles the number of bins but leaves each bin's relative error at 100%. This is *inconsistency*: more data give finer resolution, not lower variance. Variance falls only by averaging, across segments (Welch), tapers (multitaper), neighboring frequencies (smoothing), or trials.

29.6 Bias, variance, and degrees of freedom

Averaging K approximately independent periodogram-like estimates gives an estimate distributed approximately as $S(f)\chi^2_{2K}/(2K)$. Its relative standard deviation is $1/\sqrt{K}$, and a 95% confidence interval for $S(f)$ is

$$\Big[\frac{2K\hat{S}(f)}{\chi^2_{2K,0.975}},\ \frac{2K\hat{S}(f)}{\chi^2_{2K,0.025}}\Big],$$

which on a logarithmic (dB) scale has a width independent of $S(f)$. This is one reason to plot spectra in decibels. Overlapping Welch segments are correlated; with 50% overlap and a Hann window the effective number of independent segments is somewhat less than the nominal number.

29.7 Slepian tapers

Among all sequences of length N with unit energy, which one concentrates the most of its spectral energy inside the band $[-W, W]$? The answer is the first discrete prolate spheroidal (Slepian) sequence, the leading eigenvector of the $N \times N$ matrix

$$A_{mn} = \frac{\sin(2\pi W(m-n))}{\pi(m-n)} \quad (m \neq n), \qquad A_{nn} = 2W,$$

with W in cycles per sample. The eigenvalues $\lambda_0 > \lambda_1 > \cdots$ are the fractions of energy inside the band. Approximately $2NW$ of them are close to one and the rest fall rapidly to zero; the corresponding eigenvectors are mutually orthogonal. The multitaper estimate averages the $K \approx 2NW - 1$ well-concentrated eigenspectra, achieving K degrees of freedom (in pairs) with leakage controlled at resolution $2W$. A multitaper analysis therefore has two parameters with physical meaning: the window length $T = N/f_s$ and the half-bandwidth $W f_s$ in hertz. Their product fixes K.

29.8 The aperiodic component

Write a parametric spectral model as

$$\log_{10} S(f) = b - \log_{10}(k + f^{\chi}) + \sum_j G_j(f),$$

with offset b, optional "knee" k, exponent χ, and Gaussian peaks G_j in log-power. When $k = 0$, the aperiodic part is a straight line of slope $-\chi$ in log–log coordinates. Two consequences follow. First, a change in χ rotates the spectrum and changes band power in opposite directions at low and high frequencies, without any oscillation changing. Second, the ratio of two band powers (e.g. theta/delta) depends on χ even when no peak exists. The exponent has been related to the balance of excitation and inhibition in models, but that link is model-dependent and not a theorem about tissue.

Worked example 29.4 (Resolution, variance, and record length)**.** A 4-s LFP segment sampled at 1000 Hz has DFT resolution 0.25 Hz. With multitaper half-bandwidth $W = 1$ Hz, $NW = 4$ and $K = 7$ tapers give 14 degrees of freedom; the 95% confidence interval spans roughly -2.7 to $+4.0$ dB around the estimate. Doubling W to 2 Hz yields $K = 15$ and an interval of about -2.0 to $+2.5$ dB, at the cost of blurring peaks narrower than 4 Hz. Neither choice is "correct"; the report must state it.

Exercises

1. Explain the meaning of time-bandwidth product in multitaper estimation.
2. Simulate a signal whose $1/f$ slope changes but whose oscillatory peak remains fixed.
3. Explain why a spectral peak detector needs a null or baseline definition.
4. Describe two ways spectrogram edge artifacts can mislead interpretation.
5. Show that the periodogram of white noise has the same relative variance for every record length.
6. Compute the 95% confidence interval in dB for $K = 1$ and $K = 10$ independent averaged periodograms.
7. For $T = 2$ s and half-bandwidth 3 Hz, how many Slepian tapers are well concentrated?

8. Show that a change in aperiodic exponent with fixed value at $f = 20\,\text{Hz}$ increases 4–8 Hz power and decreases 30–80 Hz power.

Chapter 30

Analytic signals, circular statistics, and cross-frequency coupling

Reading goal 30.1. Connect complex phase to the analytic signal and understand exactly what phase-amplitude coupling measures.

30.1 The analytic signal

For a real-valued signal $x(t)$, define its Hilbert transform $\mathcal{H}\{x\}(t)$ by a principal-value convolution with $1/(\pi t)$, under suitable function-space assumptions. The analytic signal is

$$z(t) = x(t) + i\mathcal{H}\{x\}(t) = a(t)e^{i\phi(t)},$$

where $a(t) = |z(t)|$ is amplitude and $\phi(t) = \arg z(t)$ is phase. In Fourier space, analytic-signal construction suppresses negative frequencies and doubles positive frequencies (with appropriate DC and Nyquist conventions). These operations acquire a clean oscillatory interpretation for a reasonably narrowband component. Phase can be unstable when instantaneous amplitude is very small.

30.2 Filtering comes before interpretation

Often the signal is band-pass filtered to $x_b(t)$ before Hilbert transformation. The resulting $a_b(t)$ and $\phi_b(t)$ describe the *filtered component*, not a unique property of the raw potential. Choice of passband, filter order, padding, and transitions can change the estimates. For nonsinusoidal waves, harmonics introduced by waveform shape can appear as high-frequency power linked to the low-frequency phase [13].

30.3 Circular means and concentration

For phases $\phi_1, \ldots, \phi_n$, set $R = n^{-1}\sum_j e^{i\phi_j}$. The mean direction is $\arg R$ when $R \neq 0$, and the mean resultant length $|R|$ measures concentration between zero and one. Classical tests such as Rayleigh's test target nonuniformity in directions under specified assumptions. They do not automatically correct for temporal autocorrelation or repeated units from the same animal [4].

30.4 Phase locking and phase-amplitude dependence

The phase-locking value of phase differences is PLV $= |n^{-1}\sum_j e^{i(\phi_{1j}-\phi_{2j})}|$. It measures concentration of phase differences, not amplitude modulation. A PAC statistic such as the mean vector length $|n^{-1}\sum_j A_{\text{high}}(t_j)e^{i\phi_{\text{low}}(t_j)}|$ quantifies phase dependence of a high-band amplitude; its scale and biases require normalization and surrogates. The modulation index based on divergence of binned amplitude distributions is a different statistic. These quantities should not be conflated.

30.5 The interpretation trap

A significant PAC value need not represent two physiologically distinct oscillators interacting. Common nonstationarity, sharp transients, harmonics, or filtering artifacts can generate cross-frequency relationships [2]. Good practice checks waveform shape, aperiodic structure, trial boundaries, and meaningful surrogate null models. Early PAC observations in human cortex motivate the question, but they do not eliminate the general identifiability issues [24].

Worked example 30.2 (One nonsinusoidal oscillator)**.** Consider the following waveform:

$$x(t) = \cos(2\pi f_0 t) + 0.5\cos(4\pi f_0 t).$$

Its higher-frequency harmonic is phase-determined by the same fundamental oscillator. Filtering separately near f_0 and $2f_0$ produces components whose phase relationship is stable. It would be misleading to infer two interacting independent biological oscillators from this constructed example.

30.6 The Hilbert transform in the frequency domain

With the convention $\hat{x}(f) = \int x(t)e^{-2\pi i f t}dt$, the Hilbert transform acts as the multiplier

$$\widehat{\mathcal{H}x}(f) = -i\,\mathrm{sgn}(f)\,\hat{x}(f).$$

It shifts every positive-frequency component by $-\pi/2$ and every negative-frequency component by $+\pi/2$, leaving magnitudes unchanged. Hence $\mathcal{H}\cos = \sin$ and $\mathcal{H}\sin = -\cos$, the analytic signal of $\cos(2\pi f_0 t)$ is $e^{2\pi i f_0 t}$, and $\widehat{z}(f) = 2\hat{x}(f)\mathbf{1}_{f>0} + \hat{x}(0)\mathbf{1}_{f=0}$. Because the multiplier has modulus one, $\mathcal{H}$ preserves energy and $\mathcal{H}(\mathcal{H}x) = -x$ for signals without a DC component.

30.7 When is amplitude-phase decomposition meaningful?

Any real signal has an analytic representation $a(t)e^{i\phi(t)}$, so the decomposition always exists. The question is whether a and ϕ have the intended interpretation as "envelope" and "phase of a rhythm." Bedrosian's theorem gives a sufficient condition: if $a(t)$ has spectrum in $[-B, B]$ and the carrier frequency exceeds B, then the analytic signal of $a(t)\cos(2\pi f_0 t + \theta)$ is $a(t)e^{i(2\pi f_0 t+\theta)}$, so the envelope is recovered exactly. In words: the amplitude must vary slowly compared with the oscillation. A theta-band filter of width 4 Hz centred at 8 Hz approximately satisfies this; a 1–20 Hz "broadband" filter does not, and its "phase" mixes several rhythms.

The *instantaneous frequency* $\frac{1}{2\pi}\phi'(t)$ is a derivative of phase and inherits noise amplification. When $a(t)$ passes near zero the phase can jump by nearly π, producing large positive or negative instantaneous frequencies that have no physiological meaning. Such points should be masked using an amplitude threshold.

30.8 Distributions on the circle

The *von Mises* distribution with mean direction μ and concentration $\kappa \geq 0$ has density

$$p(\theta) = \frac{e^{\kappa \cos(\theta - \mu)}}{2\pi I_0(\kappa)},$$

where I_0 is a modified Bessel function. It plays the role of the normal distribution on the circle: $\kappa = 0$ is uniform, and large κ approximates $\mathcal{N}(\mu, 1/\kappa)$. Its expected resultant length is $A(\kappa) = I_1(\kappa)/I_0(\kappa)$.

Proposition 30.3 (Bias of the resultant length)**.** *If $\phi_1, \ldots, \phi_n$ are iid uniform on the circle, then $\mathbb{E}\big[|R|^2\big] = 1/n$ for $R = \frac{1}{n}\sum_j e^{i\phi_j}$.*

Proof. $|R|^2 = \frac{1}{n^2}\sum_{j,k} e^{i(\phi_j - \phi_k)}$. The n diagonal terms equal 1; for $j \neq k$, $\mathbb{E}[e^{i(\phi_j - \phi_k)}] = \mathbb{E}[e^{i\phi_j}]\mathbb{E}[e^{-i\phi_k}] = 0$. □

Hence $|R| \approx 1/\sqrt{n}$ for pure noise: 25 spikes give an expected PLV near 0.2 without any locking. Comparisons of PLV across conditions with different spike counts are biased. The *pairwise phase consistency* $\mathrm{PPC} = \frac{2}{n(n-1)}\sum_{j<k}\cos(\phi_j - \phi_k)$ removes the diagonal terms and is an unbiased estimator of the squared population resultant length, $\mathrm{PPC} = (n|R|^2 - 1)/(n-1)$. The Rayleigh test uses $Z = n|R|^2$, which is approximately exponential with mean one under uniformity, so $P(Z > z) \approx e^{-z}$ for large n.

30.9 Two PAC estimators and their assumptions

Let $\phi(t)$ be a low-frequency phase and $A(t)$ a high-frequency amplitude.

1. The *mean vector length* $\mathrm{MVL} = |\frac{1}{T}\sum_t A(t)e^{i\phi(t)}|$ depends on the overall scale of A; it is usually normalized by surrogates.
2. The *modulation index* bins ϕ into J bins, computes the mean amplitude $\bar{A}_j$ in each, normalizes to a distribution $p_j = \bar{A}_j / \sum_k \bar{A}_k$, and reports $\mathrm{MI} = D_{\mathrm{KL}}(p \,\|\, u)/\log J$, where $D_{\mathrm{KL}}(p \,\|\, u) = \log J - H(p)$ is the KL divergence from the uniform distribution. It is invariant to scaling A but depends on J and on the number of cycles.

Both estimators are positive under the null and grow with recording length differently, so they must be compared with surrogates that preserve the autocorrelation of each signal (e.g. cutting A at a random point and swapping the two pieces), not with zero.

Worked example 30.4 (Envelope of two close sinusoids)**.** Let $x(t) = \cos(2\pi \cdot 8t) + \cos(2\pi \cdot 9t) = 2\cos(\pi t)\cos(2\pi \cdot 8.5t)$. Its analytic signal is $e^{2\pi i 8t} + e^{2\pi i 9t}$, with amplitude $|2\cos(\pi t)|$, a 1 Hz beat, and phase advancing at 8.5 Hz but jumping by π every second when the amplitude passes through zero. An envelope fluctuation and a "phase reset" appear although two perfectly stationary rhythms are present. Amplitude modulation and phase discontinuities in a band-limited signal are therefore not, by themselves, evidence of transient events.

Exercises

1. Show why the arithmetic mean of angles near $\pm\pi$ is unreliable.
2. Distinguish PLV from PAC with explicit input variables.
3. Propose a surrogate method and specify exactly what structure it preserves.

4. Explain why zero-phase filters cannot be interpreted as real-time causal processing.
5. Show that $\mathcal{H}\{\cos(2\pi f_0 t)\} = \sin(2\pi f_0 t)$ for $f_0 > 0$ using the frequency-domain multiplier.
6. Compute the expected PLV for 100 uniformly distributed spike phases, and compare with the value 0.2 for 25 phases.
7. Prove the identity $\mathrm{PPC} = (n|R|^2 - 1)/(n - 1)$.
8. Show that the modulation index is zero exactly when the binned amplitude distribution is uniform.
9. In the beat example, find the times at which the phase jumps and the size of the jump.

Chapter 31

Cross-spectra, coherence, and Granger prediction

Reading goal 31.1. Derive dependence measures for multivariate signals and understand why prediction is not intervention.

31.1 From autocovariance to cross-covariance

For weakly stationary processes X_t and Y_t, define $C_{xy}(\tau) = E[(X_t - \mu_x)(Y_{t+\tau} - \mu_y)]$. Its Fourier transform is the cross-spectral density $S_{xy}(f)$. The autospectra S_{xx}, S_{yy} describe each process; the cross-spectrum is complex and contains magnitude and phase information about their lagged relationship.

Magnitude-squared coherence is

$$\mathcal{C}_{xy}(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f)S_{yy}(f)},$$

when the denominator is positive. For valid spectral-density matrices, $0 \le \mathcal{C}_{xy}(f) \le 1$. Coherence is non-directional. Magnitude-only summaries discard phase and cannot by themselves indicate whether one observed signal drives another.

31.2 Common input and reference contamination

Suppose $x = s + n_x$ and $y = s + n_y$ share a signal s while n_x, n_y are independent noise processes. Then x and y can exhibit substantial coherence even though no direct interaction between their unique sources exists. Similarly, a reference electrode that carries electrical activity can insert a common component into many recordings. Bipolar referencing or spatial derivatives may reduce some effects but introduce others; one must evaluate the measurement geometry and signal-to-noise ratio [3, 12].

31.3 Autoregressive models and direction of prediction

A vector autoregressive model of order p for $z_t \in \mathbb{R}^d$ is

$$z_t = \sum_{k=1}^{p} A_k z_{t-k} + \varepsilon_t.$$

Consider predicting Y_t from its own past versus predicting it from both its own past and the past of X_t. A classical Gaussian linear Granger statistic compares residual variances:

$$F_{X\to Y} = \log \frac{\sigma^2_{\text{restricted}}}{\sigma^2_{\text{full}}}.$$

When the full model predicts better, the statistic is positive. The arrow denotes an improvement in prediction under the candidate model, not proof that manipulating X would change Y. Latent common causes, instantaneous mixing, nonstationarity, model order, and noise can alter apparent direction. Spectral Granger extensions decompose predictive relationships by frequency under additional factorization assumptions [9].

31.4 Transfer entropy and conditional dependence

Transfer entropy is a conditional information measure, often written $I(Y_t; X_{t-1}^{(p)} \mid Y_{t-1}^{(q)})$. It compares how informative the past of X is about Y_t once the past of Y is known. Nonlinear estimators can be useful but require much data and careful treatment of bias. Prediction-based asymmetry is not a universal causal-identification guarantee.

Worked example 31.2 (A shared driver)**.** Let s_t be a stochastic signal, $x_t = s_t + \eta_t$, and $y_t = s_{t-1} + \xi_t$ with independent observation noises. The lag creates directional predictability from x to y, but the actual source is s, not a connection from the observed x channel to the observed y channel. This is a constructive counterexample to an unrestricted causal interpretation.

31.5 The spectral density matrix

For a d-dimensional weakly stationary process with absolutely summable cross-covariances, the *spectral density matrix* $S(f)$ has entries $S_{jk}(f)$, the cross-spectra. For every f it is Hermitian ($S_{kj} = \overline{S_{jk}}$) and positive semidefinite: for any $a \in \mathbb{C}^d$, $a^* S(f) a$ is the spectral density of the scalar process $\sum_j \bar{a}_j X_j$ and is therefore nonnegative.

Proposition 31.3. $0 \le \mathcal{C}_{xy}(f) \le 1$ *whenever* $S_{xx}(f) S_{yy}(f) > 0$.

Proof. The 2×2 matrix $\begin{pmatrix} S_{xx} & S_{xy} \\ S_{yx} & S_{yy} \end{pmatrix}$ is positive semidefinite, so its determinant $S_{xx}S_{yy} - |S_{xy}|^2$ is nonnegative. □

31.6 Coherence under a common source

Let $x = s + n_x$, $y = s + n_y$ with s, n_x, n_y mutually uncorrelated and spectra S_s, S_x^n, S_y^n. Then $S_{xy} = S_s$, $S_{xx} = S_s + S_x^n$, $S_{yy} = S_s + S_y^n$, and

$$\mathcal{C}_{xy}(f) = \frac{S_s^2}{(S_s + S_x^n)(S_s + S_y^n)} = \frac{1}{(1 + 1/\text{SNR}_x)(1 + 1/\text{SNR}_y)},$$

with $\text{SNR}_x = S_s / S_x^n$. Coherence here is entirely a function of signal-to-noise ratios. A condition that increases the power of the shared signal (e.g. running increases theta in all electrodes) increases coherence without any change in interaction. Conversely, adding independent noise to one channel decreases coherence without any change in interaction. Comparisons of coherence across conditions are only interpretable alongside the corresponding power changes.

31.7 Bias of estimated coherence

Coherence is estimated from K independent trials or tapers by $\hat{\mathcal{C}} = |\sum_k X_k \bar{Y}_k|^2 / (\sum_k |X_k|^2 \sum_k |Y_k|^2)$. When the true coherence is zero, $\mathbb{E}\hat{\mathcal{C}} = 1/K$ for Gaussian data, so ten tapers give a baseline near 0.1, not 0. The same $1/K$ bias that affects the PLV affects coherence. Confidence intervals are usually constructed on the Fisher-transformed coherency $\operatorname{atanh}|\hat{c}|$ or by jackknife over tapers or trials.

31.8 The imaginary part of coherency

Coherency is the complex quantity $c_{xy} = S_{xy}/\sqrt{S_{xx}S_{yy}}$. Under instantaneous linear mixing, for example volume conduction in a quasistatic medium where $x = a_1 s_1 + a_2 s_2$ and $y = b_1 s_1 + b_2 s_2$ with real coefficients and independent sources, the cross-spectrum is $S_{xy} = a_1 b_1 S_{s_1} + a_2 b_2 S_{s_2}$, which is *real*. Hence:

Proposition 31.4. *If two channels are real instantaneous mixtures of mutually uncorrelated sources, then $\Im c_{xy}(f) = 0$ for every f.*

A nonzero imaginary part therefore requires a time lag between the channels' contributions and cannot be produced by zero-lag mixing alone. This motivates the imaginary coherence and the (weighted) phase-lag index. The converse inference fails: a genuine zero-lag interaction is invisible to these measures, and lags due to differing filter delays or reference effects produce nonzero imaginary parts without interaction.

31.9 Spectral Granger causality

Write a stationary bivariate VAR in moving-average form $Z(f) = H(f)E(f)$ with innovation covariance Σ. If Σ is diagonal, $S_{yy}(f) = |H_{yy}(f)|^2\Sigma_{yy} + |H_{yx}(f)|^2\Sigma_{xx}$, and Geweke's frequency-domain measure is

$$f_{X\to Y}(f) = \log \frac{S_{yy}(f)}{|H_{yy}(f)|^2 \Sigma_{yy}},$$

the log ratio of total power of Y at f to the part of its power driven by its own innovations. (When Σ has off-diagonal terms a normalizing transformation is applied first.) Under regularity conditions, $\int_{-1/2}^{1/2} f_{X\to Y}(f)\, df$ equals the time-domain statistic $F_{X\to Y}$. Non-parametric estimates compute H and Σ from the spectral matrix by Wilson's spectral factorization rather than fitting a VAR. All the cautions of the time-domain measure carry over, and filtering the data before estimating Granger causality can create or destroy spectral structure in it; filtering is therefore discouraged.

Worked example 31.5 (Coherence change without coupling change)**.** Two electrodes record $x = s + n_x$, $y = s + n_y$ with noise spectra equal to 1 at 8 Hz. At rest the shared theta spectrum is $S_s = 1$, so $\mathcal{C} = 1/4$. During running $S_s = 4$, so $\mathcal{C} = (4/5)^2 = 0.64$. The mixing and the noise are unchanged; only the shared source power has grown. A report that "theta coherence increased during running" is accurate; a report that "coupling increased" is not supported.

Exercises

1. State why coherence is unchanged by a fixed nonzero scaling of one signal.
2. Distinguish phase locking, coherence, and Granger prediction.
3. Construct an example of common reference that raises coherence without coupling.

4. Describe checks for VAR residual autocorrelation and stability.
5. Show that the spectral density matrix is Hermitian.
6. Compute coherence for the common-source model with $\mathrm{SNR}_x = \mathrm{SNR}_y = 1$ and with $\mathrm{SNR}_x = 9, \mathrm{SNR}_y = 1$.
7. Show that a pure delay $y(t) = x(t - \tau)$ gives coherency $e^{-2\pi i f \tau}$ and determine when its imaginary part vanishes.
8. Explain why a common reference electrode with zero-lag activity contributes only to the real part of coherency.

Chapter 32

Extracellular potentials and source reconstruction

Reading goal 32.1. Connect physical current sources, volume conduction, referencing, and the limitations of current-source-density analysis.

32.1 Potential is a field

An extracellular electric potential $\phi(\mathbf{r}, t)$ is a scalar field. The electric field is $\mathbf{E} = -\nabla\phi$. Electrode voltages are differences in potential; changing the reference affects the data. In an idealized homogeneous isotropic quasistatic conductor, a point current source contributes potential proportional to $I/(4\pi\sigma r)$, where σ is conductivity and r the distance to the source. This approximation illustrates spatial spreading but does not accurately describe every layered or anisotropic brain tissue.

32.2 Current source density

Conservation of charge and Ohm's law under a quasistatic conductor model lead to relationships between current source density and spatial derivatives of potential. Under constant scalar conductivity, the extracellular current-source-density convention can be written $\mathrm{CSD} = -\sigma\nabla^2\phi$ (the sign depends on the source/sink convention). A laminar one-dimensional approximation replaces the Laplacian by the second depth derivative; a discrete central difference is

$$\left.\frac{\partial^2\phi}{\partial z^2}\right|_{z_j} \approx \frac{\phi_{j+1} - 2\phi_j + \phi_{j-1}}{(\Delta z)^2}.$$

This formula assumes uniform spacing and sufficiently smooth fields. Differentiation amplifies high-spatial-frequency noise, often making smoothing or regularization necessary.

32.3 Assumptions should travel with the result

The physics depends on conductivities, boundaries, geometry, electrode positions, and reference schemes. A depth reversal in LFP phase may be consistent with a laminar generator but does not alone locate a unique set of synaptic currents. Modeling and validation with synthetic sources help distinguish plausible mechanisms [8, 16].

32.4 From Maxwell's equations to the quasistatic approximation

In a conductor with conductivity σ and permittivity ε, the current density is $\mathbf{J} = \sigma\mathbf{E} + \mathbf{J}^s$, where $\mathbf{J}^s$ is the transmembrane source current density. Charge conservation reads $\nabla \cdot \mathbf{J}_{\text{total}} = -\partial\rho/\partial t$. For brain tissue at frequencies below a few kilohertz, three approximations are standard: (i) capacitive currents are small compared with ohmic ones, because $\omega\varepsilon/\sigma \ll 1$; (ii) magnetic induction is negligible, so $\mathbf{E} = -\nabla\phi$; and (iii) propagation delays are negligible over centimetres. Under these *quasistatic* assumptions, conservation of current becomes

$$\nabla \cdot (\sigma \nabla \phi) = \nabla \cdot \mathbf{J}^s =: I_{\text{CSD}},$$

a Poisson equation relating the potential to the current source density. When σ is constant, $\sigma\nabla^2\phi = I_{\text{CSD}}$, and the CSD convention of the earlier section follows (up to the sign convention for sources and sinks). Because time enters only as a parameter, each instant of the potential is determined by the instantaneous sources; filtering by tissue is, in this approximation, frequency independent.

32.5 The point-source solution and superposition

Proposition 32.2. *In an infinite homogeneous medium, the potential of a point current source I at the origin is $\phi(\mathbf{r}) = I/(4\pi\sigma|\mathbf{r}|)$.*

Proof. By symmetry the current flows radially and the total current through a sphere of radius r is I, so $|\mathbf{J}| = I/(4\pi r^2)$. Ohm's law gives $|\mathbf{E}| = I/(4\pi\sigma r^2)$, and integrating $-\partial\phi/\partial r = |\mathbf{E}|$ with $\phi(\infty) = 0$ yields the claim. □

Since the Poisson equation is linear, potentials of several sources add. Every neuron has as much current leaving as entering, so the simplest realistic source is a *dipole*: $+I$ at $\mathbf{r}_+$ and $-I$ at $\mathbf{r}_-$ separated by d. Far away ($r \gg d$),

$$\phi(\mathbf{r}) \approx \frac{I d \cos\theta}{4\pi\sigma r^2},$$

where θ is the angle between the dipole axis and $\mathbf{r}$. Dipole potentials decay as r^{-2}, faster than monopoles, and change sign across the plane perpendicular to the dipole: this is the polarity reversal seen across cortical layers. Spatially aligned dipoles (pyramidal cells with parallel apical dendrites) add constructively; randomly oriented ones (many interneurons) largely cancel. The LFP is therefore weighted towards geometrically aligned populations, not towards the most active ones.

32.6 Uniqueness and nonuniqueness

Theorem 32.3 (Uniqueness for the forward problem)**.** *Let Ω be a bounded domain with smooth boundary, $\sigma > 0$ smooth, and sources given. Then the solution of $\nabla \cdot (\sigma\nabla\phi) = I_{\text{CSD}}$ with prescribed boundary values (or prescribed normal current and a fixed reference) is unique.*

Proof. The difference w of two solutions satisfies $\nabla \cdot (\sigma\nabla w) = 0$ with zero boundary data. Multiply by w and integrate by parts: $\int_\Omega \sigma|\nabla w|^2 = 0$, so $\nabla w = 0$ and w is constant, hence zero (or fixed by the reference). □

The *inverse* problem has no such uniqueness. Helmholtz already observed that a current distribution inside a volume conductor can be altered by any "silent" source configuration,

one that produces zero potential outside a region, without changing the measurements made outside it. A closed shell of dipoles pointing radially (a double layer of constant strength) is an explicit example. Any source estimate from extracellular potentials therefore rests on constraints (anatomy, sparsity, smoothness, a fixed number of dipoles) that are not determined by the data.

32.7 Inverse CSD and the role of boundaries

The standard three-point second difference estimates the CSD only at interior contacts and assumes infinite, laterally homogeneous sheets of current. When the activated region has finite lateral extent, the second difference reports spurious sources at the edges of the probe. *Inverse* CSD methods instead assume a parametric source distribution (e.g. discs of finite radius R at each contact depth), compute the forward matrix F from sources to potentials, and estimate the CSD as $F^{-1}\phi$, possibly regularized. Kernel CSD performs the same task with a smoothness prior in a reproducing-kernel Hilbert space. Each method is the pseudoinverse or Tikhonov estimator of Chapter 21 with a different forward model and prior.

Worked example 32.4 (How fast does a spike's field decay?)**.** Model the extracellular action potential of a soma near the electrode as a monopole-dominated near field and a dipole far field. With $\sigma = 0.3\,\mathrm{S/m}$ and a peak net current of $1\,\mathrm{nA}$, the monopole formula gives $\phi \approx 1\times10^{-9}/(4\pi\cdot0.3\cdot r)\,\mathrm{V}$; at $r = 50\,\mu\mathrm{m}$ this is about $5.3\,\mu\mathrm{V}$ per nanoampere, and spike amplitudes of tens to hundreds of microvolts imply currents of several to tens of nanoamperes. Beyond about 100–150 μm the dipolar r^{-2} decay dominates and the amplitude falls below typical noise levels, which bounds the radius from which a single electrode can isolate units.

Exercises

1. Derive the units of $\sigma\nabla^2\phi$ in SI units.
2. Explain why a second spatial derivative is sensitive to electrode noise.
3. State two conditions needed for a one-dimensional laminar CSD approximation.
4. Derive the far-field dipole formula by expanding the two point-source potentials to first order in d/r.
5. Check that $\omega\varepsilon/\sigma \ll 1$ at 1 kHz for $\sigma = 0.3\,\mathrm{S/m}$ and $\varepsilon = 10^5\varepsilon_0$ (with $\varepsilon_0 \approx 8.85\times10^{-12}\,\mathrm{F/m}$).
6. Show that adding a constant to the potential leaves the CSD unchanged, and explain what this implies about referencing.
7. Explain why randomly oriented dipoles contribute little to the LFP but may contribute to multi-unit activity.

Chapter 33

Bridges from finite sequences to spectra

Reading goal 33.1. Establish why discrete Fourier coefficients, filtering, power, phase, and aliasing behave as they do before treating recorded LFP samples as continuous-time signals.

Bridge 43: Complex exponentials form an orthogonal finite family

Proposition 33.2. *For integers $k, \ell \in \{0, \dots, N-1\}$, $N \geq 1$,*

$$\sum_{n=0}^{N-1} e^{2\pi i(k-\ell)n/N} = \begin{cases} N, & k = \ell, \\ 0, & k \neq \ell. \end{cases}$$

Proof. For $k = \ell$, every term is one. Otherwise set $r = e^{2\pi i(k-\ell)/N}$; then $r^N = 1$ but $r \neq 1$. The finite geometric sum equals $(1 - r^N)/(1 - r) = 0$. □

Example. This finite orthogonality is the algebraic reason a DFT separates integer-grid Fourier bins; an off-grid sinusoid leaks across bins under a finite window.

Bridge 44: The DFT admits an exact finite inverse

Proposition 33.3. *For $x_0, \dots, x_{N-1} \in \mathbb{C}$, set $X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi ikn/N}$. Then*

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{2\pi ikn/N}.$$

Proof. Insert the definition of X_k into the inverse expression and exchange two *finite* sums. The coefficient of x_m is $N^{-1} \sum_k e^{2\pi ik(n-m)/N}$, which equals one for $m = n$ and zero otherwise by the previous bridge. Only x_n remains. □

Example. An inverse DFT exactly recovers the original finite samples up to numerical roundoff. It does not prove recovery of the unknown continuous biological signal between sample times.

Bridge 45: Finite Fourier energy has an exact accounting identity

Proposition 33.4. *With the DFT normalization above, $\sum_n |x_n|^2 = \frac{1}{N} \sum_k |X_k|^2$.*

Proof. Insert the inverse DFT into $\sum_n x_n \bar{x}_n$. Expand the finite sums and group terms by pairs (k, ℓ). The inner sum over n vanishes whenever $k \neq \ell$ by orthogonality and equals N when $k = \ell$, leaving $N^{-1} \sum_k |X_k|^2$. □

Example. Periodogram normalization should respect energy units and sampling interval. "Power" and "power spectral density" differ by frequency-bin widths.

Bridge 46: Circular convolution becomes multiplication under the DFT

Proposition 33.5. *If* $(x \circledast h)_n = \sum_{m=0}^{N-1} x_m h_{(n-m) \bmod N}$, *then* $\mathrm{DFT}(x \circledast h)_k = X_k H_k$.

Proof. Substitute the circular sum into the DFT, exchange finite sums, and for each m set $j = (n - m) \bmod N$. Periodicity gives $e^{-2\pi i k n/N} = e^{-2\pi i k m/N} e^{-2\pi i k j/N}$. The two sums factor into X_k and H_k. □

Example. Multiplying two equal-length FFTs implements *circular* convolution. To obtain linear finite convolution one must pad sufficiently or handle boundaries separately; otherwise opposite edges mix.

Bridge 47: Time shifts rotate Fourier phase

Proposition 33.6. *For a circular shift* $y_n = x_{(n-n_0) \bmod N}$, *the DFT is* $Y_k = e^{-2\pi i k n_0/N} X_k$.

Proof. In the DFT sum set $m = (n - n_0) \bmod N$. Periodicity preserves the summation domain and extracts the multiplier $e^{-2\pi i k n_0/N}$. The multiplier has modulus one, so magnitudes are preserved and phases change. □

Example. Two copies of the same oscillation with a constant temporal shift can have identical PSD yet different cross-spectral phase. PSD alone cannot recover relative timing.

Bridge 48: Uniform samples cannot distinguish aliased sinusoids

Proposition 33.7. *For sample rate* $F_s > 0$, *every* $f \in \mathbb{R}$ *and integer* m *satisfy*

$$e^{2\pi i (f + m F_s) n/F_s} = e^{2\pi i f n/F_s} \quad \text{for all integers } n.$$

Proof. The ratio of the two exponentials equals $e^{2\pi i m n} = 1$ for every integer n. Hence their sample sequences coincide exactly, although their continuous frequencies differ. □

Example. A 90-Hz tone sampled at 100 Hz can appear at a frequency of 10 Hz after folding; no digital filter applied afterward can establish which continuous sinusoid generated the samples. Acquisition requires appropriate analog anti-aliasing.

Exercises

1. Work out the DFT and inverse DFT of $[1, 0, 0, 0]$ by hand.
2. Numerically compare circular convolution with linear convolution for a short impulse near an array boundary.

3. Demonstrate DFT magnitude invariance under circular shifts and the corresponding phase change.

Part VII

Point processes and neuronal populations

Chapter 34

Spike trains as point processes

Reading goal 34.1. Model spike timestamps without pretending they are uniformly sampled voltages.

34.1 Events rather than amplitudes

A spike train is a locally finite collection of event times $\{t_1, \dots, t_K\}$. Its counting process is $N(t) = \#\{j : t_j \le t\}$. It is nondecreasing, integer valued, and jumps at event times. In generalized-function notation $dN(t)/dt = \sum_j \delta(t - t_j)$, but the counting process is the more elementary construction.

34.2 Poisson models and their scope

For a homogeneous Poisson process with rate $\lambda > 0$, counts in disjoint intervals are independent and $N(b) - N(a) \sim \text{Poisson}(\lambda(b-a))$. The waiting times are exponentially distributed, and the memoryless model has no refractory period. Real neurons often violate some of these assumptions. A time-inhomogeneous Poisson model admits $\lambda(t)$, with integrated intensity $\int_a^b \lambda(s)ds$ over an interval.

A conditional intensity $\lambda(t \mid \mathcal{H}_t)$ is a limiting event rate given the observed history $\mathcal{H}_t$. Roughly,

$$\lambda(t \mid \mathcal{H}_t) = \lim_{\Delta \downarrow 0} \frac{P(N(t+\Delta) - N(t) = 1 \mid \mathcal{H}_t)}{\Delta},$$

when the limit is defined and multiple jumps are negligible at small scales. History dependence permits refractory behavior, bursting, and coupling covariates.

34.3 Binning and the loss of exact timing

Dividing time into bins of width Δ creates counts $n_k = N((k+1)\Delta) - N(k\Delta)$. This may simplify statistics but loses information about positions inside bins. Narrower bins increase temporal precision yet yield sparse counts; wider bins stabilize rate estimates but obscure precise coincidence. A precise synchrony claim therefore needs a time-resolution convention and an explicit null model.

Worked example 34.2 (Poisson probability of a silent window)**.** A 10-spikes/s homogeneous Poisson process has expected count $\lambda\Delta = 0.1$ in 10 ms. The chance of observing zero spikes is $e^{-0.1} \approx 0.905$. Frequent silent bins are compatible with a nonzero underlying intensity.

34.4 Constructing the homogeneous Poisson process

Three descriptions of the homogeneous Poisson process with rate λ are equivalent.

1. *Counting description*: $N(0) = 0$; counts in disjoint intervals are independent; $N(t+s) - N(t) \sim \text{Poisson}(\lambda s)$.
2. *Interval description*: inter-event intervals $\tau_1, \tau_2, \ldots$ are iid exponential with mean $1/\lambda$, and $t_k = \tau_1 + \cdots + \tau_k$.
3. *Infinitesimal description*: $P(N(t+h) - N(t) = 1) = \lambda h + o(h)$ and $P(N(t+h) - N(t) \geq 2) = o(h)$, with independent increments.

Proposition 34.3. *Under the counting description, the first event time is exponential with rate λ.*

Proof. $P(t_1 > t) = P(N(t) = 0) = e^{-\lambda t}$. □

The interval description gives the simplest simulation: draw $\tau_k = -\log(U_k)/\lambda$ for iid uniform U_k and accumulate. The *conditional uniformity* property is also useful: given $N(T) = n$, the n event times are distributed as the order statistics of n iid uniform points on $[0, T]$. This property underlies spike-time randomization nulls; it holds for homogeneous Poisson processes and fails for any process with refractoriness or bursting.

34.5 Inhomogeneous processes, thinning, and time rescaling

For an inhomogeneous Poisson process with intensity $\lambda(t)$, define the cumulative intensity $\Lambda(t) = \int_0^t \lambda(s)\, ds$. Counts in disjoint intervals are independent and $N(b) - N(a) \sim \text{Poisson}(\Lambda(b) - \Lambda(a))$.

Theorem 34.4 (Time rescaling for Poisson processes)**.** *If $t_1 < t_2 < \cdots$ are the events of an inhomogeneous Poisson process with continuous positive intensity, then $\Lambda(t_1), \Lambda(t_2), \ldots$ are the events of a homogeneous Poisson process with rate one. Equivalently, the rescaled intervals $\Lambda(t_k) - \Lambda(t_{k-1})$ are iid exponential with mean one.*

Proof. The map $t \mapsto \Lambda(t)$ is continuous and strictly increasing. The number of rescaled events in $(\Lambda(a), \Lambda(b)]$ equals $N(b) - N(a)$, which is Poisson with mean $\Lambda(b) - \Lambda(a)$, the length of the rescaled interval; independence over disjoint intervals is preserved. □

The general theorem replaces $\lambda(t)$ by the conditional intensity, so that any correctly specified history-dependent model also maps spikes to a unit-rate Poisson process. Testing whether the rescaled intervals look exponential (via $u_k = 1 - e^{-(\Lambda(t_k) - \Lambda(t_{k-1}))}$, which should be uniform) is the basis of KS plots for spike-train models.

Thinning simulates an inhomogeneous process: generate a homogeneous process at rate $\lambda_{\max} \geq \sup_t \lambda(t)$ and keep each event at time t independently with probability $\lambda(t)/\lambda_{\max}$. The kept events form a Poisson process of intensity $\lambda(t)$.

34.6 Renewal processes and interval statistics

A *renewal process* has iid inter-event intervals with a general distribution on $(0, \infty)$. It can express refractoriness (a gap near zero) and regularity, but not dependence between successive intervals. Two summary statistics are standard:

- the *coefficient of variation* $\mathrm{CV} = \mathrm{sd}(\tau)/\mathbb{E}(\tau)$, equal to 1 for the Poisson process, below 1 for regular firing, and above 1 for bursty firing;
- the *Fano factor* $F(T) = \mathrm{Var}\, N(T)/\mathbb{E}N(T)$ of counts in windows of length T.

For a stationary renewal process, $F(T) \to \mathrm{CV}^2$ as $T \to \infty$. For finite T, both are biased by rate fluctuations across a session, so the local measure $\mathrm{CV}_2 = \frac{2|\tau_{k+1}-\tau_k|}{\tau_{k+1}+\tau_k}$, averaged over k, is often preferred: it compares adjacent intervals and is insensitive to slow rate drift. A gamma-distributed renewal process with shape κ has $\mathrm{CV} = 1/\sqrt{\kappa}$.

34.7 Second-order statistics: autocorrelograms and the Bartlett spectrum

For a stationary point process with rate λ, the *conditional intensity given an event at* 0, $\lambda\, g(\tau)$, describes how likely another event is at lag τ. Its histogram estimate is the autocorrelogram. For a homogeneous Poisson process, $g \equiv 1$: the autocorrelogram is flat. A dip near zero reflects refractoriness; a peak at ± 125 ms reflects 8 Hz rhythmicity. The Fourier transform of the covariance density gives the point-process spectrum (Bartlett spectrum), which tends to λ at high frequencies for any simple process; a spike train is therefore never "white" or "band-limited" in the way a continuous signal can be.

Worked example 34.5 (Refractoriness changes the counts)**.** A neuron firing at 20 spikes/s with an absolute refractory period of $d = 3$ ms can be modeled as a dead-time-modified Poisson process: intervals are d+Exponential(μ) with $d + 1/\mu = 50$ ms. Then $\mathrm{CV} = (1/\mu)/(d + 1/\mu) = 47/50 = 0.94$, and the long-window Fano factor is $0.94^2 \approx 0.88$. Refractoriness alone makes counts sub-Poisson, so a measured Fano factor slightly below one is not evidence of a precise temporal code.

Exercises

1. Derive the probability of at least one spike in a window under a homogeneous Poisson model.
2. Explain why spike sorting uncertainty can affect autocorrelograms.
3. Distinguish an observed spike count from conditional intensity.
4. Give a point-process model that excludes spikes for a fixed period after each event.
5. Show that the sum of two independent Poisson processes with rates λ_1, λ_2 is a Poisson process with rate $\lambda_1 + \lambda_2$.
6. Prove that thinning a homogeneous process with constant retention probability p gives a Poisson process with rate $p\lambda$.
7. Compute CV for a gamma renewal process with shape $\kappa = 4$.
8. Explain why the time-rescaling theorem requires the intensity model to use only past information.

Chapter 35

Spike-train GLMs, resampling, and conditional hypotheses

Reading goal 35.1. Build spike prediction models and valid null tests from conditional probabilities instead of selecting statistics first.

35.1 A Poisson generalized linear model

A GLM relates the expected count to covariates through a link function. For binned counts n_k over exposure Δ_k, one specification is

$$n_k \mid x_k \sim \text{Poisson}(\mu_k), \qquad \log \mu_k = \log \Delta_k + \beta_0 + x_k^\top \beta.$$

The offset $\log \Delta_k$ ensures that changing bin exposure has the correct interpretation in rate units. Under this model, $\lambda_k = \exp(\beta_0 + x_k^\top \beta)$ is the conditional event rate. For sufficiently fine bins a Bernoulli model with appropriate link can also be used, but the likelihood assumptions differ.

35.2 History filters and coupling

A point-process GLM can augment predictors with spike-history bases or other units' recent spikes. Negative recent-self-history weights may express refractoriness; positive interactions may express predictive coupling. The estimated coefficients depend on basis choice, observation artifacts, unobserved common inputs, and regularization. Coupling filters are model parameters and not direct measurements of synaptic transmission.

35.3 Model assessment rather than fit alone

Inspect residual or time-rescaled-event structure, held-out log-likelihood, calibration, stability across bin widths, and sensitivity to covariates. Time-rescaling transforms integrated intensities between consecutive events; a correctly specified conditional-intensity model yields rescaled intervals with a known reference distribution. Failure suggests misspecification but does not identify the correct alternative automatically.

35.4 Jitter and conditional resampling

A resampling null model may fix the spike count inside each jitter window and randomize event locations within specified allowed regions. Exactness of conditional tests depends on the null

distribution used for resampling. Generic "jitter the spikes" is not a statistical procedure until its preserved quantities, randomized quantities, boundary handling, and sampling distribution are fully specified. Conditional modeling clarifies the hypotheses supported by interval jitter and related tests [1].

If T_{obs} is a test statistic and $T_1^*, \dots, T_B^*$ are exchangeable null replicates under the test construction, a common Monte Carlo tail estimate is

$$\hat{p} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{T_b^* \geq T_{\mathrm{obs}}\}}{B+1}.$$

The plus-one convention avoids zero estimated p-values, but validity also requires the simulation to represent the intended null.

Worked example 35.2 (What is preserved?)**.** Suppose one window has three spikes. A conditional jitter method preserving this count cannot test whether the process increased its number of spikes in that window; it can test aspects of fine timing within the window. Rejection must be interpreted relative to preserved coarse structure, not as evidence for a particular synapse.

35.5 Concavity of the Poisson GLM log-likelihood

With $\log \mu_k = \log \Delta_k + x_k^\top \beta$ (absorbing β_0 into β with a constant regressor), the log-likelihood is

$$\ell(\beta) = \sum_k \left[n_k x_k^\top \beta - \Delta_k e^{x_k^\top \beta} \right] + \text{const.}$$

Proposition 35.3. *The gradient and Hessian are*

$$\nabla \ell(\beta) = \sum_k (n_k - \mu_k) x_k, \qquad \nabla^2 \ell(\beta) = -\sum_k \mu_k x_k x_k^\top = -X^\top W X,$$

with $W = \mathrm{diag}(\mu_k)$. Hence ℓ is concave, and strictly concave if X has full column rank.

Proof. Differentiate term by term. For any v, $v^\top \nabla^2 \ell \, v = -\sum_k \mu_k (x_k^\top v)^2 \leq 0$, with equality only if $Xv = 0$. □

Concavity means every local maximum is global, and Newton's method converges reliably from reasonable starting points. The Newton step can be written as a weighted least-squares problem with working response $\tilde{y}_k = x_k^\top \beta + (n_k - \mu_k)/\mu_k$: this is *iteratively reweighted least squares*. The MLE may fail to exist if the data are *separable*, for example if a regressor is positive in every bin with spikes and zero otherwise: the likelihood then increases without bound as the coefficient grows. Regularization (a ridge penalty $-\frac{\alpha}{2}\|\beta\|^2$ keeps concavity) resolves this, at the cost of bias.

35.6 Basis functions for filters

A stimulus or history filter of length L bins has L free coefficients. Representing it as $h = Bc$ with a basis $B \in \mathbb{R}^{L \times J}$, $J \ll L$, reduces parameters and imposes smoothness. Raised-cosine bumps with logarithmically stretched time are common for history filters because they give fine resolution at short lags (refractoriness) and coarse resolution at long lags. The basis choice is a prior: features not representable in the span of B cannot be estimated, and the apparent shape of a fitted filter partly reflects the basis.

35.7 Goodness of fit with time rescaling

Given a fitted conditional intensity $\hat{\lambda}(t \mid \mathcal{H}_t)$ and spike times t_k, compute $z_k = \int_{t_{k-1}}^{t_k} \hat{\lambda}(s \mid \mathcal{H}_s)\, ds$ and $u_k = 1 - e^{-z_k}$. If the model is correct, the u_k are iid uniform. A KS plot compares their sorted values with uniform quantiles, with approximate 95% bands $\pm 1.36/\sqrt{K}$ for K spikes. Two practical caveats apply. In discrete time, the integral must be computed with a correction for the bin width, or the test rejects correct models at high rates. And with many spikes the bands become narrow enough that every model is rejected; the KS plot is then more useful as a diagnostic of *where* the model fails than as a binary test.

35.8 Validity of Monte Carlo tests

Proposition 35.4. *Suppose that under H_0 the values $T_{\text{obs}}, T_1^*, \dots, T_B^*$ are exchangeable and almost surely distinct. Then $P_{H_0}(\hat{p} \le \alpha) \le \alpha$ for every $\alpha \in (0,1)$, where $\hat{p} = (1 + \#\{b : T_b^* \ge T_{\text{obs}}\})/(B+1)$.*

Proof. By exchangeability, the rank of T_{obs} among the $B+1$ values is uniformly distributed on $\{1, \dots, B+1\}$. The event $\hat{p} \le \alpha$ says that T_{obs} is among the $\lfloor \alpha(B+1) \rfloor$ largest, which has probability $\lfloor \alpha(B+1) \rfloor/(B+1) \le \alpha$. □

The hypothesis is exchangeability *of the observed data with the surrogates under the null*. Interval jitter achieves it exactly when the null is "given the counts in each window, spike positions within windows are uniform and independent": the observed train is then one draw from the same conditional distribution that generates the surrogates. Shuffling procedures that do not have this property yield p-values whose validity is unknown.

Worked example 35.5 (A coupling filter that is really a common input)**.** Simulate two neurons that are independent given a shared slow input $s(t)$, with intensities $\exp(a + s(t))$ for each. Fit a GLM for neuron 2 that includes neuron 1's recent spikes but omits s. The fitted coupling filter from 1 to 2 is positive at short lags, because neuron 1's spikes are a noisy proxy for s. Including an estimate of s, for instance a smoothed population rate or the LFP, shrinks the coupling filter towards zero. The coefficient measured the predictive value of neuron 1 given the regressors *in the model*; it changes whenever the model changes.

Exercises

1. Derive the score of the Poisson log-likelihood with respect to β.
2. State the null hypothesis and exact conditioning in a 20-ms interval jitter test.
3. Explain why randomizing every timestamp independently across a session may destroy behavioral modulation.
4. Design a split-trial validation plan that avoids training and testing on overlapping windows from the same trial.
5. Derive the IRLS working response from Newton's method for the Poisson GLM.
6. Construct a tiny dataset (three bins) for which the Poisson GLM MLE does not exist.
7. Show that adding a ridge penalty keeps the log-likelihood strictly concave.
8. Prove that, under exchangeability, $P(\hat{p} \le \alpha) = \lfloor \alpha(B+1) \rfloor/(B+1)$ when ties have probability zero.

Chapter 36

Neuronal covariance and the geometry of population responses

Reading goal 36.1. Distinguish signal correlation, noise correlation, factor structure, and the information represented in population covariance.

36.1 One neuron versus a population

Let $r(t) = (r_1(t), \dots, r_N(t))^\top$ represent firing-rate estimates for N units. A time series $r(t)$ traces a curve through $\mathbb{R}^N$. Neurons' responses may change together because of shared inputs, latent states, task drive, or interactions. The covariance matrix captures second-order cofluctuation but does not identify which explanation generated it.

In repeated-stimulus experiments, *signal correlation* measures similarity in mean tuning across stimuli; *noise correlation* measures trial-to-trial cofluctuation after controlling the stimulus or condition. These are distinct statistics, and their estimates depend on temporal window, firing rate, spike sorting, and brain state [11].

36.2 Shared gain and overdispersion

A modulated Poisson count model can be written $N \mid G \sim \text{Poisson}(G\mu)$, where $G > 0$ is a random gain with $E[G] = 1$. Then $E[N] = \mu$ and

$$\text{Var}(N) = \mu + \mu^2 \,\text{Var}(G).$$

The second term represents fluctuations in the latent gain; the decomposition follows directly from the law of total variance. For two neurons with gains G_1, G_2, shared gain fluctuations can generate positive count covariance. This does not require a direct synaptic link between their spike generators [20].

36.3 Latent factors and factor analysis

A linear observation model $r_t = Cz_t + d + \varepsilon_t$ represents high-dimensional responses through lower-dimensional latent z_t. Factor analysis separates low-rank shared covariance from modeled observation noise. GPFA adds smooth temporal covariance priors to latent trajectories. Identifying the effective latent dimensionality requires validation and attention to scaling, stationarity, and neuron subsampling [21, 18].

36.4 Random-matrix caution

For a high-dimensional sample covariance, random eigenvalue fluctuations can be large even when the true covariance is isotropic. In a simple independent-Gaussian null with feature-to-sample ratio q, the Marčenko–Pastur bulk has support $\sigma^2(1 \pm \sqrt{q})^2$ under the standard asymptotic assumptions. This is not a universal biological noise threshold; correlated trials, heterogeneous variances, and temporal smoothing violate simple null assumptions.

36.5 Covariance from a latent factor model

Under the factor model $r = Cz + d + \varepsilon$ with $z \sim \mathcal{N}(0, I_k)$ independent of $\varepsilon \sim \mathcal{N}(0, \Psi)$, Ψ diagonal, the covariance of r is

$$\Sigma = CC^\top + \Psi.$$

The *shared* covariance $CC^\top$ has rank k; the *private* variance Ψ is neuron-specific. PCA treats all variance alike and therefore mixes the two: a neuron with large private variance attracts a principal component of its own. Factor analysis estimates C and Ψ by maximum likelihood (usually with the EM algorithm). The factor loadings are identifiable only up to rotation, since $C \mapsto CQ$ for orthogonal Q leaves Σ unchanged; the *shared subspace* $\mathrm{col}(C)$ is identifiable, the axes within it are not.

36.6 Noise correlations and coding

Consider a population encoding a scalar stimulus s through mean responses $\mu(s)$ with stimulus-independent covariance Σ. The *linear Fisher information* is

$$I_{\mathrm{lin}}(s) = \mu'(s)^\top \Sigma^{-1} \mu'(s).$$

It bounds the precision of any locally unbiased linear decoder: $\mathrm{Var}(\hat{s}) \geq 1/I_{\mathrm{lin}}$. Whether noise correlations help or hurt depends on their *direction* relative to the signal direction $\mu'(s)$, not on their average size.

Proposition 36.2. *If $\Sigma = \Sigma_0 + \epsilon\, \mu'\mu'^\top$ with Σ_0 positive definite and $\epsilon > 0$, then*

$$I_{\mathrm{lin}} = \frac{I_0}{1 + \epsilon I_0}, \qquad I_0 = \mu'^\top \Sigma_0^{-1} \mu',$$

which is bounded by $1/\epsilon$ no matter how many neurons are recorded.

Proof. By the Sherman–Morrison formula, $\Sigma^{-1} = \Sigma_0^{-1} - \epsilon \Sigma_0^{-1}\mu'\mu'^\top\Sigma_0^{-1}/(1 + \epsilon I_0)$. Hence $\mu'^\top\Sigma^{-1}\mu' = I_0 - \epsilon I_0^2/(1 + \epsilon I_0) = I_0/(1 + \epsilon I_0)$. □

These *differential* or *information-limiting* correlations are small in any single pair but cap the population's information. They are indistinguishable from upstream noise in the stimulus itself, which no downstream circuit can remove.

36.7 Estimating correlation from spike counts

The Pearson correlation of spike counts depends on the counting window and on firing rate even when the underlying dependence is fixed. For two neurons driven by a common Gaussian input and spiking by thresholding, the count correlation grows with window length and with firing rate. Comparing r_{sc} between conditions with different rates therefore requires matching rates or modeling the dependence. Short windows produce correlations close to zero for almost any mechanism, because counts are mostly 0 or 1.

36.8 Signal versus noise correlation from data

Given trials $r_j^{(s)}$ for stimulus s, the noise covariance is estimated within each stimulus and pooled; the signal covariance is the covariance of the stimulus-conditioned means across stimuli. Shuffling trial order independently for each neuron within a stimulus destroys noise correlations while preserving signal correlations, giving a direct test of whether noise correlations affect decoding. With few trials, the estimated means contain noise, which inflates the apparent signal covariance by approximately $\Sigma_{\text{noise}}/n_{\text{trials}}$; this bias should be removed or cross-validated.

Worked example 36.3 (Information saturation)**.** Let N neurons have $\mu_i' = 1$, private variances 1, and differential correlations of strength $\epsilon = 0.01$. Then $I_0 = N$ and $I_{\text{lin}} = N/(1 + 0.01N)$. For $N = 10$, $I_{\text{lin}} \approx 9.1$; for $N = 100$, 50; for $N = 1000$, ≈ 91; the limit is 100. The pairwise correlation coefficient implied is about $0.01/1.01 \approx 0.01$, below what most experiments can resolve reliably, yet it halves the information of a 100-neuron population.

Exercises

1. Derive the gain-modulated variance formula using total variance.
2. Explain how simultaneous recording differs from assembling pseudo-populations across sessions.
3. Give a counterexample where noise correlation increases without a change in the direct connection between neurons.
4. Explain why a large first eigenvalue is not automatically a genuine latent neural mode.
5. Show that C and CQ give the same covariance in the factor model for orthogonal Q.
6. Verify the Sherman–Morrison formula by multiplying Σ by the claimed inverse.
7. Compute I_{lin} for two neurons with $\mu' = (1, 1)$, unit variances, and correlation ρ; for which sign of ρ is information larger?
8. Explain why trial shuffling within a stimulus preserves signal correlations.

Chapter 37

State-space dynamics, neural manifolds, and rotations

Reading goal 37.1. Connect dynamical evolution, latent inference, and geometric structure while keeping descriptive and mechanistic claims separate.

37.1 A state is more than a snapshot

A discrete linear dynamical system has $z_{t+1} = Az_t + Bu_t + w_t$ and observations $y_t = Cz_t + v_t$. The hidden state z_t evolves according to a transition rule while observations mix it with noise. If the model is linear Gaussian, Kalman filtering estimates the posterior state sequentially. Smoothing uses future as well as past observations and is not equivalent to an online estimate.

37.2 Continuous dynamics and stability

A continuous linear system obeys $\dot{z} = Az + Bu(t)$. The eigenvalues of A govern local behavior: negative real parts give asymptotic decay for unforced linear systems, while complex eigenvalues generate damped rotational components. More generally $\dot{z} = f(z, u, t)$ can represent nonlinear dynamics. Stable equilibria and attractors are properties of the chosen model; inferring them from noisy, partial observations requires care.

37.3 jPCA and rotational population trajectories

Principal-component analysis selects directions of large variance, not necessarily directions displaying dynamics. The jPCA approach first reduces dimensionality, then fits an approximately skew-symmetric linear flow for the derivatives of the reduced trajectories, highlighting rotations [10]. The existence of a rotational projection is a description of population activity under chosen preprocessing; attributing that rotation to a particular recurrent mechanism requires additional evidence.

37.4 Neural manifolds and coordinate nonuniqueness

A low-dimensional manifold is a set locally parameterized by a smaller number of coordinates than its ambient dimension. If observations obey $y = g(z)$ with $z \in \mathbb{R}^d$, many coordinate systems for z can describe the same observed trajectories. Thus latent axes may not have unique

biological labels. A fitted low-dimensional model can be useful for decoding even when its coordinates are not uniquely identifiable. Reviews of movement-related manifolds emphasize dynamical and task constraints [17].

37.5 Nonlinear latent models

Sequential autoencoders and recurrent neural networks can infer nonlinear latent dynamics from single trials, under training objectives and architecture assumptions. Models such as LFADS use latent generative dynamics for population spike data. Flexible models can also overfit. Appropriate controls include held-out likelihood, behavior decoding, simulation-based identifiability tests, and stability across recording sessions.

Worked example 37.2 (A two-dimensional rotation)**.** Let $A = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix}$. Then $\dot z = Az$ has solutions $z_1(t) = R\cos(\omega t + \phi)$ and $z_2(t) = R\sin(\omega t + \phi)$. The trajectory is a circle of radius R. A change of orthonormal coordinates rotates the picture but leaves its essential frequency unchanged.

37.6 Solving linear systems of differential equations

For $\dot z = Az$ with $z(0) = z_0$, the solution is $z(t) = e^{tA}z_0$, where the matrix exponential $e^{tA} = \sum_{k\ge 0} t^k A^k/k!$ converges for every A. If $A = V\Lambda V^{-1}$ is diagonalizable, $e^{tA} = Ve^{t\Lambda}V^{-1}$ and each mode evolves as $e^{\lambda_j t}$. A complex pair $\lambda = \alpha \pm i\omega$ produces oscillations at angular frequency ω with envelope $e^{\alpha t}$. Hence:

Theorem 37.3. *The equilibrium $z = 0$ of $\dot z = Az$ is asymptotically stable if and only if every eigenvalue of A has negative real part. For the discrete system $z_{t+1} = Az_t$, the condition is that every eigenvalue has modulus less than one.*

For nonlinear systems $\dot z = f(z)$, the *linearization* at an equilibrium z^* ($f(z^*) = 0$) uses the Jacobian $J = Df(z^*)$. If every eigenvalue of J has negative real part, z^* is locally asymptotically stable; if one has positive real part, it is unstable; eigenvalues on the imaginary axis leave the question to higher-order terms (Hartman–Grobman applies only to hyperbolic equilibria).

37.7 Planar systems

For 2×2 real A with trace τ and determinant Δ, the eigenvalues are $\lambda = \big(\tau \pm \sqrt{\tau^2 - 4\Delta}\big)/2$. The equilibrium is a saddle if $\Delta < 0$; a node if $\Delta > 0$ and $\tau^2 \ge 4\Delta$; a focus (spiral) if $\Delta > 0$ and $\tau^2 < 4\Delta$; stable iff $\tau < 0$ and $\Delta > 0$; and a center if $\tau = 0 < \Delta$ (rotation without decay). This trace–determinant classification is the basis for reading two-dimensional neuron models such as FitzHugh–Nagumo, where a change in input current moves the equilibrium from a stable focus to an unstable one surrounded by a limit cycle (a Hopf bifurcation), producing periodic firing.

37.8 The Kalman filter

For the linear Gaussian model $z_{t+1} = Az_t + w_t$, $y_t = Cz_t + v_t$ with $w_t \sim \mathcal{N}(0, Q)$, $v_t \sim \mathcal{N}(0, R)$, the posterior $p(z_t \mid y_{1:t})$ is Gaussian with mean $\hat{z}_{t|t}$ and covariance $P_{t|t}$, computed recursively:

$$\begin{aligned} \hat{z}_{t|t-1} &= A\hat{z}_{t-1|t-1}, & P_{t|t-1} &= AP_{t-1|t-1}A^\top + Q, \\ K_t &= P_{t|t-1}C^\top(CP_{t|t-1}C^\top + R)^{-1}, & & \\ \hat{z}_{t|t} &= \hat{z}_{t|t-1} + K_t(y_t - C\hat{z}_{t|t-1}), & P_{t|t} &= (I - K_tC)P_{t|t-1}. \end{aligned}$$

The update is the Gaussian conditioning formula applied to the joint distribution of z_t and y_t given the past. The innovation $y_t - C\hat{z}_{t|t-1}$ is white under a correct model, which provides a diagnostic. In brain–computer interfaces, the Kalman filter decodes kinematics z from firing rates y; its gain K_t balances the prior dynamics against the new neural evidence.

37.9 Observability and identifiability

The state is *observable* if the initial state can be determined from noiseless outputs. For the linear system this holds iff the observability matrix $[C^\top, (CA)^\top, \dots, (CA^{n-1})^\top]^\top$ has rank n. Observability is about states for known (A, C). *Identifiability* of (A, C) from data is weaker: the substitution $z \mapsto Mz$ for invertible M replaces (A, C, Q) by $(MAM^{-1}, CM^{-1}, MQM^\top)$ with identical output distributions. Eigenvalues of A (time constants and rotation frequencies) are invariant under this change and are therefore estimable; the individual latent axes are not.

Worked example 37.4 (Rotations are invariant; axes are not)**.** Fit the rotation model of the previous example with an arbitrary invertible M: $A' = MAM^{-1}$. If M is not orthogonal, the trajectory in the new coordinates is an ellipse rather than a circle, but A' still has eigenvalues $\pm i\omega$, and the rotation frequency is unchanged. Statements about "the rotation frequency" are identifiable; statements about "the second latent dimension encodes velocity" require additional constraints, such as aligning axes to behavioral variables, before they are meaningful.

Exercises

1. Determine eigenvalues of the rotation matrix above.
2. Describe a dynamical system with a stable fixed point.
3. Explain why PCA dimensions and dynamical dimensions need not coincide.
4. Name two empirical checks required before claiming a neural manifold controls behavior.
5. Compute e^{tA} for the rotation generator $A = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix}$ using its power series.
6. Classify the equilibrium of $\dot{z} = Az$ for $A = \begin{bmatrix} -1 & -4 \\ 1 & -1 \end{bmatrix}$.
7. Derive the scalar Kalman filter for $z_{t+1} = z_t + w_t$, $y_t = z_t + v_t$, and find the steady-state gain.
8. Show that the eigenvalues of MAM^{-1} equal those of A.

Part VIII

Scientific inference and reproducible methods

Chapter 38

Hypothesis testing, resampling, and uncertainty

Reading goal 38.1. Understand what a test proves, how uncertainty is quantified, and why a null model must preserve the right structure.

38.1 What is a hypothesis test?

A test begins by specifying a null family H_0 of possible data-generating mechanisms and a test statistic $T(D)$. The *null distribution* describes the variation of T under that family. A p-value is the probability, under the stipulated null model, of observing a statistic at least as extreme as the realized one according to the prespecified rule. It is not the posterior probability that the null is true, nor is it the probability the result occurred "by chance" without further qualification.

38.2 Type I error, power, and effect size

A level-α test controls the chance of rejecting a true null, provided its assumptions hold. Power is the rejection probability under a specified alternative. Statistical significance alone says little about effect magnitude. A coherence difference that is tiny but precisely estimated may be statistically significant, while a large effect in a small dataset may remain uncertain. Report the statistic, units, effect size, sampling unit, interval estimate, and model assumptions.

38.3 Resampling methods

A bootstrap approximates an estimator's sampling variability by resampling observational units from the data under specified conditions. The units might be animals or independent trials; sampling every LFP time point independently typically destroys serial dependence. A permutation test requires an exchangeability justification under the null, often by randomizing treatment labels under a randomized design. Surrogate time shifts can test certain synchrony claims only if their shift mechanism preserves the appropriate slow structure while breaking the tested fine-time association. These methods are not interchangeable.

38.4 Hierarchical data and pseudoreplication

Samples nested within a trial, trials within sessions, and sessions within animals are rarely independent. Treating every overlapping 100-ms analysis window as a fresh biological replicate inflates nominal sample size. A hierarchical model, cluster bootstrap, or analysis at the appropriate experimental unit can address some dependencies. Repeated statistical tests across channel pairs, frequency bins, or time bins require a multiple-comparison strategy adapted to the intended claim.

38.5 Bias, generalization, and validation

Preprocessing fitted on all the data before cross-validation can leak information from test folds into training. Selection of frequency bands after inspecting the outcome similarly biases later inference unless validated separately. Hold out whole animals or sessions when the scientific goal is cross-animal generalization. Predictive success on held-out data supports a model's useful regularities, not necessarily the literal truth of its latent biological explanation.

Worked example 38.2 (An exact conditional question)**.** To test whether two cells exhibit synchrony below a 5-ms scale beyond fixed 20-ms window spike counts, define windows, fixed spike counts, a synchrony statistic, and a resampling mechanism that randomizes fine timing conditional on the fixed counts. Any rejection should refer to the particular 20-ms coarse information held fixed, as emphasized in [1].

38.6 The Neyman–Pearson lemma

Theorem 38.3 (Neyman–Pearson)**.** *For testing a simple null $H_0 : X \sim p_0$ against a simple alternative $H_1 : X \sim p_1$, the test that rejects when the likelihood ratio $p_1(X)/p_0(X)$ exceeds a threshold c (chosen so that the size is α) has the greatest power among all tests of size at most α.*

Proof. Let $\phi^* = \mathbf{1}\{p_1 > cp_0\}$ and let ϕ be any test with $\mathbb{E}_0\phi \le \alpha = \mathbb{E}_0\phi^*$. Pointwise, $(\phi^* - \phi)(p_1 - cp_0) \ge 0$, since $\phi^* = 1$ exactly where $p_1 - cp_0 > 0$. Integrating gives $\mathbb{E}_1\phi^* - \mathbb{E}_1\phi \ge c(\mathbb{E}_0\phi^* - \mathbb{E}_0\phi) \ge 0$. □

The lemma explains why likelihood-ratio statistics are the default: they are optimal for the simplest problem and asymptotically well behaved more generally. Under regularity, Wilks' theorem states that $2\log$ of the maximized likelihood ratio between nested models differing by q parameters is approximately χ^2_q under the null. Nested GLMs (e.g. with and without a coupling term) are compared this way; non-nested models require cross-validation or information criteria instead.

38.7 Confidence intervals and their duality with tests

A $(1-\alpha)$ confidence set $C(X)$ satisfies $P_\theta(\theta \in C(X)) \ge 1-\alpha$ for every θ. Given a family of level-α tests of $H_0 : \theta = \theta_0$, the set of θ_0 not rejected is a $(1-\alpha)$ confidence set, and conversely. The statement concerns the *procedure*: before the data are seen, the random set covers the fixed parameter with the stated probability. A realized interval either contains θ or does not.

38.8 Multiple comparisons

For m tests, let V be the number of false rejections and R the number of rejections. Two error rates are common:

- the *family-wise error rate* $\mathrm{FWER} = P(V \geq 1)$, controlled by Bonferroni (reject $p_i \leq \alpha/m$) or the uniformly more powerful Holm step-down procedure;
- the *false discovery rate* $\mathrm{FDR} = \mathbb{E}[V/\max(R,1)]$, controlled by the Benjamini–Hochberg procedure.

Theorem 38.4 (Benjamini–Hochberg)**.** *Order the p-values $p_{(1)} \leq \cdots \leq p_{(m)}$, let k be the largest i with $p_{(i)} \leq i\alpha/m$, and reject the hypotheses with the k smallest p-values. If the null p-values are independent of each other and of the non-null p-values, then* $\mathrm{FDR} \leq m_0\alpha/m \leq \alpha$*, where m_0 is the number of true nulls.*

Under arbitrary dependence, replacing α by $\alpha/\sum_{i\leq m} 1/i$ (Benjamini–Yekutieli) restores control. For maps over time, frequency, or channels, *cluster-based permutation tests* control the FWER for the statement "there is some effect somewhere" by comparing the largest observed cluster statistic with its permutation distribution. They do not license claims about the exact extent or location of a cluster.

38.9 The bootstrap

Given iid data $X_1, \ldots, X_n$ and an estimator $\hat\theta$, the nonparametric bootstrap draws n observations with replacement, recomputes $\hat\theta^*$, and repeats B times. The spread of $\hat\theta^*$ estimates the sampling spread of $\hat\theta$. The percentile interval $[\hat\theta^*_{\alpha/2}, \hat\theta^*_{1-\alpha/2}]$ is simple; bias-corrected and accelerated (BCa) intervals improve coverage for skewed estimators. The bootstrap fails, or needs modification, for extremes (e.g. the maximum), for very small samples, and for dependent data, where whole blocks of consecutive observations (block bootstrap) or whole clusters (cluster bootstrap) must be resampled.

Worked example 38.5 (Thirty electrodes, one hundred frequencies)**.** A study tests coherence change at 30 channel pairs and 100 frequency bins, 3000 tests. With uncorrected $\alpha = 0.05$ and no true effect, about 150 false positives are expected. Bonferroni requires $p \leq 1.7 \times 10^{-5}$; Benjamini–Hochberg adapts to the number of true effects; a cluster test over frequency within each pair exploits the fact that true spectral effects span neighboring bins. The three answers address different claims: "no false positive anywhere", "few false positives among the reported ones", and "some effect exists in this pair".

Exercises

1. State the difference between an effect-size estimate and a hypothesis-test decision.
2. Propose a hierarchical resampling unit for a dataset containing animals, sessions, trials, and time windows.
3. Explain why using the test set to choose a frequency band biases held-out performance.
4. Describe a null distribution that would be inappropriate for testing phase-amplitude coupling under strong signal nonstationarity.
5. Apply Holm and Benjamini–Hochberg at $\alpha = 0.05$ to the p-values $0.001, 0.008, 0.012, 0.04, 0.2$.

6. Derive the likelihood-ratio test for a Poisson rate $\lambda = \lambda_0$ versus $\lambda = \lambda_1 > \lambda_0$, and show it rejects for large counts.

7. Explain why bootstrapping individual time points of an autocorrelated LFP envelope underestimates uncertainty.

8. Show that the set of θ_0 not rejected by level-α tests is a $(1-\alpha)$ confidence set.

Chapter 39

Numerical analysis and reproducible scientific computation

Reading goal 39.1. Translate mathematical equations into algorithms without overlooking finite precision, memory layout, and validation.

39.1 A formula is not yet an algorithm

A mathematical expression can define an exact quantity while computer code approximates it. Floating-point arithmetic rounds values and is not always associative: $(a+b)+c$ and $a+(b+c)$ can differ. An apparently exact bin count computed by $\lfloor T/\Delta \rfloor$ can be off by one if T/Δ is represented just below an integer. Implementations must state endpoint conventions and test boundary cases rather than rely on a universal ad hoc epsilon.

39.2 Conditioning and algorithmic stability

A well-posed problem has a solution that exists, is unique, and depends continuously on the data in an appropriate sense. An ill-conditioned problem may magnify small perturbations even under perfect implementation. A numerically unstable algorithm may magnify errors beyond what conditioning requires. Computing least-squares coefficients via QR or SVD is typically more stable than directly inverting $X^\top X$. Regularization stabilizes some inverse problems by choosing a constrained or penalized solution, at the cost of explicit modeling assumptions.

39.3 Memory and complexity

A dense sample covariance among N neurons requires $O(N^2)$ storage, while a full matrix decomposition can be computationally expensive for large N. When sample count T is much smaller than N, methods operating on a $T \times T$ Gram matrix can be more efficient. Signal operations can frequently be evaluated chunk by chunk, although filtering chunks requires state continuity or overlap handling. Memory mapping avoids loading full arrays but does not eliminate I/O costs.

39.4 Testing the mathematical contract

Good numerical tests use known synthetic inputs: a sinusoid with known frequency for PSD estimation; a complex exponential for phase operations; a Poisson simulation for count statistics;

a low-rank signal with controlled noise for PCA; and a known VAR for Granger analysis. Property-based tests can verify constraints such as covariance symmetry, nonnegative variance, PLV $\in [0, 1]$, and consistency of concatenated block processing with full-array processing within a stated tolerance.

39.5 Scientific metadata as part of the result

An estimate needs sampling rate, time origin, measurement units, reference configuration, filtering parameters, electrode and trial identifiers, quality masks, random seeds, and software versions. Metadata are part of the mathematical map between the array and its real-world interpretation. A floating-point matrix detached from its coordinates can be mathematically valid yet scientifically unusable.

39.6 Floating-point error analysis

The standard model of floating-point arithmetic states that for each basic operation $\circ \in \{+, -, \times, /\}$, $\mathrm{fl}(a \circ b) = (a \circ b)(1 + \delta)$ with $|\delta| \le u = \epsilon_{\mathrm{mach}}/2$. Errors in a long computation are analysed by propagating these factors.

Proposition 39.2 (Error of recursive summation)**.** *Computing $s = \sum_{i=1}^{n} x_i$ left to right gives $\hat{s}$ with $|\hat{s} - s| \le (n-1)u \sum_i |x_i| + O(u^2)$.*

The bound grows linearly in n and with $\sum |x_i|$, not $|s|$: summing many terms of mixed sign whose total nearly cancels gives large *relative* error. Pairwise (tree) summation reduces the factor $n - 1$ to $\lceil \log_2 n \rceil$, and compensated (Kahan) summation reduces it to a constant.

Catastrophic cancellation. Subtracting two nearly equal rounded numbers exposes their rounding errors. The one-pass variance formula $\frac{1}{n}\sum x_i^2 - \bar{x}^2$ suffers from this when the mean is large relative to the standard deviation: for a signal with mean $10^4\,\mu$V and standard deviation $1\,\mu$V, the two terms agree to eight digits and the computed variance retains only about eight correct digits in double precision, or none in single precision. The two-pass formula $\frac{1}{n}\sum (x_i - \bar{x})^2$, or Welford's streaming update, avoids the problem.

39.7 Backward stability

An algorithm is *backward stable* if its computed answer is the exact answer for slightly perturbed data, with perturbation of order u. Combined with the condition number, this gives the rule of thumb

$$\text{relative forward error} \lesssim \kappa \times \text{backward error} \approx \kappa u.$$

Householder QR and the SVD are backward stable for least squares, so the error is governed by $\kappa(X)$. Solving the normal equations is governed by $\kappa(X)^2$. For $\kappa(X) = 10^6$ in double precision, QR may deliver about ten correct digits and the normal equations about four.

39.8 Complexity of core algorithms

Operation	Cost
FFT of length N	$O(N \log N)$
Direct convolution, signal N, filter L	$O(NL)$
FFT convolution (overlap–add)	$O(N \log L)$
Covariance of $T \times N$ data	$O(TN^2)$
Full SVD of $T \times N$, $T \geq N$	$O(TN^2)$
Top-k SVD by randomized methods	$O(TNk)$
Pairwise cross-correlograms, N units	$O(N^2)$ pairs

For $N = 1000$ units, all pairwise cross-correlograms involve about 5×10^5 pairs; at one millisecond per pair the job takes about eight minutes, and storing 201 lag bins per pair in double precision requires about 0.8 GB. Estimating cost before running a full analysis is part of numerical design.

39.9 Randomness and reproducibility

Pseudorandom number generators are deterministic sequences initialized by a seed. Recording the seed and generator type makes a resampling analysis exactly reproducible. Monte Carlo p-values carry their own sampling error: with B surrogates and true p, the estimate has standard error about $\sqrt{p(1-p)/B}$. Near $p = 0.05$, $B = 1000$ gives a standard error near 0.007, so a reported $p = 0.048$ is not distinguishable from 0.052. When a decision depends on the third decimal, increase B.

Worked example 39.3 (Counting bins without floating-point surprises)**.** Spike times are stored in seconds and binned at $\Delta = 0.001$ s. Computing the bin index as `floor(t/0.001)` for $t = 0.003$ may return 2, because neither 0.003 nor 0.001 is exactly representable and their quotient can round to 2.9999999999999996. Storing times as integer sample counts at the acquisition rate, or computing indices as `floor(round(t*fs)/ (fs*Delta))` with integer arithmetic, removes the ambiguity. The mathematical definition of a bin is exact; only its floating-point implementation is fragile.

Exercises

1. Explain why zero-padding an FFT does not improve physical spectral resolution.
2. Design a test for chunked FIR filtering that detects boundary discontinuities.
3. List the metadata necessary to interpret a Hilbert-envelope moment series from 10-second windows.
4. Show that the two-pass variance formula is algebraically equal to the one-pass formula, and explain why they differ numerically.
5. Implement Welford's update $\bar{x}_k = \bar{x}_{k-1} + (x_k - \bar{x}_{k-1})/k$, $M_k = M_{k-1} + (x_k - \bar{x}_{k-1})(x_k - \bar{x}_k)$, and verify that $M_n/(n-1)$ is the sample variance.
6. Estimate the time and memory needed to compute a 2048-point FFT spectrogram of a 2-hour, 384-channel recording at 2.5 kHz with 50% overlap.
7. How many surrogates are needed for a Monte Carlo p-value near 0.01 to have standard error 0.001?

Chapter 40

A complete path from first definitions to real datasets

Reading goal 40.1. Practice the complete mathematical chain from a statement in English to a reproducible conclusion, with simulations preceding claims about physiology.

40.1 Project I: two signals that look coupled but are not

Start from the English claim: "two channels have high coherence because the associated neurons communicate." Translate the measurable component to a statistic $\mathcal{C}_{xy}(f)$ and identify the unsupported causal clause. Construct independent Gaussian noises n_x, n_y and a common oscillatory source s, then define $x = s+n_x$ and $y = s+n_y$. Calculate coherence, add measurement noise, and vary reference contamination. Compare estimates with and without the common source. Show that coherence may be high without direct communication. This project is motivated by [3, 12].

40.2 Project II: what a theta peak actually means

Simulate a process with a $1/f$-like background and a genuine narrowband theta oscillator. Next simulate only the aperiodic background, vary its exponent, and compute band power in both cases. Apply periodogram, Welch, and multitaper estimators; then compare parametric aperiodic fitting and an IRASA-type separation. Quantify false positives and biases across simulated repetitions before trying the methods on LFP. Here the hypotheses must distinguish a local peak *over background* from integrated power over a fixed canonical band [15, 19].

40.3 Project III: a manufactured phase-amplitude relationship

Simulate a nonsinusoidal slow oscillator with harmonic content, and separately simulate a high-frequency oscillator whose amplitude truly depends on slow phase. Filter both into low and high bands, compute analytic signals, and compare PAC measures. Repeat with sharp transient waveforms and phase surrogates that preserve the important nuisance structure. Identify which simulated condition actually contains two distinct interacting generators. The validation target is the generation mechanism, not merely a significant PAC statistic [2, 13].

40.4 Project IV: spikes, jitter, and conditional inference

Simulate independent Poisson spike trains sharing a slow time-dependent firing-rate modulation. A naive cross-correlogram may show coincidence peaks from rate cofluctuations. Define a conditional interval-jitter hypothesis holding coarse counts fixed and re-evaluate fine-time synchrony. Add precisely coordinated spikes and measure power across timescales and window widths. The lesson is to connect every resampling decision to an explicit null hypothesis [1].

40.5 Project V: latent neuronal dynamics

Simulate $z_{t+1} = Az_t + w_t$ for a two-dimensional rotating latent state, then observe N neurons through $r_t = Cz_t + \varepsilon_t$. Compare PCA, factor analysis, and a fitted linear state-space model. Vary the number of observed neurons, number of trials, observation noise, and dimensionality. Track held-out prediction and recovery of the latent subspace up to rotational ambiguity; avoid claiming exact recovery of nonidentifiable coordinate axes [10, 21, 17].

40.6 A research conclusion requires four layers

For each study, the final report should distinguish (1) the original English claim; (2) the formal mathematical target and its assumptions; (3) the numerical algorithm and validation outcomes; and (4) the physiological interpretation and its remaining alternatives. A clean implementation establishes only the third layer. A statistically significant result speaks to the second within its null model. A mechanistic conclusion requires evidence beyond either one in isolation.

40.7 Project VI: a point-process model of a place cell

Simulate a neuron whose intensity depends on position $x(t)$ along a linear track through a Gaussian tuning curve, $\lambda(t) = \lambda_0 + \lambda_1 \exp(-(x(t)-c)^2/(2w^2))$, together with a refractory history term. Fit (i) an occupancy-normalized rate map, (ii) a Poisson GLM with a spline basis in position, and (iii) the same GLM with a history filter. Compare estimates of c and w, compute KS plots by time rescaling, and quantify how many laps are needed for a given precision. Then make the true field drift slowly across laps and identify which fits report this drift and which average it away. The mathematical tools are Chapters 34 and 35; the scientific lesson is that "the place field" is an estimand defined by a model.

40.8 Project VII: source mixing and laminar profiles

Place three dipolar sources at different depths of a homogeneous medium and compute the potential at a 32-contact linear probe with $50\,\mu$m spacing using the formulas of Chapter 32. Give each source an independent time course. Compute the second-difference CSD, an inverse CSD with finite-radius discs, and a PCA of the laminar profiles. Vary the lateral extent of the sources and add independent noise at each contact. Determine when each method recovers the number, depth, and time courses of the sources, and when it creates spurious sinks at probe edges. The point of the project is to know the forward model before trusting any inverse.

40.9 Project VIII: population information and correlations

Simulate N neurons with tuning curves to orientation and noise covariance built from three components: private Poisson-like variability, a shared multiplicative gain, and a small differential-correlation term along $\mu'(s)$. Estimate linear Fisher information with bias correction as a function of N and of the number of trials, and compare with the value obtained after shuffling trials within stimulus. Identify which covariance component limits information and whether a realistic number of trials could detect the limit. Compare the results with the formulas of Chapter 36.

40.10 A checklist for each project

1. State the English claim and the formal estimand, with units.
2. List every assumption of the generative model and of the estimator.
3. Write down the null hypothesis, what it preserves, and what it randomizes.
4. Simulate data where the answer is known, including at least one case designed to fool the method.
5. Estimate bias, variance, coverage, and false-positive rate across many simulated datasets.
6. Only then apply the method to recordings, reporting the same quantities that were validated.
7. Separate in the conclusion what was established mathematically, numerically, statistically, and physiologically.

Each item corresponds to a part of this book: language and logic for (1), models and probability for (2) and (3), numerical analysis for (4) and (5), and the scientific chapters for (6) and (7).

Exercises

1. Draft a preregistered null hypothesis for one project, including the sampling unit, statistic, and exclusion rule.
2. Create a matrix matching five methods to their scientific estimands, key assumptions, and primary failure modes.
3. Design one negative control and one positive control for your chosen method.
4. For Project VI, derive the expected spike count in a lap from the intensity and the animal's trajectory.
5. For Project VII, compute the potential at a contact $100\,\mu$m above a dipole of moment $Id = 10^{-12}\,\mathrm{A\,m}$ aligned with the probe.
6. For Project VIII, state one result that would falsify the hypothesis that differential correlations limit information in your simulation.

Appendix A

Notation, assumptions, and units

Symbol	Meaning
$\in, \subseteq$	Membership in a set; inclusion of one set in another
$\forall, \exists, \neg, \Rightarrow$	For all; there exists; not; logical implication
$\mathbb{N}_0, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	Natural including zero, integer, rational, real, complex numbers
$\|x\|_2, \langle x, y \rangle$	Euclidean norm and inner product
$E[X], \mathrm{Var}(X), \mathrm{Cov}(X, Y)$	Expectation, variance, covariance
$\hat{\theta}, \theta$	Estimator/estimate and model parameter
$f_s, \Delta t, N$	Sampling frequency (Hz), interval (s), sample count
$x[n], x(t)$	Discrete sampled signal and continuous-time signal
$X[k], S_{xx}(f)$	DFT coefficient; power spectral density
$z(t), a(t), \phi(t)$	Analytic signal, amplitude envelope, phase
$N(t), \lambda(t \mid \mathcal{H}_t)$	Point-process count and conditional intensity
$\mathcal{C}_{xy}(f)$	Magnitude-squared coherence
$X \in \mathbb{R}^{T \times N}$	Population matrix: times by neurons
$\mathcal{X} \in \mathbb{R}^{N \times T \times R}$	Tensor: neurons by time by trials

Explicit conventions

PSD values have signal-units-squared per Hz under a density convention; firing rate uses spikes/s; phase is measured in radians unless labeled otherwise. Distinguish the sample count N from the point-process count $N(t)$ using context. The symbol T may denote a recording duration in signal chapters but number of observations in matrix expressions when explicitly defined. We use log for natural logarithm. Complex inner products are linear in their first argument by the convention stated in Chapter 17. A reported result must disclose its observational unit and any filtering or binning convention.

Appendix B

Provisional mathematical map to `matlab/path`

This is a **conceptual map, not a verified file inventory**. The GitHub repository and the actual `matlab/path` tree were not available during compilation. Names below describe analysis families and do not assert that particular MATLAB files are implemented.

Method family	Mathematical prerequisites	Checks before interpreting results
Binning and firing rates	Natural numbers; functions; point processes	Window edges, exposure, refractory period
FFT and PSD	Complex numbers; sequences; convolution	Sample rate, leakage, normalization, 1/f background
Hilbert envelopes	Complex phase; Fourier; filtering	Narrowband validity, edges, amplitude near zero
Sliding moments	Probability; expectation; central moments	Window overlap, variance conventions, autocorrelation
Coherence and PLV	Cross-spectra; circular statistics	Reference, common input, sample size, null design
PAC	Analytic signals; nonlinear statistics	Harmonics, transients, nonstationarity, surrogates
Granger and VAR	Linear algebra; stochastic processes; likelihood	Lag order, stationarity, latent common input
Spike synchrony	Point processes; conditional probability	Coarse counts, jitter window, precise null
PCA and SVD	Linear maps; covariance; spectral theorem	Scaling, noise, held-out stability
State-space models	EDOs; conditional inference; linear algebra	Observability, latent-coordinate ambiguity
CSD and inverse models	Multivariable derivatives; forward physics	Geometry, conductivity, regularization

For a future verified edition, each row should link to the exact MATLAB function, an explicit function signature, the corresponding mathematical equation, a reference, a tested example, and at least one synthetic negative control. An automatically generated index from the real repository would be preferable to copying guessed filenames.

Appendix C

Selected answers, proof sketches, and further problems

C.1 Selected answers

Why empty collections matter (Chapter 1). The statement $\forall x \in \varnothing\, P(x)$ is true because no element provides a counterexample. This is not evidence that any spike occurred: it is a logical consequence of quantifying over an empty domain.

Conditional order (Chapter 2). "Every cell fires at some time" has the form $\forall u \exists t F(u, t)$. "There exists a time when every cell fires" is $\exists t \forall u F(u, t)$. The second implies the first for a nonempty common time domain, but not conversely.

Count normalization (Chapter 10). Twenty spikes in 5 s gives 4/s; thirty spikes in 15 s gives 2/s. Their unweighted mean is 3/s; the pooled rate is $50/20 = 2.5$/s. The different answers correspond to different weighting rules.

Projection (Chapter 19). $P^2 = UU^\top UU^\top = UIU^\top = P$. Also $P^\top = P$. For any v in the column span of U, $Pv = v$.

Gain model (Chapter 36). Because $N \mid G$ is Poisson with mean $G\mu$, total variance yields $\operatorname{Var} N = E[G\mu] + \operatorname{Var}(G\mu) = \mu + \mu^2 \operatorname{Var} G$ when $EG = 1$.

C.2 Capstone proof and analysis problems

1. Start from the definition of a function and develop the meaning of a sampled time series, including its domain, range, units, and missing data.
2. Prove that the covariance matrix of a finite centered real data matrix is positive semidefinite. Identify when it fails to be positive definite.
3. From a linear state-space model, construct two different latent coordinate systems that generate identical observations.
4. Compare a phase-locking measure with an amplitude-weighted PAC measure on signals with very low amplitude. Characterize instability.

5. Write a full conditional resampling test for spike synchrony, explicitly stating what remains fixed under the null.

6. Derive the memory footprint and main arithmetic complexity of a covariance-based versus an SVD-based approach for $T \ll N$.

7. Design and report a blinded synthetic benchmark where an algorithm is required to distinguish an oscillation from a steep aperiodic spectrum.

These problems are deliberately cumulative. They require assembling definitions from several parts rather than treating methods as isolated formulas.

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